Probability Theory: The Language of Econometrics

A Capacity-Building Handout, from Measure-Theoretic Foundations to Asymptotic Theory

Eliab Luvanda

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Objective

The objective of this handout is to build capacity in econometric theory among staff and graduate students in the School of Economics, University of Dar es Salaam. Probability is the language in which econometric assumptions, estimator properties, and inferential claims are expressed; mastering that language makes the logic of econometrics substantially more transparent.

My aim throughout is therefore not to present probability theory for its own sake, but to develop precisely those ideas — probability spaces, random variables, expectation, the moment inequalities, the modes of convergence, and the limit theorems — upon which the asymptotic theory of econometrics rests, and to keep their econometric purpose in view at every stage. The intended reader is a member of staff or a graduate student who has some acquaintance with intermediate statistics and calculus, but who need not possess a strong background in real analysis or measure theory.

This handout is based on the sketchy notes that I have accumulated over the years and used to teach probability theory and statistical inference in the MA and PhD courses of the School of Economics, University of Dar es Salaam. In preparing it, I have tried as much as possible to include examples set in the context of Tanzania — drawing on household consumption, agricultural yields, rainfall shocks, and related settings — in order to motivate the understanding of the concepts and to root abstract ideas in problems that are familiar to the reader.

The exposition is measure-theoretic in its foundations yet deliberately accessible in its manner. Each principal result is accompanied by a discussion of its logic and intuition and by a remark on its relevance to econometric practice, so that the reader sees not only *that* a result holds but *why* it matters for empirical work. An appendix offers an informal, diagram-based introduction to Lebesgue measure and integration for those who wish to glimpse the underlying machinery, and a dedicated section employs Monte Carlo simulation in R to render the law of large numbers and the central limit theorem concrete rather than merely formal.

The Introduction gives the subjective interpretation somewhat more space because it provides the foundation for Bayesian econometrics, which is often less visible than classical and frequentist methods in the standard curriculum. The aim is not to rank interpretations of probability, but to prepare readers for Bayesian work in macroeconomics, forecasting, state-space models, and hierarchical analysis.

To support continued learning beyond the handout, the accompanying materials include the standalone R script used for the Monte Carlo simulations and a documented explanation of the code. Readers are encouraged to use these resources to modify the simulation settings, distributions, sample sizes, and number of replications, rerun the experiments, and continue exploring the law of large numbers and the central limit theorem independently.

In the interest of transparency, I disclose the assistance of Anthropic's Claude AI agent in preparing this handout and in writing, testing, debugging, and refactoring the R code used to run the Monte Carlo simulations. The selection and framing of the material, the judgements of emphasis and exposition, and any errors that remain are my own.

How to Use This Handout

The handout assumes familiarity with calculus, elementary statistics, and basic matrix algebra. It is organised as a cumulative sequence:

1. **Foundations (Sections 1–3):** interpretations of probability, probability spaces, events, and random variables.
2. **Distributions and moments (Sections 4–6):** distribution functions, independence, expectation, and moment inequalities.
3. **Asymptotic tools (Section 7):** modes of convergence, mapping results, laws of large numbers, and central limit theorems.
4. **Computation and foundations (Section 8 and the appendices):** Monte Carlo illustrations, an informal guide to Lebesgue integration, and reproducible R code.

On completing the handout, the reader should be able to formulate a probability model, distinguish the principal modes of stochastic convergence, state the assumptions behind the main LLNs and CLTs, and explain how those results establish consistency and asymptotic normality in econometric applications. Readers seeking an econometrics-first treatment may use Hansen (2022) and Bierens (2004) alongside Sections 6–8; readers seeking fuller measure-theoretic foundations may consult Billingsley (1995), Durrett (2019), or Pollard (2002).

1 Introduction

1.1 Probability and Uncertainty

In a world of perfect information, economic analysis would reduce to a deterministic calculus: households would allocate resources with certainty, firms would optimise production against known prices and costs, and governments would design policy in full knowledge of its consequences. The real world, however, is irreducibly uncertain. Outcomes that matter enormously to economic agents — harvests, prices, wages, health shocks, interest rates, policy changes — are not known in advance and cannot be controlled.

Probability theory provides the standard rigorous language for reasoning about such uncertainty. By assigning numerical measures to the likelihood of events before they are realised, probability transforms vague intuitions about chance into a precise, internally consistent calculus that can be embedded in optimisation problems, tested against data, and used to derive welfare implications. Without this foundation, core constructs of modern economics — expected utility, risk aversion, insurance, option pricing, hypothesis testing, and identification in causal inference — would lose their usual formal basis.

The importance of probability in economics goes beyond mere bookkeeping of likelihoods. It provides the structural scaffolding for rational decision-making under uncertainty. When an agent faces a decision whose consequences depend on an unknown state of the world, the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ formalises the environment: Ω lists the states that might obtain, $\mathcal{F}$ specifies which events are observable and contractible, and $\mathbb{P}$ encodes beliefs about how likely each state is.

The agent's objective — typically the maximisation of expected utility — is then a probability-weighted average of utilities across states, and the entire machinery of constrained optimisation can be brought to bear. Crucially, probability also disciplines empirical work: econometric estimators, confidence intervals, hypothesis tests, and identification strategies are all defined in terms of probability limits, distributions, and measures. The probability space is therefore not an abstract luxury but the invisible infrastructure on which both economic theory and empirical practice rest.

To make these ideas concrete, consider agents operating in environments representative of the Tanzanian and broader developing-country context. The following examples illustrate how probability structures the decision problems of a smallholder farmer, a rural household, and a firm, each facing a distinct but equally fundamental form of uncertainty.

Example 1.1 (Tanzanian Smallholder Farmer and Weather Uncertainty)**.** Consider a smallholder farmer in a village in the Dodoma region of Tanzania, preparing planting decisions ahead of the long rains season. The farmer cannot observe next season's weather in advance, but recognises three possible states of nature based on historical experience and knowledge of local climate patterns:

$$\Omega = \{\text{good weather},\ \text{flood},\ \text{drought}\},$$

with probabilities assigned on the basis of historical climatological frequencies:

$$\mathbb{P}(\{\text{good weather}\}) = 0.55, \quad \mathbb{P}(\{\text{flood}\}) = 0.20, \quad \mathbb{P}(\{\text{drought}\}) = 0.25.$$

Each state maps to a distinct crop yield and hence a distinct level of household food security and income. Under good weather the farmer harvests y_g kilograms of maize; under flood, $y_f < y_g$; under drought, $y_d < y_f$. The farmer's decision problem — how much land to cultivate, whether to purchase drought-resistant seed varieties, and whether to participate in a weather-indexed insurance scheme — is formally an optimisation over expected utility:

$$\max_{c,\,q} \;\; 0.55\, u(y_g - c - \pi q) + 0.20\, u(y_f - c - \pi q) + 0.25\, u(y_d + q - c - \pi q),$$

where c denotes input expenditure, q is the quantity of insurance purchased, and π is the insurance premium per unit. The premium πq is paid ex ante and therefore reduces resources in every state, while the indemnity q is paid only in the drought state. Without an explicit probability space $(\Omega, \mathcal{F}, \mathbb{P})$, this expected-utility formulation cannot be stated precisely, let alone solved or estimated from household survey data.

Example 1.2 (Rural Household and Income Uncertainty)**.** Consider a rural household in Tanzania whose total income is drawn from two sources: on-farm crop income, which is subject to weather risk, and off-farm casual labour income, which fluctuates with local demand conditions. Let the household's income in state $\omega \in \Omega$ be denoted $Y(\omega)$, a random variable defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The household's central concern is the distribution of Y across states, and in particular its lower tail: the probability of falling below a subsistence threshold $\underline{y}$ is

$$\mathbb{P}(Y < \underline{y}) = \sum_{\omega:\, Y(\omega) < \underline{y}} \mathbb{P}(\{\omega\}),$$

which is the formal expression of poverty risk. The household manages this risk through a portfolio of strategies — precautionary saving, diversification of income sources, informal risk-sharing arrangements with neighbours, and participation in formal insurance or social protection programmes. Each of these strategies can be evaluated and ranked only by comparing the probability distributions they induce over income outcomes, making the probability space the indispensable foundation of household welfare analysis and the design of social protection policy.

Example 1.3 (Firm and Price Uncertainty)**.** Consider a firm operating in an agricultural output market in Tanzania, deciding how much to invest in processing capacity before the harvest price $\tilde{p}$ is realised. The price is a random variable drawn from a distribution $\mathbb{P}$ over a Borel-measurable subset of $\mathbb{R}_+$, reflecting uncertainty about domestic supply, export demand, and government price interventions. The firm's profit in state ω is

$$\pi(\omega) = \tilde{p}(\omega) \cdot q - C(q),$$

where q is the chosen capacity and $C(q)$ is the cost of that capacity. The firm maximises expected profit

$$\max_{q \geq 0} \;\; \mathbb{E}_{\mathbb{P}}[\tilde{p}] \cdot q - C(q) = \max_{q \geq 0} \;\; \int_{\Omega} \tilde{p}(\omega)\, d\mathbb{P}(\omega) \cdot q - C(q),$$

where the expectation is a Lebesgue integral with respect to the probability measure $\mathbb{P}$. A risk-averse firm, by contrast, maximises the expected utility of profit, placing additional weight on unfavourable price realisations. In either case, the probability measure $\mathbb{P}$ over price states is the primitive object of the firm's decision problem, and its accurate estimation from market data is the central task of an empirical industrial organisation or agricultural economics study. The probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is thus not an abstract formalism but the direct structural input into firm-level investment, production, and contracting decisions.

In economics, uncertainty is central to decision-making, from consumption choices to financial markets and policy design. Probability theory gives these problems a formal structure, transforming vague intuitions about chance into a precise, internally consistent calculus that can be embedded in optimisation problems, tested against data, and used to derive welfare implications.

1.2 Probability Theory as the Foundation of Econometric Theory

Econometrics is, at its foundational level, an application of probability theory to the empirical analysis of economic relationships. This insight was stated with remarkable clarity and force by Trygve Haavelmo in his landmark 1944 monograph *The Probability Approach in Econometrics*, for which he was awarded the Nobel Memorial Prize in Economic Sciences in 1989. Haavelmo's central argument was that economic models should not be treated as deterministic algebraic relationships fitted to data by minimising discrepancies, but should instead be conceived from the outset as *probability models* — complete specifications of the joint probability distribution of all variables entering the analysis. Modern probability-based treatments of econometrics develop this same connection systematically (see Bierens 2004; Hansen 2022).

On this view, the observed data are not a fixed object to be described but a single realisation drawn from a well-defined probability space $(\Omega, \mathcal{F}, \mathbb{P})$, and the task of econometrics is to use that realisation to learn about the probability measure $\mathbb{P}$ that generated it. Without an explicit probabilistic framework, the concepts of estimation, consistency, efficiency, and statistical inference are simply undefined, and the applied researcher has no principled basis for distinguishing signal from noise or for quantifying the uncertainty that surrounds any empirical conclusion.

To see why the probability framework is not merely convenient but logically indispensable, consider the classical consumption function

$$C_t = \alpha + \beta Y_t + u_t,$$

where C_t is household consumption in period t, Y_t is disposable income, α and β are structural parameters, and u_t is an unobservable disturbance. In a purely algebraic reading, this equation is simply a line with an error attached, and estimation reduces to a curve-fitting exercise. Haavelmo's probability approach transforms this interpretation entirely.

Because we work with observational rather than experimental data, income Y_t is not set by the researcher but is itself generated by an underlying economic process — it is a random variable with its own marginal distribution, reflecting the aggregate interactions of labour markets, policy, technology, and household decisions that lie outside the model. Conditional on the realised value of Y_t and the true but unobservable structural parameter β — which encodes the rational marginal propensity to consume of a utility-maximising household — we observe a further random variable C_t, whose stochastic variation is governed by u_t. The full probability model is therefore a specification of the joint distribution

$$(C_t, Y_t) \sim F_{\alpha,\beta,\sigma^2},$$

parameterised by the structural parameters $(\alpha, \beta, \sigma^2)$. The population is this distribution; the sample $\{(C_t, Y_t)\}_{t=1}^T$ is a sequence of realisations drawn from it. Probability theory operates on the population side, characterising the data-generating process; econometrics operates on the sample side, using the observed realisations to form estimators — such as the ordinary least squares estimator $\hat{\beta}$ — and to evaluate their properties.

Under the relevant regularity and exogeneity conditions, the consistency of $\hat{\beta}$, meaning that $\hat{\beta} \xrightarrow{p} \beta$ as $T \to \infty$, is a theorem about probability limits, proved using laws of large numbers. Its asymptotic normality, which underpins t-tests and confidence intervals, is a theorem about convergence in distribution, proved using central limit theorems. Every inferential statement an econometrician makes is therefore a statement about the probability model, and its validity depends on whether the assumed probability space and identifying assumptions are adequate representations of the true data-generating process.

This intimate relationship between probability theory and econometrics has a direct and consequential implication for how econometrics should be learned and practised. Econometric methods are not a self-contained toolkit that can be mastered independently of the probabilistic foundations on which they rest; they are theorems and procedures derived from, and interpretable only within, a probability-theoretic framework.

A researcher who does not understand the probability space underlying a model cannot assess whether its assumptions are satisfied in a given application, cannot diagnose the source of an inconsistency when an estimator fails, and cannot evaluate whether a proposed extension or modification of a standard method is internally coherent. Conversely, a researcher with a thorough command of probability theory can reconstruct the logic of any standard estimator from first principles, detect violations of identifying assumptions by reasoning about the underlying distributions, and engage critically with the frontier of econometric theory as it evolves.

Remark (Probability Theory as a Core Foundation). A thorough understanding of probability theory is a *necessary* foundation for a deep understanding of econometric theory, though it is not by itself sufficient: econometrics also draws on economic theory, identification arguments, optimization, linear algebra, and substantive knowledge of the data-generating environment. Probability is necessary because every core concept in econometrics — the sampling distribution of an estimator, the consistency and asymptotic normality of extremum estimators, the size and power of hypothesis tests, the construction of confidence sets, and the identification of structural parameters from observational data — is defined and proved within a probability-theoretic framework.

An econometrician who lacks this foundation is reduced to memorising formulas without understanding their scope, their assumptions, or the conditions under which they break down. A researcher who has fully internalised the probability space, random variables, convergence concepts, and limit theorems is well placed to derive the properties of estimators, construct valid tests, and assess identification from first principles, while still relying on the other mathematical and economic tools that econometric analysis requires.

Remark (Probability Theory as a Foundation for Responsible Empirical Practice). Beyond theoretical understanding, a solid grounding in probability theory is essential for the informed and responsible application of econometrics in empirical research. Many of the most consequential errors in applied economic research — inappropriate use of asymptotic approximations in small samples, failure to account for clustering or heteroskedasticity, misinterpretation of p-values and confidence intervals, and the conflation of statistical significance with economic significance — are not errors of computation but errors of probabilistic reasoning.

A researcher who understands that a confidence interval is a statement about the sampling distribution of an estimator across hypothetical repetitions of the data-generating process, not a statement about the probability that the true parameter lies in a given range, will not misreport their results. A researcher who understands that a p-value is the probability, under the null hypothesis and the assumed probability model, of observing a test statistic at least as extreme as the one obtained, will not treat it as a measure of the probability that the null hypothesis is true. These distinctions are probabilistic in nature, and their mastery is a precondition for credible empirical work.

Remark (Haavelmo's Legacy and the Modern Structural Approach). Haavelmo's probability approach is not a historical curiosity but the living foundation of modern structural econometrics. The potential outcomes framework of Rubin and the structural approach of the Cowles Commission are both, at their core, specifications of probability models for the data-generating process under alternative interventions or counterfactuals.

The identification problem — the central preoccupation of modern applied econometrics — is fundamentally a question about whether the probability distribution of observables uniquely determines the structural parameters of interest. Instrumental variables, regression discontinuity, difference-in-differences, and synthetic control methods are all strategies for using observable variation in the joint distribution of (Y, X, Z) to recover parameters that are not directly identified from the marginal distribution of Y alone. Understanding why any of these strategies works, and when it fails, requires exactly the probabilistic reasoning that Haavelmo placed at the centre of econometric methodology eight decades ago.

1.3 Approaches to Probability

1.3.1 Classical

The classical (or Laplacean) approach to probability defines probability in terms of equally likely outcomes within a finite sample space. Formally, let Ω be a finite sample space with $|\Omega| < \infty$, and suppose that all elementary outcomes $\omega \in \Omega$ are equally likely. Then, for any event $A \subseteq \Omega$, the probability of A is defined as

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|},$$

where $|A|$ denotes the number of favorable outcomes in A, and $|\Omega|$ is the total number of possible outcomes. This approach relies critically on the principle of insufficient reason (or indifference), which assigns equal probabilities to all outcomes in the absence of further information.

While the classical definition provides an intuitive and mathematically convenient foundation for probability theory—particularly in games of chance such as coin tosses, dice rolls, and card draws—it is limited to settings with finite and symmetric outcome spaces, and does not readily extend to situations involving infinite sample spaces or outcomes that are not equally likely.

1.3.2 Frequentist

The frequentist approach to probability defines the probability of an event in terms of its long-run relative frequency in repeated, identical trials of a random experiment. Formally, consider a sequence of independent and identically distributed trials, and let A be an event of interest. Let $N_n(A)$ denote the number of occurrences of A in the first n trials. Then, the probability of A is defined as the limiting relative frequency:

$$\mathbb{P}(A) = \lim_{n\to\infty} \frac{N_n(A)}{n},$$

provided that this limit exists. This definition is grounded in the law of large numbers, which ensures that, under suitable regularity conditions, the sample frequency converges to a stable value as the number of trials becomes large.

The frequentist interpretation is particularly well-suited for empirical and experimental contexts where repeatability is meaningful; however, it is conceptually limited when dealing with unique or non-repeatable events, and it presumes the existence of an underlying stochastic mechanism generating the observed outcomes.

1.3.3 Subjective

The subjective (or Bayesian) approach to probability defines probability as a degree of belief or confidence that an individual assigns to the occurrence of an event, based on their personal judgment, information, and prior experience. Formally, let A be an event, and let $\mathbb{P}(A)$ represent the subjective probability assigned to A. This interpretation is inherently personal and can vary between individuals, as it reflects their unique perspectives and information sets. The subjective approach is particularly useful in decision-making under uncertainty, where probabilities are used to guide choices in the absence of objective frequencies or symmetries. However, it can be criticized for its lack of objectivity and potential for inconsistency across different individuals.

1.3.3.1 The Subjective Approach as the Foundation of Bayesian Econometrics

The subjective interpretation of probability is not merely a philosophical position — it is the operational foundation of an entire approach to statistical inference known as **Bayesian econometrics** (or Bayesian statistics). The central mechanism that converts subjective prior beliefs into updated posterior beliefs in light of data is **Bayes' theorem**, first stated by the Reverend Thomas Bayes in 1763. For an event A and observed data D,

$$\mathbb{P}(A \mid D) = \frac{\mathbb{P}(D \mid A)\, \mathbb{P}(A)}{\mathbb{P}(D)}.$$

In an econometric context, let θ denote the parameter of interest (a regression coefficient, an elasticity, a structural parameter) and let y denote the observed data. Bayes' theorem gives the **posterior distribution** of θ given the data:

$$\underbrace{p(\theta \mid y)}_{\text{posterior}} \propto \underbrace{p(y \mid \theta)}_{\text{likelihood}} \times \underbrace{p(\theta)}_{\text{prior}}.$$

This equation is the engine of Bayesian inference. It says: start with a prior distribution $p(\theta)$ that quantifies your beliefs about θ before seeing the data; update those beliefs using the likelihood $p(y \mid \theta)$, which measures how probable the observed data would be under each possible value of θ; the result is the posterior distribution $p(\theta \mid y)$, which represents your updated beliefs about θ after incorporating the data. All Bayesian inference — point estimates, interval estimates, hypothesis tests, predictions — is derived from the posterior.

1.3.3.2 The Essence of Bayesian Econometrics

Bayesian econometrics treats model parameters as **random variables** that have probability distributions, rather than as fixed but unknown constants. This shift in ontology has several concrete implications.

Prior distributions. Before observing any data, the researcher specifies a prior distribution $p(\theta)$ over the parameter space. This distribution encodes existing knowledge, theoretical restrictions, or expert judgment. A prior can be **informative** — reflecting substantive prior knowledge, such as the belief from prior studies that an income elasticity is likely between 0.5 and 1.5 — or **diffuse** (sometimes called weakly informative or non-informative), such as a flat or very wide distribution that exerts little influence on the posterior, effectively letting the data speak. In practice, the choice of prior is a modelling decision, analogous to choosing a functional form or instrument set in classical econometrics, and should be made transparently and defended on substantive grounds.

The likelihood. The likelihood function $p(y \mid \theta)$ is common to both Bayesian and classical approaches. It summarises the information that the observed data carry about θ under a given model.

Posterior distributions and inference. Once the prior and likelihood are specified, the posterior $p(\theta \mid y)$ is fully determined by Bayes' theorem. All inferential statements are then statements about the posterior. A **point estimate** is typically the posterior mean, median, or mode. A **credible interval** — the Bayesian counterpart of a confidence interval — is a region that contains the parameter with a specified posterior probability, e.g.

$$\mathbb{P}(\theta \in [L, U] \mid y) = 0.95.$$

This statement has the direct probabilistic interpretation that analysts often mistakenly attribute to classical confidence intervals: given the observed data, there is a 95% probability that θ lies in $[L, U]$. Bayesian hypothesis testing proceeds by comparing posterior probabilities or Bayes factors, rather than p-values.

Prediction. Bayesian econometrics produces predictive distributions for out-of-sample outcomes by integrating over the posterior uncertainty in θ:

$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \theta)\, p(\theta \mid y)\, d\theta.$$

This **posterior predictive distribution** propagates parameter uncertainty into forecasts, yielding prediction intervals that honestly reflect both sampling variation and parameter uncertainty.

Remark (Bayes' Theorem and Learning). Bayes' theorem formalises the intuitive process of learning from evidence. The prior represents beliefs before the data; the likelihood represents the evidential weight of the data; and the posterior represents updated beliefs after incorporating the data. As more data accumulate, the posterior becomes increasingly concentrated, and under regularity conditions the posterior converges to a point mass at the true parameter value — a Bayesian analogue of consistency. This sequential updating property is particularly valuable in settings where data arrive incrementally, such as real-time macroeconomic forecasting or adaptive policy evaluation.

1.3.3.3 How Bayesian Econometrics Differs from Classical Econometrics

Classical (frequentist) econometrics and Bayesian econometrics share the same mathematical foundation — probability theory and statistical models — but they differ fundamentally in their interpretation of probability, their treatment of parameters, and their inferential targets.

The nature of parameters. In classical econometrics, the parameter θ is a fixed but unknown constant. It has a true value; it simply is not observed. Probability statements can only be made about estimators — random functions of the data — not about θ itself. In Bayesian econometrics, θ is a random variable with a probability distribution. Probability statements about θ are well-defined and are the primary output of inference.

The interpretation of probability. Classical econometrics uses the frequentist interpretation: probability refers to the long-run frequency of outcomes in repeated hypothetical experiments. A confidence interval at the 95% level means that, if the estimation procedure were repeated on many independent samples from the same data-generating process, 95% of the resulting intervals would contain the true parameter. The interval from any particular sample either contains θ or it does not — there is no probability statement to be made about the specific realised interval. Bayesian econometrics uses the subjective interpretation: the 95% credible interval directly means that, conditional on the observed data, θ lies in the interval with probability 0.95.

Prior information. Classical econometrics, at least in its standard forms, does not incorporate prior information about parameters formally into the estimation procedure. All information flows from the likelihood. Bayesian econometrics formalises prior knowledge through the prior distribution, enabling the systematic incorporation of evidence from previous studies, theoretical restrictions, or calibrated structural knowledge. This is a significant advantage in small-sample settings, where the data alone may be too sparse to identify parameters precisely, and in structural models, where economic theory places strong restrictions on the parameter space.

Estimation and inference. Classical estimators — OLS, IV, GMM, MLE — are constructed to minimise a loss function or maximise a likelihood, and their sampling properties (unbiasedness, consistency, efficiency) are evaluated over hypothetical repeated samples. Bayesian estimators are derived from the posterior distribution and their properties are evaluated conditionally on the observed data. Classical hypothesis tests produce p-values — the probability, under the null hypothesis and the assumed model, of observing a test statistic as extreme as or more extreme than the one obtained. Bayesian hypothesis tests produce posterior probabilities or Bayes factors, which directly compare the posterior probability of competing hypotheses given the data.

Computational demands. Classical estimation is often analytically tractable or numerically straightforward (OLS has a closed-form solution; MLE requires a single optimisation). Bayesian estimation requires computing the posterior distribution $p(\theta \mid y)$, which is typically a high-dimensional integral with no closed form except in special cases (conjugate prior-likelihood pairs). In practice, Bayesian inference relies on simulation methods, principally **Markov Chain Monte Carlo (MCMC)** algorithms such as the Gibbs sampler and the Metropolis-Hastings algorithm, which draw samples from the posterior without requiring it to be computed analytically. Advances in computing power and the development of probabilistic programming languages (Stan, PyMC, JAGS) have made Bayesian methods increasingly practical for complex structural models.

Remark (Bayesian and Classical Econometrics in Practice). The choice between Bayesian and classical approaches is not purely philosophical — it is often dictated by the structure of the problem. Classical methods remain the dominant paradigm in applied microeconometrics, partly for their computational simplicity and partly because large datasets make the choice of prior largely inconsequential (the likelihood overwhelms the prior when n is large). Bayesian methods are particularly valuable in structural macroeconometrics (Dynamic Stochastic General Equilibrium models, Vector Autoregressions with sign restrictions), small-sample settings (cross-country regressions, historical data), models with latent variables (mixture models, state-space models), and hierarchical panel data models where partial pooling across units is desired. In all of these settings, the ability to encode prior knowledge formally and to make direct probability statements about parameters provides advantages that frequentist methods cannot easily replicate.

1.3.4 Axiomatic (Measure-Theoretic)

Probability is defined as a measure on a measurable space.

The axiomatic (measure-theoretic) approach to probability, formalized by *Andrei Kolmogorov*, defines probability as a measure on a structured set, thereby providing a rigorous and general foundation for both finite and infinite probabilistic models. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, where Ω is the sample space, $\mathcal{F}$ is a σ-algebra of subsets of Ω (the collection of events), and $\mathbb{P} : \mathcal{F} \to [0, 1]$ is a probability measure.

The measure $\mathbb{P}$ satisfies three axioms: (i) non-negativity, $\mathbb{P}(A) \geq 0$ for all $A \in \mathcal{F}$; (ii) normalization, $\mathbb{P}(\Omega) = 1$; and (iii) countable additivity, such that for any countable sequence of disjoint events $\{A_i\}_{i=1}^{\infty} \subset \mathcal{F}$,

$$\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

This framework unifies probability with measure theory and allows probabilistic reasoning to be conducted with full mathematical rigor.

A key strength of the axiomatic approach is its ability to handle infinite and uncountable sample spaces, which arise naturally in economics and stochastic processes. For example, when $\Omega = \mathbb{R}$ (e.g., modeling continuous variables such as income, time, or prices), it is not meaningful to assign equal probability to individual outcomes. Instead, the σ-algebra $\mathcal{F}$ (typically the Borel σ-algebra) restricts attention to well-behaved subsets of $\mathbb{R}$, and probabilities are assigned via measures such as the Lebesgue measure.

In this setting, probabilities are defined over intervals and more complex measurable sets, rather than points, thereby resolving the conceptual limitations of classical probability. The use of countable additivity ensures that probabilities remain consistent even when dealing with infinite collections of events, while avoiding paradoxes associated with uncountable unions that are not measurable.

The structure of a probability space can be visualized as a rectangle representing the sample space Ω, whose total area is normalized to 1. The σ-algebra $\mathcal{F}$ is not a region within Ω but rather a *collection of admissible subsets* of Ω — those subsets to which the probability measure $\mathbb{P}$ is permitted to assign a value. An event $A \in \mathcal{F}$ is one such admissible subset, depicted as the shaded region, and its probability $\mathbb{P}(A)$ corresponds to the ratio of the shaded area to the total area of Ω:

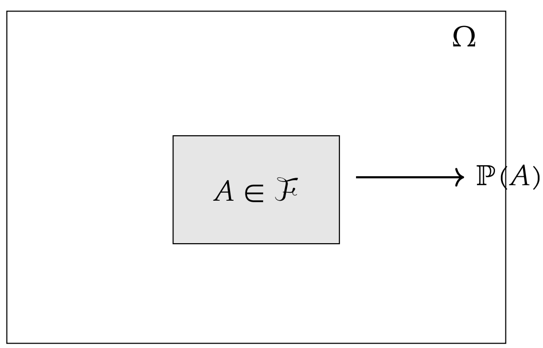

Figure 1: Structure of a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The outer rectangle represents the sample space Ω, normalized so that its total area equals 1. The shaded region is an event $A \in \mathcal{F}$, and $\mathbb{P}(A)$ is the ratio of the shaded area to the total area of Ω. The σ-algebra $\mathcal{F}$ is not depicted as a region but is understood as the family of all subsets of Ω — such as A — to which $\mathbb{P}$ can be legitimately assigned.

This diagram highlights that $\mathbb{P}(A)$ is always measured relative to the whole of Ω, consistently with the requirement $\mathbb{P}(\Omega) = 1$. The role of $\mathcal{F}$ is not to carve out a sub-region of Ω but to specify which subsets are *measurable* — a distinction that becomes essential in uncountable spaces. For standard continuous models, one typically works with a rich σ-algebra such as the Borel or Lebesgue σ-algebra rather than the set of all subsets, because some pathological sets cannot be assigned probabilities while preserving the intended properties of the measure. By embedding probability within measure theory, the axiomatic approach provides a coherent and flexible framework capable of supporting modern econometrics, stochastic calculus, and advanced statistical inference.

2 Probability Spaces

A probability space is the fundamental mathematical structure that underlies all of modern probability theory, providing a rigorous foundation for reasoning about uncertainty. Formally introduced by Andrei Kolmogorov in his 1933 *Grundbegriffe der Wahrscheinlichkeitsrechnung*, it is defined as a triple $(\Omega, \mathcal{F}, \mathbb{P})$ consisting of three objects that work in concert: a sample space Ω, a σ-algebra $\mathcal{F}$, and a probability measure $\mathbb{P}$.

The power of this construction lies not in any single component but in the disciplined way the three interact — Ω specifies what can happen, $\mathcal{F}$ specifies what can be meaningfully measured, and $\mathbb{P}$ assigns to each measurable event a number in $[0, 1]$ that quantifies its likelihood. Together, they elevate probability from an informal intuition about chance to a branch of measure theory, granting it the full deductive machinery of mathematical analysis and enabling a unified treatment of both discrete and continuous phenomena within a single, coherent framework.

2.1 Sample Space

The sample space is the starting point of any probabilistic model: before assigning likelihoods to outcomes, one must first declare precisely what outcomes are possible. Intuitively, it is an exhaustive list of everything that could happen in a given experiment or situation, with no ambiguity about what constitutes a distinct outcome.

Definition 2.1 (Sample Space)**.** The **sample space**, denoted Ω, is the set of all possible outcomes of a random experiment. Each element $\omega \in \Omega$ is called a **sample point** or **elementary outcome**, and the collection must be mutually exclusive and collectively exhaustive — that is, exactly one $\omega \in \Omega$ will be realized on any given run of the experiment.

The two conditions embedded in this definition carry real logical weight. Mutual exclusivity requires that the outcomes be defined finely enough that no two can occur simultaneously: if two outcomes overlap, the model is underspecified. Collective exhaustiveness requires that the sample space be defined broadly enough to contain every conceivable realization: if an outcome occurs that is not in Ω, the model is incomplete. In practice, the analyst constructs Ω by a deliberate act of abstraction — retaining only those features of reality that are relevant to the question at hand and discarding the rest. A sample space may be finite, countably infinite, or uncountable, and this distinction has profound consequences for the subsequent mathematical machinery required to assign probabilities.

Example 2.2 (Tossing a Coin)**.** A fair coin is tossed once. The two possible outcomes are a head and a tail, so the sample space is

$$\Omega = \{H, T\}.$$

This is the simplest non-trivial sample space: finite, with two mutually exclusive and collectively exhaustive elementary outcomes.

Example 2.3 (Tossing a Die)**.** A standard six-sided die is rolled once. Each face shows a distinct number of pips, yielding six possible outcomes, so the sample space is

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

Any event of practical interest — such as rolling an even number — corresponds to a subset of Ω, for instance $A = \{2, 4, 6\}$.

Example 2.4 (Weather Conditions and Farm Output)**.** A farmer operates in a region subject to three mutually exclusive and collectively exhaustive seasonal weather regimes: favorable conditions, flooding, and drought. From the perspective of an econometrician modeling agricultural output or crop insurance take-up, the relevant sample space is

$$\Omega = \{\text{good weather}, \ \text{flood}, \ \text{drought}\}.$$

Each $\omega \in \Omega$ triggers a distinct production outcome — and hence a distinct realization of farm revenue — making Ω the primitive object over which a structural model of decision-making under uncertainty is built. Extensions of this framework allow each weather state to be associated with a continuous yield distribution, in which case Ω would be enriched to accommodate a continuum of possible output realizations.

2.2 Sigma-Algebra

Where the sample space Ω catalogues every possible outcome of a random experiment, the σ-algebra refines the model by specifying precisely which *collections* of outcomes are admissible objects of probabilistic reasoning. Not every subset of Ω need be included among the events, and in standard uncountable models some pathological subsets are deliberately excluded to preserve the intended properties of the probability measure. The σ-algebra is the mathematical device that enforces this discipline.

Definition 2.5 (σ-Algebra)**.** Let Ω be a sample space. A **σ-algebra** (or σ-field) on Ω is a collection $\mathcal{F}$ of subsets of Ω satisfying the following three axioms:

1. **Contains the sample space:** $\Omega \in \mathcal{F}$.
2. **Closed under complementation:** If $A \in \mathcal{F}$, then $A^c = \Omega \setminus A \in \mathcal{F}$.
3. **Closed under countable unions:** If $A_1, A_2, A_3, \ldots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

The pair $(\Omega, \mathcal{F})$ is called a **measurable space**, and any set $A \in \mathcal{F}$ is called a **measurable event**.

Each axiom has a precise logical purpose. The first ensures that the certain event — something happens — is always assignable a probability, namely $\mathbb{P}(\Omega) = 1$. The second ensures coherence under negation: if one can speak meaningfully about the probability of an event A occurring, one must equally be able to speak about the probability of A *not* occurring, and that complement must also be a measurable event. The third axiom extends this coherence to countably infinite sequences of events, ensuring that asking whether *at least one* of a sequence of events occurs is always a well-posed question. Taken together, the three axioms guarantee that $\mathcal{F}$ is closed under all the set-theoretic operations — complementation, countable union, and hence countable intersection by De Morgan's laws — that arise naturally in probabilistic arguments.

Remark (Why the Sample Space Alone Is Not Enough). One might reasonably ask why Ω is insufficient on its own and why a σ-algebra is needed at all. Consider the simple experiment of tossing a fair coin, where $\Omega = \{\text{H}, \text{T}\}$. Here Ω is finite, and there is no difficulty in assigning probabilities to every subset of Ω: the power set $2^{\Omega} = \{\emptyset, \{\text{H}\}, \{\text{T}\}, \Omega\}$ serves perfectly well as the σ-algebra, and one can assign $\mathbb{P}(\{\text{H}\}) = \mathbb{P}(\{\text{T}\}) = \frac{1}{2}$ without any contradiction. In this finite setting, the σ-algebra is a formality. The necessity becomes acute when Ω is uncountable — as it is whenever outcomes are modeled as real numbers, for instance when a crop yield, an asset return, or a wage is drawn from a continuous distribution.

In such cases, constructions such as the Vitali set show that one cannot assign Lebesgue length to *all* subsets of $\mathbb{R}$ while preserving countable additivity, translation invariance, and the usual lengths of intervals.

The σ-algebra solves this problem by restricting attention to a well-behaved subfamily of subsets — large enough to contain the events of ordinary analytical and empirical interest, but small enough to admit the intended measure. For the real line, a standard choice is the Borel σ-algebra $\mathcal{B}(\mathbb{R})$, generated by all open intervals, which contains all intervals, open and closed sets, and the usual sets encountered in econometric applications.

Example 2.6 (σ-Algebra for a Coin Toss)**.** Let $\Omega = \{\text{H}, \text{T}\}$. The unique σ-algebra that contains all subsets of Ω is the power set

$$\mathcal{F} = \{\emptyset,\ \{\text{H}\},\ \{\text{T}\},\ \Omega\}.$$

One can verify all three axioms directly: $\Omega \in \mathcal{F}$; the complement of each set is also in $\mathcal{F}$; and every union of elements of $\mathcal{F}$ remains in $\mathcal{F}$. In this finite setting the power set is the natural and only sensible choice of σ-algebra, and it permits probabilities to be assigned to every conceivable event.

Example 2.7 (σ-Algebra for a Die Roll)**.** Let $\Omega = \{1, 2, 3, 4, 5, 6\}$. The power set 2^{Ω} contains $2^6 = 64$ subsets and constitutes the maximal σ-algebra on Ω, allowing probabilities to be assigned to every event. A strictly coarser σ-algebra can be constructed by grouping outcomes: for instance, defining $E = \{2, 4, 6\}$ (even) and $O = \{1, 3, 5\}$ (odd) yields

$$\mathcal{F} = \{\emptyset,\ E,\ O,\ \Omega\},$$

which is a valid σ-algebra that supports only the question of whether the outcome is even or odd, discarding all finer information about the individual face. This illustrates that the choice of σ-algebra encodes the *information structure* of the model.

Example 2.8 (σ-Algebra for Weather States)**.** Let the sample space be

$$\Omega = \{\text{good},\ \text{flood},\ \text{drought}\},$$

representing mutually exclusive and exhaustive seasonal weather outcomes faced by a farmer.

Define the σ-algebra $\mathcal{F}$ as the power set of Ω, i.e.

$$\mathcal{F} = 2^{\Omega} = \big\{\emptyset, \quad \{\text{good}\}, \{\text{flood}\}, \quad \{\text{drought}\}, \{\text{good}, \text{flood}\}, \quad \{\text{good}, \text{drought}\}, \{\text{flood}, \text{drought}\}, \quad \Omega\big\}.$$

Then $(\Omega, \mathcal{F})$ is a measurable space. Any probability measure $\mathbb{P} : \mathcal{F} \to [0, 1]$ satisfying $\mathbb{P}(\Omega) = 1$ completes the probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Interpretation. Each element $A \in \mathcal{F}$ represents a well-defined event. For example,

$$\mathbb{P}(\{\text{flood}, \text{drought}\})$$

is the probability of an adverse weather outcome. In economic terms, such events may define trigger conditions for crop insurance contracts or government relief programs.

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is the probability of an adverse weather outcome. In economic terms, such events may define trigger conditions for crop insurance contracts or government relief programs. \end{example}

Remark. In a finite sample space, the σ-algebra generated by Ω is typically taken to be the full power set, ensuring that *all* subsets are measurable. This guarantees that any economically meaningful event — whether simple (e.g. "flood") or composite (e.g. "not good weather") — can be assigned a probability.

In more general (infinite) settings, however, not every subset of Ω can be assigned a probability in a logically consistent way: such subsets are said to be *non-measurable*, and no σ-algebra can contain them without destroying the coherence of the probability measure. The σ-algebra therefore delineates precisely the boundary between sets that are amenable to probabilistic measurement and those that are not — the latter being not merely excluded by convention but genuinely uncontrollable within any coherent probability framework.

2.3 Probability Measure

With the sample space Ω enumerating all possible outcomes and the σ-algebra $\mathcal{F}$ identifying which events are admissible objects of measurement, the final ingredient of a probability space is the probability measure $\mathbb{P}$ — the function that completes the triple $(\Omega, \mathcal{F}, \mathbb{P})$ by assigning to each measurable event a number that quantifies its likelihood. The probability measure is not defined arbitrarily; it must satisfy a set of axioms, known as the Kolmogorov axioms, that encode the minimal logical requirements for a coherent calculus of chance.

Definition 2.9 (Probability Measure)**.** Let $(\Omega, \mathcal{F})$ be a measurable space. A **probability measure** is a function $\mathbb{P} : \mathcal{F} \to [0, 1]$ satisfying the following three axioms:

1. **Non-negativity:** For every event $A \in \mathcal{F}$,

$$\mathbb{P}(A) \geq 0.$$

2. **Normalization:** The probability of the certain event is one,

$$\mathbb{P}(\Omega) = 1.$$

3. **Countable additivity (σ-additivity):** For any countable collection of mutually exclusive events $A_1, A_2, A_3, \ldots \in \mathcal{F}$ satisfying $A_i \cap A_j = \emptyset$ for all $i \neq j$,

$$\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a **probability space**.

Each axiom distills a fundamental requirement of rational belief about uncertainty. Non-negativity reflects the irreducibly non-negative nature of likelihood: no event can be less probable than impossible, and impossibility itself is represented by $\mathbb{P}(\emptyset) = 0$, which follows immediately from the other axioms. Normalization anchors the scale: by convention, certainty is assigned the value one, so that all probabilities are expressed as fractions of the whole.

It is this axiom that gives the geometric interpretation its force — $\mathbb{P}(A)$ is literally the proportion of the total probability mass, normalized to unity, that is allocated to A. Countable additivity is the most substantive of the three: it asserts that when events are mutually exclusive, their probabilities *add*, and that this additivity extends to countably infinite collections of events, not merely finite ones.

This extension to the countably infinite case is what distinguishes σ-additivity from the weaker requirement of finite additivity, and it is precisely what enables the passage from sums to integrals and from discrete to continuous probability distributions. Together the three axioms imply all the familiar rules of probability — the complement rule $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$, the addition rule $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, and monotonicity $A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$ — as purely logical consequences, requiring no further assumptions.

Example 2.10 (Probability Measure for a Coin Toss)**.** Let $\Omega = \{\mathrm{H}, \mathrm{T}\}$ and $\mathcal{F} = \{\emptyset, \{\mathrm{H}\}, \{\mathrm{T}\}, \Omega\}$. For a fair coin, the probability measure $\mathbb{P}$ is defined by assigning

$$\mathbb{P}(\emptyset) = 0, \quad \mathbb{P}(\{\mathrm{H}\}) = \tfrac{1}{2}, \quad \mathbb{P}(\{\mathrm{T}\}) = \tfrac{1}{2}, \quad \mathbb{P}(\Omega) = 1.$$

One verifies immediately that all three Kolmogorov axioms are satisfied: probabilities are non-negative, they sum to one over the whole space, and because $\{\mathrm{H}\}$ and $\{\mathrm{T}\}$ are mutually exclusive, $\mathbb{P}(\{\mathrm{H}\} \cup \{\mathrm{T}\}) = \mathbb{P}(\{\mathrm{H}\}) + \mathbb{P}(\{\mathrm{T}\}) = 1 = \mathbb{P}(\Omega)$, as required.

Example 2.11 (Probability Measure for a Die Roll)**.** Let $\Omega = \{1, 2, 3, 4, 5, 6\}$ and let $\mathcal{F} = 2^{\Omega}$ be the full power set. For a fair die, each elementary outcome is assigned equal probability:

$$\mathbb{P}(\{\omega\}) = \frac{1}{6} \quad \text{for each } \omega \in \Omega.$$

By σ-additivity, the probability of any event $A \subseteq \Omega$ is

$$\mathbb{P}(A) = \frac{|A|}{6},$$

where $|A|$ denotes the cardinality of A. For instance, the probability of rolling an even number is $\mathbb{P}(\{2, 4, 6\}) = \frac{3}{6} = \frac{1}{2}$, and the probability of rolling a number greater than four is $\mathbb{P}(\{5, 6\}) = \frac{2}{6} = \frac{1}{3}$.

Example 2.12 (Probability Measure for Weather States)**.** Let $\Omega = \{\text{good, flood, drought}\}$ and let $\mathcal{F} = 2^\Omega$ be the full power set. Suppose that historical agronomic data for a given region yield the following climatological frequencies:

$$\mathbb{P}(\{\text{good}\}) = 0.60, \quad \mathbb{P}(\{\text{flood}\}) = 0.25, \quad \mathbb{P}(\{\text{drought}\}) = 0.15.$$

These values are non-negative and sum to one, satisfying the Kolmogorov axioms. By σ-additivity, the probability of any compound event is obtained by summing over its constituent states; for example, the probability of an adverse weather realization — the event that triggers a crop insurance payout — is

$$\mathbb{P}(\{\text{flood, drought}\}) = \mathbb{P}(\{\text{flood}\}) + \mathbb{P}(\{\text{drought}\}) = 0.25 + 0.15 = 0.40.$$

An econometrician estimating a structural model of farmer behavior under uncertainty would treat these probabilities as primitives of the decision problem, using them to compute expected utilities, optimal insurance coverage, and the welfare effects of weather-indexed transfer programs.

The three Kolmogorov axioms are not merely definitional conveniences; they are sufficiently powerful to imply a rich set of properties that govern the behaviour of any probability measure. The following theorem collects the four most fundamental of these derived properties.

Theorem 2.13 (Derived Properties of a Probability Measure)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Then the following properties hold for all $A, B \in \mathcal{F}$:*

1. ***Probability of the empty set:***
$$\mathbb{P}(\emptyset) = 0.$$

2. ***Complement rule:***
$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

3. ***Monotonicity:***
$$A \subseteq B \implies \mathbb{P}(A) \leq \mathbb{P}(B).$$

4. ***Addition rule (inclusion-exclusion):***
$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Proof. **Property 1.** Since Ω and $\emptyset$ are mutually exclusive and $\Omega \cup \emptyset = \Omega$, countable additivity gives

$$\mathbb{P}(\Omega) = \mathbb{P}(\Omega \cup \emptyset) = \mathbb{P}(\Omega) + \mathbb{P}(\emptyset).$$

Subtracting $\mathbb{P}(\Omega)$ from both sides yields $\mathbb{P}(\emptyset) = 0$. □

Property 2. For any $A \in \mathcal{F}$, the sets A and A^c are mutually exclusive and satisfy $A \cup A^c = \Omega$. By countable additivity,

$$\mathbb{P}(A \cup A^c) = \mathbb{P}(A) + \mathbb{P}(A^c).$$

Since $\mathbb{P}(\Omega) = 1$ by normalization, it follows immediately that

$$\mathbb{P}(A) + \mathbb{P}(A^c) = 1 \implies \mathbb{P}(A^c) = 1 - \mathbb{P}(A). \quad \square$$

Property 3. Suppose $A \subseteq B$. Then B can be decomposed into two mutually exclusive sets:

$$B = A \cup (B \setminus A), \quad A \cap (B \setminus A) = \emptyset.$$

By countable additivity,

$$\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A).$$

Since $\mathbb{P}(B \setminus A) \geq 0$ by non-negativity, it follows that $\mathbb{P}(A) \leq \mathbb{P}(B)$. $\square$

Property 4. Any union $A \cup B$ can be written as the union of three mutually exclusive sets:

$$A \cup B = (A \setminus B) \cup (A \cap B) \cup (B \setminus A).$$

By countable additivity,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A \setminus B) + \mathbb{P}(A \cap B) + \mathbb{P}(B \setminus A).$$

Observe that $A = (A \setminus B) \cup (A \cap B)$ and $B = (B \setminus A) \cup (A \cap B)$ are also disjoint decompositions, so

$$\mathbb{P}(A) = \mathbb{P}(A \setminus B) + \mathbb{P}(A \cap B), \quad \mathbb{P}(B) = \mathbb{P}(B \setminus A) + \mathbb{P}(A \cap B).$$

Adding these two expressions and substituting back gives

$$\mathbb{P}(A) + \mathbb{P}(B) = \mathbb{P}(A \setminus B) + \mathbb{P}(B \setminus A) + 2\,\mathbb{P}(A \cap B) = \mathbb{P}(A \cup B) + \mathbb{P}(A \cap B),$$

from which

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B). \quad \square$$

$\blacksquare$

Each of these four properties illuminates a distinct facet of probabilistic reasoning. Property~1 establishes that impossibility is represented by zero probability: the empty set $\emptyset$ contains no outcomes, so no probability mass can be allocated to it, and this follows as a theorem rather than an additional assumption. Property~2 formalises the intuition that certainty is conserved: since A and its complement A^c together account for all possibilities, their probabilities must sum to one, and knowing $\mathbb{P}(A)$ immediately determines $\mathbb{P}(A^c)$. This is one of the most frequently used tools in applied probability, since it is often far easier to compute the probability of an event *not* occurring and subtract from one than to compute its probability directly.

Property~3 encodes a basic requirement of logical consistency: if every outcome in A is also an outcome in B, then B is at least as likely as A, and the probability measure must respect this ordering. In econometric applications, monotonicity underpins the coherence of nested hypothesis tests and the nesting of information sets in dynamic models. Property~4, the inclusion-exclusion principle, corrects for double-counting: when two events overlap, summing their individual probabilities counts the intersection twice, and subtracting $\mathbb{P}(A \cap B)$ restores the correct total. This property generalises to arbitrarily many events and is the foundation of combinatorial probability calculations, as well as of the union bound widely used in asymptotic theory and statistical learning.

2.4 Continuity Properties of Probability Measures

Imagine watching a sequence of events slowly shrink or expand toward some limiting event. For instance, picture a nested sequence of intervals on the real line, each one contained in the previous, all collapsing toward a single point. Intuitively, the probability of landing in any one of those intervals should gradually approach the probability of landing exactly at the limiting point.

This natural and reassuring behavior—that probabilities *follow* their events to the limit—is precisely what the **Continuity of Probability** theorem guarantees. Far from being an abstract technicality, this result is the bridge between the combinatorial, finite world where probability is easy to compute and the analytic, infinite world where calculus and limits live. It tells us that a probability measure, much like a continuous function, does not jump erratically: as sets converge, so do their probabilities. This idea can be formally stated in a proposition.

Proposition 2.14 (Continuity of Probability)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.*

(*i*) ***Continuity from below.*** *If $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$ is an increasing sequence of events in $\mathcal{F}$, then*

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mathbb{P}(A_n).$$

(*ii*) ***Continuity from above.*** *If $A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots$ is a decreasing sequence of events in $\mathcal{F}$, then*

$$\mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mathbb{P}(A_n).$$

Proof. (**i**) **Continuity from below.** Define disjoint sets by "peeling off" successive layers:

$$B_1 := A_1, \qquad B_n := A_n \setminus A_{n-1} \quad \text{for } n \geq 2.$$

By construction the B_n are pairwise disjoint, each $B_n \in \mathcal{F}$, and

$$\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} A_n, \qquad \bigcup_{k=1}^{n} B_k = A_n \quad \text{for all } n.$$

Applying countable additivity of $\mathbb{P}$ and then finite additivity yields

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} B_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(B_n) = \lim_{n\to\infty} \sum_{k=1}^{n} \mathbb{P}(B_k) = \lim_{n\to\infty} \mathbb{P}\left(\bigcup_{k=1}^{n} B_k\right) = \lim_{n\to\infty} \mathbb{P}(A_n).$$

(**ii**) **Continuity from above.** Consider the complementary sequence A_n^c. Since $A_1 \supseteq A_2 \supseteq \cdots$, we have $A_1^c \subseteq A_2^c \subseteq \cdots$, an increasing sequence. By part (i):

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n^c\right) = \lim_{n\to\infty} \mathbb{P}(A_n^c).$$

By De Morgan's law, $\bigcup_{n=1}^{\infty} A_n^c = \left(\bigcap_{n=1}^{\infty} A_n\right)^c$. Using complementation $\mathbb{P}(E^c) = 1 - \mathbb{P}(E)$:

$$1 - \mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty}(1 - \mathbb{P}(A_n)) = 1 - \lim_{n\to\infty} \mathbb{P}(A_n),$$

and rearranging gives the result. ∎

2.4.1 Intuition and Logic

The deep idea here is that a probability measure behaves like a *flow of mass*: every outcome $\omega \in \Omega$ carries a tiny parcel of probability, and when we accumulate outcomes into a set we collect the corresponding mass. For an increasing sequence of sets, each new set merely *adds* more outcomes and hence more mass; the running total $\mathbb{P}(A_n)$ is monotone non-decreasing and bounded above by 1, so it must converge.

The key algebraic trick in the proof—decomposing the union into disjoint "rings" B_n—lets us cash in countable additivity, the cornerstone axiom, and turn an infinite union into a convergent series. Continuity from above follows almost for free by duality: complementation flips a decreasing family into an increasing one, and we recycle the first result. Together, the two statements say that the probability measure $\mathbb{P}$ is *sequentially continuous* with respect to monotone set limits, in perfect analogy with how a continuous real-valued function commutes with limits of sequences.

Remark (Necessity of the finiteness condition). In continuity from above, the hypothesis $\mathbb{P}(A_1) < \infty$ is automatically satisfied since we are in a probability space ($\mathbb{P}(\Omega) = 1$). In the more general setting of σ-finite measures, the analogous result *requires* $\mathbb{P}(A_1) < \infty$; without it, the statement can fail. A classical counterexample in measure theory is $A_n = [n, \infty)$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$ with Lebesgue measure: $\lambda(A_n) = \infty$ for all n, yet $\bigcap_n A_n = \emptyset$ has measure zero.

Remark (Equivalent formulation via σ-additivity). Continuity from below is in fact *equivalent* to countable additivity given finite additivity and the other Kolmogorov axioms. Some axiomatic treatments therefore adopt continuity from below as the third axiom in place of σ-additivity, yielding an equivalent but sometimes more geometrically transparent foundation.

Remark (Subadditivity and Borel–Cantelli). Continuity of probability underpins many classical results. For instance, the first Borel–Cantelli lemma—which states that if $\sum_n \mathbb{P}(A_n) < \infty$ then $\mathbb{P}(A_n \text{ i.o.}) = 0$—relies on writing the limsup event as a decreasing intersection of unions and then applying continuity from above to pass the limit through $\mathbb{P}$.

Remark (Topological perspective). From a functional-analytic viewpoint, the space of probability measures on a metric space can itself be equipped with the topology of weak convergence. Continuity of probability for monotone set sequences is a special case of the more general principle that integrals (and hence probabilities) behave continuously under dominated convergence, a result that ultimately rests on the same σ-additivity arguments presented above.

3 Random Variables

Recall that a probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is the *sample space* (the set of all possible outcomes), $\mathcal{F}$ is a σ-algebra of subsets of Ω (the collection of events to which we can assign probabilities), and $\mathbb{P} : \mathcal{F} \to [0, 1]$ is a probability measure satisfying Kolmogorov's axioms. In many applications, we are not directly interested in the raw outcomes $\omega \in \Omega$ but rather in some numerical quantity that depends on the outcome. A *random variable* X is a rule that assigns a number to each outcome—a bridge from the abstract probability space into the real line $\mathbb{R}$, where standard analytical tools become available.

Definition 3.1 (Random Variable)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be the measurable space consisting of the real line equipped with its Borel σ-algebra. A **random variable** is a function

$$X : \Omega \longrightarrow \mathbb{R}$$

that is $\mathcal{F}$*-measurable*, meaning that for every Borel set $B \in \mathcal{B}(\mathbb{R})$,

$$X^{-1}(B) \;=\; \{\, \omega \in \Omega : X(\omega) \in B \,\} \;\in\; \mathcal{F}.$$

Equivalently, X is a random variable if and only if

$$\{\, \omega \in \Omega : X(\omega) \leq x \,\} \;\in\; \mathcal{F} \qquad \text{for every } x \in \mathbb{R}.$$

The definition may appear technical at first, but its logic is both natural and necessary. The measurability condition says that for any numerical range of interest B—for instance, the outcome produces a value no greater than~5—the set of raw outcomes ω that lead to a value in B must be an *event*, that is, a member of $\mathcal{F}$. This is precisely the requirement needed for the statement X lies in B to have a well-defined probability: we need the pre-image $X^{-1}(B)$ to live inside $\mathcal{F}$ so that $\mathbb{P}$ can be applied to it.

Without measurability, the expression $\mathbb{P}(X \in B)$ would be meaningless—we would be asking for the probability of a set that the probability measure cannot evaluate. In short, a random variable is not "random" in the colloquial sense of being haphazard; it is a perfectly deterministic function. What is random is the input ω, drawn according to $\mathbb{P}$. The randomness of the outcome flows through the function X to produce a random numerical value.

Example 3.2 (Tossing a fair coin.)**.** Let $\Omega = \{H, T\}$, $\mathcal{F} = 2^{\Omega}$, and $\mathbb{P}(H) = \mathbb{P}(T) = \frac{1}{2}$. Define $X : \Omega \to \mathbb{R}$ by $X(H) = 1$ and $X(T) = 0$. Then X is a random variable that encodes "heads" as 1 and "tails" as 0. Every subset of Ω is in $\mathcal{F}$, so measurability is trivially satisfied. X follows a Bernoulli$(\frac{1}{2})$ distribution.

Example 3.3 (Tossing a fair die)**.** Let $\Omega = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{F} = 2^{\Omega}$, and $\mathbb{P}(\{\omega\}) = \frac{1}{6}$ for each ω. The identity function $X(\omega) = \omega$ is a random variable mapping each face to its numerical value. Again measurability holds trivially. More interesting random variables on the same space include $Y(\omega) = 1[\omega \text{ is even}]$ (indicator of an even outcome) or $Z(\omega) = \omega^2$.

Example 3.4 (A smallholder farmer in Tanzania)**.** Consider a small-scale farmer in a village facing uncertain rainfall during the growing season. Let Ω be the set of all possible weather scenarios over the season (combinations of rainfall amounts, timing, and temperature). Define $\mathcal{F}$ as a suitable σ-algebra of weather events and $\mathbb{P}$ as the probability measure reflecting climatological knowledge. The farmer's harvest yield Y (in kilograms) is a function of the realised weather scenario: $Y(\omega)$ is the harvest that results when weather scenario ω occurs. Since Y maps uncertain environmental outcomes to real numbers and the event $\{Y \leq y\}$ is measurable for each y, Y is a random variable. The farmer faces not just a single unknown number but an entire *distribution* of possible yields induced by $\mathbb{P}$ through Y.

Example 3.5 (A household)**.** Consider a household whose monthly consumption expenditure depends on random factors such as income shocks, health events, and price fluctuations. Let Ω be the space of all possible states of the world affecting the household, equipped with an appropriate σ-algebra $\mathcal{F}$ and probability measure $\mathbb{P}$. The household's consumption expenditure $C(\omega)$ is a random variable mapping each state of the world to a non-negative real number. Events such as $\{C(\omega) \leq c\}$—the event that the household spends no more than c—must be in $\mathcal{F}$ for C to qualify as a proper random variable, ensuring that we can compute $\mathbb{P}(C \leq c)$, the probability that consumption falls at or below a given threshold.

Example 3.6 (A firm)**.** A firm produces output whose quantity and price depend on uncertain demand conditions, input costs, and productivity shocks. Let $\omega \in \Omega$ represent a realised state of the business environment. The firm's profit

$$\Pi(\omega) \;=\; p(\omega)\, q(\omega) \;-\; c(\omega)$$

is a function of random prices p, quantities q, and costs c—each themselves random variables on the same probability space. The composite function Π is also a random variable (since measurable functions are closed under algebraic operations), enabling the analysis of expected profit, profit volatility, and the probability of loss.

3.1 Random Variables as Measurable Functions

It is worth pausing to unpack what it means, in plain terms, to say that X is a *measurable function*. Measurability is simply a *compatibility* requirement between two structures: the σ-algebra $\mathcal{F}$ on the domain Ω (which says which subsets of outcomes are observable events) and the Borel σ-algebra $\mathcal{B}(\mathbb{R})$ on the codomain $\mathbb{R}$ (which says which subsets of numbers are well-behaved sets). The condition $X^{-1}(B) \in \mathcal{F}$ for all $B \in \mathcal{B}(\mathbb{R})$ says precisely: whenever you identify a "nice" numerical set B on the output side, pulling it back through X gives you a "nice" event on the input side. The function *respects* the observable structure.

A useful way to build intuition is to think about what *fails* when measurability is violated. If $X^{-1}(B) \notin \mathcal{F}$ for some B, it means there is a perfectly sensible numerical question—does X land in B—for which no probability can be assigned. The probability measure $\mathbb{P}$ is only defined on $\mathcal{F}$, so if the pre-image falls outside $\mathcal{F}$, the question is literally unanswerable within the model.

Measurability ensures this never happens: every numerical question about X translates back into an event whose probability is defined. In practice, all functions encountered in economics and finance—continuous functions, step functions, limits of sequences of measurable functions—are automatically measurable with respect to standard σ-algebras, so measurability is rarely a binding constraint in applications.

Its importance is foundational: it guarantees that the induced distribution $\mathbb{P}_X(B) \equiv \mathbb{P}(X^{-1}(B))$, known as the *pushforward measure* or *law* of X, is itself a well-defined probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. In this sense, a random variable does not merely "produce" random numbers—it *transports* the entire probability structure from the abstract sample space to the real line.

4 Distribution Function

Once we have established that a random variable $X : \Omega \to \mathbb{R}$ is a measurable function on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a natural and practically important question arises: how is the probability *distributed* across the real line by X? We know from the definition of measurability that for any Borel set $B \in \mathcal{B}(\mathbb{R})$, the pre-image $X^{-1}(B)$ is an event in $\mathcal{F}$, so the probability $\mathbb{P}(X \in B)$ is always well-defined. Yet specifying $\mathbb{P}(X \in B)$ for every possible Borel set B would be an unwieldy way to describe the probabilistic behaviour of X.

The *distribution function*—also called the *cumulative distribution function* (CDF)—offers an elegant and complete solution: instead of tracking all Borel sets simultaneously, it suffices to track a single function of a real argument x, namely $F(x) = \mathbb{P}(X \leq x) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq x\})$. This is possible because the Borel σ-algebra $\mathcal{B}(\mathbb{R})$ is generated by half-lines of the form $(-\infty, x]$, so knowledge of $F(x)$ for all $x \in \mathbb{R}$ determines $\mathbb{P}(X \in B)$ for *every* Borel set B—the distribution function encodes the full law of X. In this sense, F is the concrete, real-line object that fully represents the pushforward measure $\mathbb{P}_X$, collapsing the entire abstract probability structure into a single, computationally tractable function of one variable.

Definition 4.1 (Distribution Function)**.** Let X be a random variable on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The **(cumulative) distribution function** of X is the function $F : \mathbb{R} \to [0, 1]$ defined by

$$F(x) \;=\; \mathbb{P}(X \leq x) \;=\; \mathbb{P}(\{\, \omega \in \Omega : X(\omega) \leq x \,\}), \qquad x \in \mathbb{R}.$$

A function $F : \mathbb{R} \to [0, 1]$ is a valid distribution function if and only if it satisfies the following three conditions:

1. (**Monotonicity**) F is non-decreasing: $x \leq y \implies F(x) \leq F(y)$.
2. (**Right-continuity**) F is right-continuous: $\lim_{t \downarrow x} F(t) = F(x)$ for all $x \in \mathbb{R}$.
3. (**Boundary conditions**) $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to +\infty} F(x) = 1$.

Meaning, Logic, and Intuition

The distribution function $F(x) = \mathbb{P}(X \leq x)$ is, at its core, a device for *accumulating* probability as we sweep the real line from left to right. Fix any threshold $x \in \mathbb{R}$ and ask: what is the total probability mass that X assigns to everything at or below x? That is precisely $F(x)$.

As x increases from $-\infty$ to $+\infty$, the set $\{\omega \in \Omega : X(\omega) \leq x\}$ grows monotonically — it can only gain new outcomes, never lose them — so F is non-decreasing by construction. At the far left, $x \to -\infty$, the event $\{X \leq x\}$ becomes the empty set and carries probability zero; at the far right, $x \to +\infty$, it becomes all of Ω and carries probability one. These are not arbitrary normalisation conventions but direct consequences of $\mathbb{P}$ being a probability measure on $(\Omega, \mathcal{F}, \mathbb{P})$.

The right-continuity condition is subtler: it reflects a convention about how probability mass sitting exactly *at* a point x is attributed. Because we write $X \leq x$ rather than $X < x$, the value $F(x)$ includes any atom at x, and right-continuity ensures that F does not jump *before* reaching that atom as we approach from the right. In short, the three defining properties of F are not additional assumptions imposed on top of the definition — they are theorems that follow inevitably from the properties of the underlying probability measure $\mathbb{P}$ and the measurability of X.

The deepest insight in the definition is that F carries *all* the probabilistic information about X, even though it is only a function of a single real argument. This is because the Borel σ-algebra $\mathcal{B}(\mathbb{R})$ is generated by half-lines $(-\infty, x]$, so any probability assigned to any Borel set can be recovered from F by taking differences, limits, and countable operations.

For instance, $\mathbb{P}(a < X \leq b) = F(b) - F(a)$, $\mathbb{P}(X > x) = 1 - F(x)$, and $\mathbb{P}(X = x) = F(x) - F(x^-)$, where $F(x^-) = \lim_{t \uparrow x} F(t)$ is the left-hand limit. The last expression is particularly revealing: if F is continuous at x, then $\mathbb{P}(X = x) = 0$ and no probability mass is concentrated at that single point (as is the case for continuous random variables such as rainfall); if F has a jump discontinuity at x, then a strictly positive mass $\mathbb{P}(X = x) > 0$ is concentrated there (as is the case for discrete random variables).

The distribution function thus unifies discrete and continuous random variables under a single mathematical object, with the shape of F — smooth, stepped, or mixed — telling us everything about the nature of the randomness in X.

In applied settings, particularly in development economics and agricultural contexts, F is not merely a theoretical object but a direct tool for welfare and risk analysis. The value $F(x)$ is the probability that an outcome falls at or below x, which can be read as a *vulnerability measure*: the fraction of probability mass exposed to conditions no better than x. Comparing distribution functions across households, regions, or time periods allows one to make rigorous probabilistic statements about relative well-being and exposure to risk, without reducing the entire distribution to a single summary statistic such as the mean.

Examples

Example 4.2 (Monthly household consumption expenditure)**.** Consider a household in rural Tanzania whose monthly consumption expenditure C (measured in Tanzanian shillings, TZS) is a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose that C is continuously distributed with distribution function F_C. Then:

$$\begin{aligned} F_C(50{,}000) &= \mathbb{P}(C \leq 50{,}000), \\ F_C(100{,}000) &= \mathbb{P}(C \leq 100{,}000), \\ F_C(200{,}000) &= \mathbb{P}(C \leq 200{,}000). \end{aligned}$$

The probability that the household's expenditure falls in the interval $(50{,}000,\ 100{,}000]$ is

$$\mathbb{P}(50{,}000 < C \leq 100{,}000) \ = \ F_C(100{,}000) - F_C(50{,}000),$$

and the probability of falling below the national poverty line $\bar{c}$ is simply $F_C(\bar{c})$. Since C is assumed continuous, F_C has no jumps, so $\mathbb{P}(C = c) = 0$ for every fixed value c — no single expenditure level carries positive probability mass.

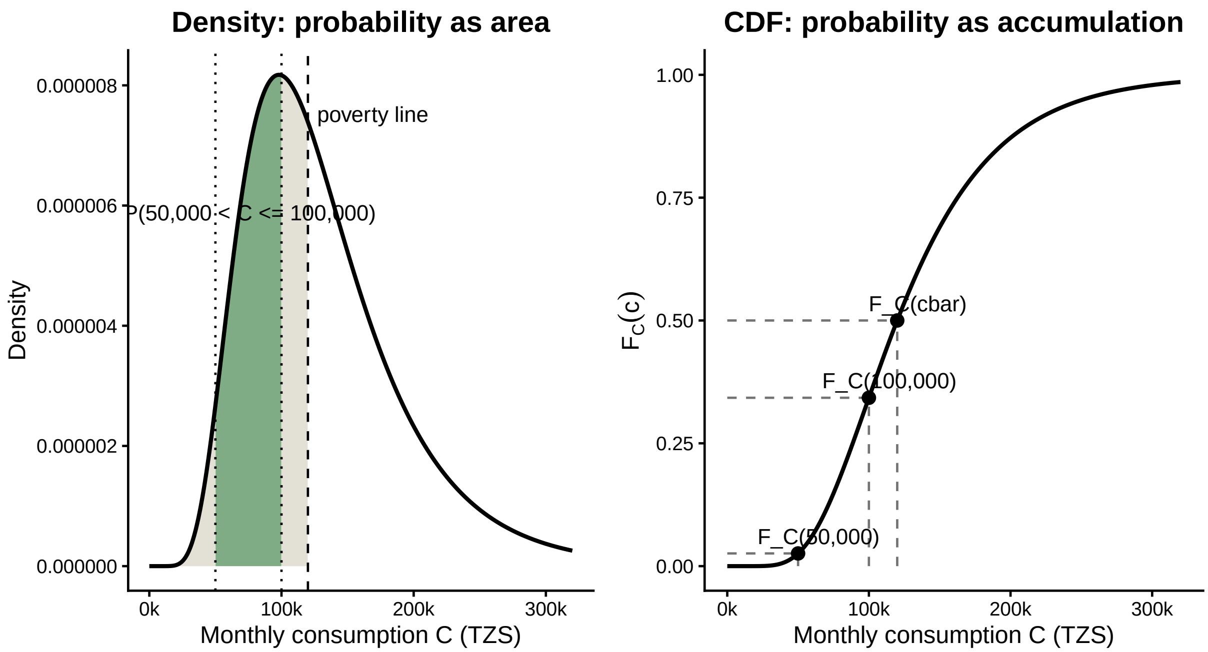

Figure 2: Monthly household consumption expenditure and probabilities from a continuous distribution.

Figure 2 translates the abstract definition of a distribution function into the concrete language of household welfare. In the left panel, probability is represented as area under the density curve: the green region is the probability that monthly consumption falls between TZS 50,000 and TZS 100,000, while the area to the left of the poverty line $\bar{c}$ is the probability that the household is poor. The right panel expresses the same information through the cumulative distribution function, where $F_C(c) = \mathbb{P}(C \leq c)$ records the total probability accumulated up to each consumption threshold. Thus, the probability of an interval is obtained by subtracting two CDF values, while the probability of poverty is read directly from the height of the CDF at $\bar{c}$.

Example 4.3 (Seasonal harvest yield of a smallholder farmer)**.** Let Y (in kilograms) denote the maize harvest of a smallholder farmer in the Dodoma region, depending on uncertain seasonal rainfall. Suppose Y is a continuous random variable with distribution function F_Y. If y^* denotes a *subsistence threshold* — the minimum yield required for the household to meet its food needs without purchasing maize on the market — then:

$$\mathbb{P}(Y \leq y^*) \; = \; F_Y(y^*)$$

is the probability of a *subsistence failure*: the probability that the farmer's harvest falls short of what the family needs. The complementary probability

$$\mathbb{P}(Y > y^*) \; = \; 1 - F_Y(y^*)$$

is the probability of achieving at least subsistence. More generally, for any interval $(a, b]$,

$$\mathbb{P}(a < Y \leq b) \; = \; F_Y(b) - F_Y(a),$$

so the entire risk profile of the farmer's harvest — the probability of any range of outcomes — is encoded in F_Y.

Example 4.4 (Discrete rainfall shock and household income)**.** Suppose that in a given agricultural season in Tanzania, rainfall can take one of three realisations: drought (ω_1), normal (ω_2), or flood (ω_3), with probabilities $\mathbb{P}(\{\omega_1\}) = 0.3$, $\mathbb{P}(\{\omega_2\}) = 0.5$, and $\mathbb{P}(\{\omega_3\}) = 0.2$. A household's net farm income X (in TZS thousands) under each realisation is $X(\omega_1) = 80$, $X(\omega_2) = 200$, and $X(\omega_3) = 120$. Then X is a discrete random variable and its distribution function is the step function:

$$F_X(x) = \begin{cases} 0 & \text{if } x < 80, \\ 0.3 & \text{if } 80 \le x < 120, \\ 0.5 & \text{if } 120 \le x < 200, \\ 1 & \text{if } x \ge 200. \end{cases}$$

The jumps of F_X occur exactly at the three possible values of X, with jump sizes equal to the corresponding probabilities: $F_X(80) - F_X(80^-) = 0.3$, $F_X(120) - F_X(120^-) = 0.2$, and $F_X(200) - F_X(200^-) = 0.5$. This illustrates clearly that right-continuity means F_X includes the jump *at* the point, and that $\mathbb{P}(X = x)$ equals the size of the jump at x.

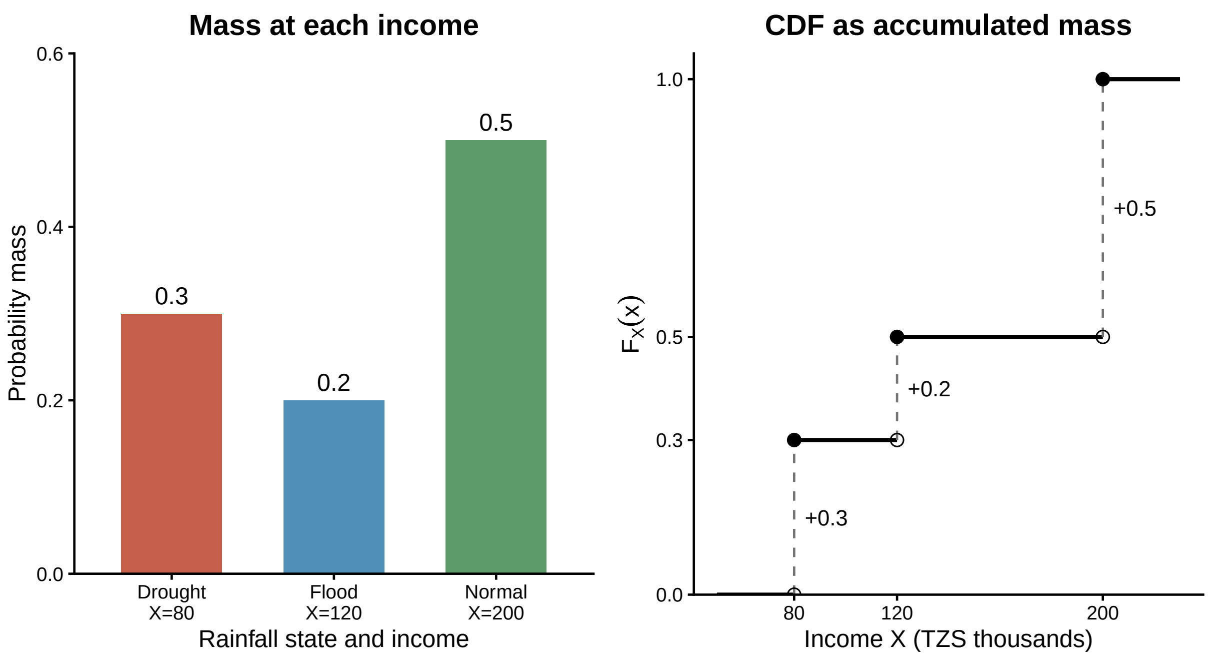

Figure 3: Discrete rainfall shocks, household income, and the induced distribution function.

The figure shows how the probability measure on rainfall states is transferred into a distribution for household income. The left panel says that the household receives TZS 80 thousand with probability 0.3 under drought, TZS 120 thousand with probability 0.2 under flood, and TZS 200 thousand with probability 0.5 under normal rainfall. The right panel then accumulates those same probabilities from left to right: the CDF jumps by 0.3 at 80, by 0.2 at 120, and by 0.5 at 200, so each vertical jump is exactly the probability mass attached to that income value.

4.1 CDF Properties

Theorem 4.5 (Properties of a Cumulative Distribution Function)**.** *Let X be a random variable with CDF $F(x) = \mathbb{P}(X \le x)$. Then F is non-decreasing and right-continuous, and satisfies*

$$\lim_{x\to-\infty} F(x) = 0, \qquad \lim_{x\to\infty} F(x) = 1.$$

Logic and Intuition. A CDF accumulates probability mass from left to right along the real line: $F(x)$ is the total mass lying at or below x. The three properties are exactly what one expects of such an accumulation. It is *non-decreasing* because moving the threshold x to the right can only sweep in additional mass, never discard any. It *tends to* 0 *at* $-\infty$ *and* 1 *at* $+\infty$ because, as the threshold recedes to the far left, no mass is captured, whereas as it advances to the far right, all of the mass — which totals 1 — has been gathered. It is *right-continuous* because the definition uses "$\le x$": the point x itself is included in the event $\{X \le x\}$, so approaching x from above (through slightly larger thresholds) does not lose the mass sitting exactly at x. Each property is therefore a statement about how the probabilities of a family of nested events behave, and the formal proof simply translates "nested events" into the *Continuity of Probability* established earlier.

Proof. Write $A_x = \{X \le x\} = \{\omega \in \Omega : X(\omega) \le x\}$, so that $F(x) = \mathbb{P}(A_x)$.

Part 1: F **is non-decreasing.** Suppose $x_1 < x_2$. Any outcome with $X(\omega) \le x_1$ also satisfies $X(\omega) \le x_2$, so $A_{x_1} \subseteq A_{x_2}$. By the monotonicity of a probability measure (a derived property established for $(\Omega, \mathcal{F}, \mathbb{P})$), event inclusion implies inequality of probabilities:

$$F(x_1) = \mathbb{P}(A_{x_1}) \le \mathbb{P}(A_{x_2}) = F(x_2).$$

Part 2: F **is right-continuous.** Fix x_0 and let (x_n) be any sequence decreasing to x_0, written $x_n \downarrow x_0$. We must show $F(x_n) \to F(x_0)$. Because the thresholds decrease, the events A_{x_n} form a *decreasing* sequence,

$$A_{x_1} \supseteq A_{x_2} \supseteq A_{x_3} \supseteq \cdots,$$

and their intersection is

$$\bigcap_{n=1}^{\infty} A_{x_n} = \{\omega : X(\omega) \le x_n \text{ for all } n\} = \{\omega : X(\omega) \le x_0\} = A_{x_0},$$

where the middle equality uses $\inf_n x_n = x_0$: an outcome satisfies $X(\omega) \le x_n$ for *every* n precisely when $X(\omega) \le x_0$. Applying continuity from above (Proposition 2.14),

$$\lim_{n\to\infty} F(x_n) = \lim_{n\to\infty} \mathbb{P}(A_{x_n}) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} A_{x_n}\right) = \mathbb{P}(A_{x_0}) = F(x_0).$$

Since this holds for every sequence $x_n \downarrow x_0$, we conclude $\lim_{x\downarrow x_0} F(x) = F(x_0)$; that is, F is right-continuous.

Part 3: The limits at $\pm\infty$**.** For the limit at $+\infty$, take any sequence $x_n \uparrow \infty$. The events A_{x_n} now *increase*,

$$A_{x_1} \subseteq A_{x_2} \subseteq \cdots, \qquad \bigcup_{n=1}^{\infty} A_{x_n} = \{\omega : X(\omega) \le x_n \text{ for some } n\} = \Omega,$$

the last equality holding because X is real-valued: every outcome ω has a finite value $X(\omega)$, which is eventually exceeded by x_n. Continuity from below (Proposition 2.14) then gives

$$\lim_{n\to\infty} F(x_n) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} A_{x_n}\right) = \mathbb{P}(\Omega) = 1.$$

For the limit at $-\infty$, take $x_n \downarrow -\infty$. The events A_{x_n} decrease, with

$$\bigcap_{n=1}^{\infty} A_{x_n} = \{\omega : X(\omega) \leq x_n \text{ for all } n\} = \emptyset,$$

since no finite value $X(\omega)$ can be less than or equal to every x_n when $x_n \to -\infty$. Continuity from above yields

$$\lim_{n\to\infty} F(x_n) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} A_{x_n}\right) = \mathbb{P}(\emptyset) = 0.$$

As both limits are independent of the chosen sequences, $\lim_{x\to\infty} F(x) = 1$ and $\lim_{x\to-\infty} F(x) = 0$. ■

Remark (Why right-continuity and not left-continuity). The asymmetry between the two one-sided limits is a direct consequence of the "$\leq$" in the definition $F(x) = \mathbb{P}(X \leq x)$. Repeating the argument of Part 2 with a sequence increasing to x_0 shows that the *left* limit is

$$\lim_{x\uparrow x_0} F(x) = \mathbb{P}(X < x_0),$$

so the size of any jump in the CDF is

$$F(x_0) - \lim_{x\uparrow x_0} F(x) = \mathbb{P}(X \leq x_0) - \mathbb{P}(X < x_0) = \mathbb{P}(X = x_0).$$

Thus F is continuous at x_0 exactly when $\mathbb{P}(X = x_0) = 0$, and it has a jump of height $\mathbb{P}(X = x_0)$ at each atom of the distribution — the discrete masses of §4 are precisely these jumps. This is the same point illustrated, in a limiting context, by the discussion of continuity points in convergence in distribution.

4.2 Independence

Definition 4.6 (Independence)**.** Random variables X and Y are **independent** if, for every pair of Borel sets $A, B \in \mathcal{B}(\mathbb{R})$,

$$\mathbb{P}(X \in A,\ Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B).$$

Equivalently, the σ-algebras generated by X and Y are independent.

Example 4.7 (Independent Bernoulli outcomes)**.** Let X indicate whether a randomly selected household receives a rainfall shock and let Y indicate whether a separate, independently selected household receives the same type of shock. If $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(Y = 1) = q$, independence implies $\mathbb{P}(X = 1, Y = 1) = pq$. The joint probability factors into its two marginal probabilities because learning the value of one variable conveys no information about the other.

Remark (Independence and zero covariance). If X and Y are independent and have finite second moments, then $\mathrm{Cov}(X, Y) = 0$. The converse is false in general: zero covariance rules out only linear association, whereas independence rules out every form of probabilistic dependence. This distinction is important in econometrics, where uncorrelated errors may still be conditionally heteroskedastic or otherwise dependent.

5 Expectations

Expectation is the fundamental concept that captures the *average* or *typical* value of a random variable, taking into account the entire distribution of outcomes. It describes the *center of mass* of the probability distribution, providing a single summary statistic that reflects the overall level of the random variable.

Definition 5.1 (Expectation)**.** Let X be a random variable on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The **expectation** of X, denoted $\mathbb{E}[X]$, is defined as the Lebesgue integral of X with respect to the probability measure $\mathbb{P}$:

$$\mathbb{E}[X] \;=\; \int_{\Omega} X(\omega)\, d\mathbb{P}(\omega).$$

The expectation is finite whenever X is integrable, meaning $\mathbb{E}[|X|] < \infty$. More generally, the extended expectation is well defined if at least one of $\mathbb{E}[X^+]$ and $\mathbb{E}[X^-]$ is finite; it is undefined when both are infinite because this would require the indeterminate difference $\infty - \infty$.

The logic and intuition of this definition are as follows. The expectation $\mathbb{E}[X]$ is a weighted average of the values that X takes, where the weights are given by the probabilities of the corresponding outcomes. The Lebesgue integral is the appropriate mathematical tool for defining this average in a rigorous way, especially when X can take infinitely many values or when the distribution of X is continuous. The expectation captures the *long-run average* value of X if we were to repeat the random experiment many times and average the results. It is a fundamental building block for more advanced concepts such as variance, covariance, and higher moments, and it plays a central role in decision theory, economics, and statistics.

Example 5.2 (Expected monthly household consumption expenditure)**.** Let C be a random variable representing a household's monthly consumption expenditure in Tanzanian shillings (TZS). The expectation $\mathbb{E}[C]$ is the average expenditure we would expect to see if we observed many months of consumption for this household. If C has a continuous distribution with density f_C, then

$$\mathbb{E}[C] = \int_0^\infty c f_C(c)\, dc,$$

which integrates the expenditure c weighted by its probability density. If C is discrete with possible values $c_1, c_2, \ldots$ and probabilities $p_1, p_2, \ldots$, then

$$\mathbb{E}[C] = \sum_i c_i p_i,$$

which adds up all possible expenditures weighted by their probabilities. In either case, $\mathbb{E}[C]$ provides an average expenditure for this household.

Example 5.3 (Expected seasonal harvest yield of a smallholder farmer)**.** Let Y be a random variable representing the maize harvest (in kilograms) of a smallholder farmer in the Dodoma region, depending on uncertain seasonal rainfall. The expectation $\mathbb{E}[Y]$ is the average harvest yield we would expect if we observed many seasons of farming under similar conditions. If Y has a continuous distribution with density f_Y, then

$$\mathbb{E}[Y] = \int_0^\infty y f_Y(y)\, dy,$$

which integrates the yield y weighted by its probability density. If Y is discrete with possible values $y_1, y_2, \ldots$ and probabilities $p_1, p_2, \ldots$, then

$$\mathbb{E}[Y] = \sum_i y_i p_i,$$

which adds up all possible yields weighted by their probabilities. In either case, $\mathbb{E}[Y]$ provides an average harvest yield for this farmer, summarising the overall productivity of the land under uncertainty.

Example 5.4 (Expected profit of a firm)**.** Let Π be a random variable representing the profit of a firm, which depends on uncertain demand conditions, input costs, and productivity shocks. The expectation $\mathbb{E}[\Pi]$ is the average profit we would expect if we observed many periods of operation under similar conditions. If Π has a continuous distribution with density f_Π, then

$$\mathbb{E}[\Pi] = \int_{-\infty}^\infty \pi f_\Pi(\pi)\, d\pi,$$

which integrates the profit π weighted by its probability density. If Π is discrete with possible values $\pi_1, \pi_2, \ldots$ and probabilities $p_1, p_2, \ldots$, then

$$\mathbb{E}[\Pi] = \sum_i \pi_i p_i,$$

which adds up all possible profits weighted by their probabilities. In either case, $\mathbb{E}[\Pi]$ provides an average profit for the firm, which can be used for decision-making, risk assessment, and performance evaluation.

Remark (Interpretation of expectation)*.* The expectation $\mathbb{E}[X]$ can be interpreted in several ways. It is the long-run average value of X if we were to repeat the random experiment many times and average the results. It is also the center of mass of the probability distribution of X, reflecting the balance point of the distribution. In decision theory, $\mathbb{E}[X]$ represents the expected payoff of a random outcome, and it is often used as a criterion for rational choice under uncertainty. However, it is important to remember that $\mathbb{E}[X]$ is not necessarily a value that X can actually take; it is a theoretical construct that summarises the distribution of X in a single number. Also, note that the expected value, $\mathbb{E}[X]$, is not a sample average or an empirical mean; it is a population parameter that can only be estimated from data, and its estimation involves sampling variability and uncertainty.

5.1 Properties

Theorem 5.5 (Linearity of Expectation)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. If $X, Y \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and $a, b \in \mathbb{R}$, then $aX + bY \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and*

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

Proof. We build the proof in three stages, following the standard Lebesgue integration construction: simple functions, then non-negative measurable functions, then general integrable functions. Each stage inherits linearity from the one before it.

Stage 1: Simple random variables.

A *simple* random variable has the canonical form

$$s = \sum_{k=1}^{m} c_k \, 1_{A_k},$$

where $c_k \in \mathbb{R}$ and $\{A_1, \dots, A_m\} \subset \mathcal{F}$ is a finite partition of Ω. Its expectation is defined as $\mathbb{E}[s] = \sum_{k=1}^{m} c_k \mathbb{P}(A_k)$.

Scalar multiplication. For any $a \in \mathbb{R}$, $as = \sum_{k=1}^{m} (ac_k) 1_{A_k}$, and

$$\mathbb{E}[as] = \sum_{k=1}^{m} ac_k \, \mathbb{P}(A_k) = a \sum_{k=1}^{m} c_k \, \mathbb{P}(A_k) = a\mathbb{E}[s].$$

Additivity. Let $s = \sum_{k=1}^{m} c_k \, 1_{A_k}$ and $t = \sum_{j=1}^{n} d_j \, 1_{B_j}$ be simple. Refine both representations to the common partition $\{A_k \cap B_j : k, j\}$; on each atom $A_k \cap B_j$, the function $s + t$ takes the constant value $c_k + d_j$. Therefore,

$$\begin{aligned}
\mathbb{E}[s + t] &= \sum_{k=1}^{m} \sum_{j=1}^{n} (c_k + d_j) \, \mathbb{P}(A_k \cap B_j) \\
&= \sum_{k,j} c_k \, \mathbb{P}(A_k \cap B_j) + \sum_{k,j} d_j \, \mathbb{P}(A_k \cap B_j) \\
&= \sum_{k=1}^{m} c_k \sum_{j=1}^{n} \mathbb{P}(A_k \cap B_j) + \sum_{j=1}^{n} d_j \sum_{k=1}^{m} \mathbb{P}(A_k \cap B_j) \\
&= \sum_{k=1}^{m} c_k \, \mathbb{P}(A_k) + \sum_{j=1}^{n} d_j \, \mathbb{P}(B_j) \\
&= \mathbb{E}[s] + \mathbb{E}[t],
\end{aligned}$$

where the penultimate equality uses the fact that $\{B_j\}$ partitions Ω (so $\sum_j \mathbb{P}(A_k \cap B_j) = \mathbb{P}(A_k)$) and symmetrically for $\{A_k\}$.

Stage 2: Non-negative integrable random variables.

Let $X, Y \geq 0$ with $\mathbb{E}[X] < \infty$ and $\mathbb{E}[Y] < \infty$, and let $a, b \geq 0$. By the measurable function approximation theorem, there exist sequences of non-negative simple random variables $s_n \nearrow X$

and $t_n \nearrow Y$ pointwise on Ω. Then $as_n + bt_n \nearrow aX + bY$ pointwise. Applying the Monotone Convergence Theorem (MCT) and Stage 1:

$$\mathbb{E}[aX+bY] = \lim_{n\to\infty} \mathbb{E}[as_n+bt_n] = \lim_{n\to\infty}(a\mathbb{E}[s_n]+b\mathbb{E}[t_n]) = a\lim_{n\to\infty} \mathbb{E}[s_n]+b\lim_{n\to\infty} \mathbb{E}[t_n] = a\mathbb{E}[X]+b\mathbb{E}[Y].$$

The MCT justifies interchanging the limit and the expectation because the sequence $as_n + bt_n$ is non-negative and non-decreasing.

Stage 3: General integrable random variables.

For any random variable X, define its positive and negative parts:

$$X^+ = \max(X, 0) \geq 0, \qquad X^- = \max(-X, 0) \geq 0.$$

Then $X = X^+ - X^-$ and $|X| = X^+ + X^-$. If $X \in L^1$ then $\mathbb{E}[|X|] < \infty$, which forces both $\mathbb{E}[X^+]$ and $\mathbb{E}[X^-]$ to be finite. By definition, $\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-]$.

Scalar multiplication. For $a \geq 0$: $(aX)^+ = aX^+$ and $(aX)^- = aX^-$, so Stage 2 gives

$$\mathbb{E}[aX] = \mathbb{E}[(aX)^+] - \mathbb{E}[(aX)^-] = a\mathbb{E}[X^+] - a\mathbb{E}[X^-] = a\mathbb{E}[X].$$

For $a < 0$: the sign flips, $(aX)^+ = |a|X^-$ and $(aX)^- = |a|X^+$, so

$$\mathbb{E}[aX] = |a|\mathbb{E}[X^-] - |a|\mathbb{E}[X^+] = a(\mathbb{E}[X^+] - \mathbb{E}[X^-]) = a\mathbb{E}[X].$$

Additivity. Let $Z = X + Y$. Writing the positive-part decomposition of each variable and rearranging gives the identity

$$Z^+ + X^- + Y^- = Z^- + X^+ + Y^+.$$

Both sides are non-negative, and all six terms are integrable (since $|Z| \leq |X| + |Y|$ implies $\mathbb{E}[|Z|] \leq \mathbb{E}[|X|] + \mathbb{E}[|Y|] < \infty$). Applying Stage 2 to both sides:

$$\mathbb{E}[Z^+] + \mathbb{E}[X^-] + \mathbb{E}[Y^-] = \mathbb{E}[Z^-] + \mathbb{E}[X^+] + \mathbb{E}[Y^+].$$

Rearranging:

$$\mathbb{E}[X + Y] = \mathbb{E}[Z^+] - \mathbb{E}[Z^-] = (\mathbb{E}[X^+] - \mathbb{E}[X^-]) + (\mathbb{E}[Y^+] - \mathbb{E}[Y^-]) = \mathbb{E}[X] + \mathbb{E}[Y].$$

General case. Combining scalar multiplication and additivity:

$$\mathbb{E}[aX + bY] = \mathbb{E}[aX] + \mathbb{E}[bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

$\blacksquare$

Remark (Why Integrability Cannot Be Dropped). The condition $X, Y \in L^1$ is essential. If $\mathbb{E}[|X|] = \infty$ or $\mathbb{E}[|Y|] = \infty$, the decomposition $\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-]$ involves the indeterminate form $\infty - \infty$ and linearity fails to hold in general. A classic example is a symmetric Cauchy random variable, which has no finite mean, so the expression $\mathbb{E}[X + (-X)] = \mathbb{E}[0] = 0$ versus the undefined $\mathbb{E}[X] + \mathbb{E}[-X]$ is not well-posed.

Remark (Econometric Relevance). Linearity of expectation is invoked silently in almost every derivation in econometrics. It underlies the unbiasedness argument for the OLS estimator ($\mathbb{E}[\hat{\beta}] = \beta$), the decomposition of mean squared error into bias and variance (MSE = Bias^2 + Var), the moment conditions that define GMM estimators, and the law of iterated expectations ($\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y|X]]$). None of these results depend on the shape of the distribution; they hold for any integrable random variable, which is precisely the generality that the Lebesgue-integral proof delivers.

6 Important Inequalities

A recurring challenge in probability theory and econometrics is that the full distribution of a random variable is rarely known. What is often available instead is partial information: a mean, a variance, or a higher moment. Moment-based inequalities convert this partial information into sharp, universally valid bounds on probabilities and expectations, without requiring knowledge of the distribution beyond the specified moment. They are therefore indispensable tools, both in theoretical arguments — consistency proofs, derivations of convergence rates, verification of regularity conditions — and in applied work, where they discipline what can be inferred from data.

The four inequalities treated in this section form a natural hierarchy. Markov's inequality is the most elementary, requiring only a non-negative mean. Chebyshev's inequality refines it by incorporating variance, yielding tighter probability bounds. Jensen's inequality connects convex transformations to expectations and underlies a vast range of results in decision theory, information theory, and econometrics. Lyapunov's inequality relates moments of different orders and provides the regularity conditions that make central limit theorems applicable. Together they constitute the foundational toolkit of moment-based reasoning.

6.1 Markov's Inequality

Theorem 6.1 (Markov's Inequality). *Let X be a non-negative random variable with finite expectation $\mathbb{E}[X] < \infty$. Then for any $a > 0$,*

$$\mathbb{P}(X \geq a) \;\leq\; \frac{\mathbb{E}[X]}{a}.$$

Proof. Since $X \geq 0$, we can bound X from below by the indicator-weighted value a:

$$X \;\geq\; a \cdot 1_{\{X \geq a\}},$$

because on the event $\{X \geq a\}$ the right-hand side equals $a \leq X$, and on the complementary event it equals $0 \leq X$. Taking expectations of both sides and using monotonicity of expectation,

$$\mathbb{E}[X] \;\geq\; \mathbb{E}\big[a \cdot 1_{\{X \geq a\}}\big] \;=\; a\,\mathbb{P}(X \geq a).$$

Dividing both sides by $a > 0$ gives $\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$. ■

Logic and Intuition. Markov's inequality rests on the most elementary fact about a non-negative random variable: it cannot spend much probability mass far above its mean without inflating that mean. Formally, the proof replaces X by the cruder lower bound $a \cdot 1_{\{X \geq a\}}$, which is zero everywhere except on the very region whose probability we want to control. Averaging this lower bound gives $a\,\mathbb{P}(X \geq a)$, which cannot exceed $\mathbb{E}[X]$. Rearranging yields the bound.

The geometric picture is equally clear. Plot the density of X on the positive half-line and shade the region to the right of a. Markov's inequality says that shaded area (the probability) times the horizontal distance a cannot exceed the total first moment (the area-weighted average position of probability mass). If the mean is small relative to a, the tail probability must be small too, because a large tail would push the mean above its actual value.

The bound is tight: the random variable X that takes value 0 with probability $1-p$ and value a with probability p has $\mathbb{E}[X] = ap$ and $\mathbb{P}(X \geq a) = p = \mathbb{E}[X]/a$, so equality holds. This also reveals the bound's weakness: it makes no use of how the distribution is shaped above a, only of the mean.

Remark (Markov's Inequality in Econometrics). Markov's inequality is rarely used directly in applied work, but it is the seed from which all moment-based probability bounds grow. Its most direct econometric application is in proofs of convergence in probability. If one can show that $\mathbb{E}[|X_n - X|] \to 0$ (convergence in mean), then Markov's inequality applied to the non-negative random variable $|X_n - X|$ gives, for any $\varepsilon > 0$,

$$\mathbb{P}(|X_n - X| \geq \varepsilon) \;\leq\; \frac{\mathbb{E}[|X_n - X|]}{\varepsilon} \;\to\; 0,$$

establishing $X_n \xrightarrow{p} X$. This argument underlies many consistency proofs, including those for method-of-moments estimators. Markov's inequality also enters the proofs of more powerful inequalities (Chebyshev's, for instance), and it is the foundational step in establishing the Borel–Cantelli lemma, which in turn is used to strengthen convergence in probability to almost sure convergence under summability conditions on the probabilities.

6.2 Chebyshev's Inequality

Theorem 6.2 (Chebyshev's Inequality)**.** *Let X be a random variable with finite mean $\mu = \mathbb{E}[X]$ and finite variance $\sigma^2 = \mathrm{Var}(X)$. Then for any $k > 0$,*

$$\mathbb{P}(|X - \mu| \geq k) \;\leq\; \frac{\sigma^2}{k^2}.$$

Equivalently, writing $k = t\sigma$ for $t > 0$,

$$\mathbb{P}(|X - \mu| \geq t\sigma) \;\leq\; \frac{1}{t^2}.$$

Proof. Apply Markov's inequality to the non-negative random variable $(X-\mu)^2$ and the threshold $a = k^2 > 0$:

$$\mathbb{P}\big((X-\mu)^2 \geq k^2\big) \;\leq\; \frac{\mathbb{E}[(X-\mu)^2]}{k^2} \;=\; \frac{\sigma^2}{k^2}.$$

Since $\{(X-\mu)^2 \geq k^2\} = \{|X-\mu| \geq k\}$ (because both sides square the same absolute deviation), the result follows. ■

Logic and Intuition. Chebyshev's inequality is Markov's inequality applied one level up: instead of bounding the probability that X is large, it bounds the probability that X *deviates* from its mean by more than k, using the variance as the controlling moment. The key move is squaring: by working with $(X-\mu)^2$ rather than $|X-\mu|$, we convert a two-sided deviation problem into a one-sided non-negativity problem to which Markov applies.

The $k = t\sigma$ form is the most interpretable. It says that regardless of the distribution of X — whether it is normal, skewed, heavy-tailed, or discrete — at most $1/t^2$ of the probability mass lies more than t standard deviations from the mean. For $t = 2$, at most 25% can lie outside two standard deviations; for $t = 3$, at most 11%. These bounds are intentionally conservative: for a normal distribution, the actual probability beyond two standard deviations is about 5%, far below the 25% ceiling. The bound is distribution-free, which is its power, but distribution-agnosticism comes at the cost of looseness.

Example 6.3 (Monthly consumption expenditure)**.** Suppose a household's monthly consumption expenditure C has mean $\mu = 120{,}000$ TZS and standard deviation $\sigma = 40{,}000$ TZS. Chebyshev's inequality bounds the probability of a deviation of at least $k = 80{,}000$ TZS from the mean (i.e., $t = 2$ standard deviations):

$$\mathbb{P}(|C - 120{,}000| \geq 80{,}000) \;\leq\; \frac{(40{,}000)^2}{(80{,}000)^2} = \frac{1}{4} = 0.25.$$

At most 25% of the probability is concentrated in the tails beyond TZS 40,000 and TZS 200,000. This bound requires no assumption about the shape of C's distribution, only the existence of its first two moments.

Remark (Chebyshev's Inequality in Econometrics). Chebyshev's inequality is among the most frequently used tools in econometric proofs. Its primary role is in establishing the Weak Law of Large Numbers (WLLN). Given an i.i.d. sequence (X_i) with mean μ and variance σ^2, the sample mean $\bar{X}_n = n^{-1}\sum_{i=1}^n X_i$ satisfies $\mathbb{E}[\bar{X}_n] = \mu$ and $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$. Chebyshev's inequality then gives, for any $\varepsilon > 0$,

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \;\leq\; \frac{\sigma^2}{n\varepsilon^2} \;\longrightarrow\; 0,$$

establishing $\bar{X}_n \xrightarrow{p} \mu$ with no assumption beyond finite variance. This argument is the template for consistency proofs of method-of-moments estimators, where the estimator is expressed as a sample average and Chebyshev's inequality converts variance shrinkage into probability convergence. Chebyshev's inequality is also used in finite-sample bounds, for instance to construct non-parametric confidence intervals that are valid regardless of distributional assumptions.

\end{remark}

6.3 Jensen's Inequality

Definition 6.4 (Convex Function)**.** A function $\phi : I \to \mathbb{R}$ defined on an interval $I \subseteq \mathbb{R}$ is **convex** if for all $x, y \in I$ and all $\lambda \in [0, 1]$,

$$\phi(\lambda x + (1 - \lambda)y) \;\le\; \lambda\,\phi(x) + (1 - \lambda)\,\phi(y).$$

Equivalently, ϕ is convex if the chord connecting any two points on its graph lies *above* the graph. If the inequality is strict for $x \neq y$ and $\lambda \in (0, 1)$, ϕ is **strictly convex**. A function ϕ is **concave** if $-\phi$ is convex.

Theorem 6.5 (Jensen's Inequality)**.** *Let X be a random variable and let $\phi : \mathbb{R} \to \mathbb{R}$ be a convex function. If $\mathbb{E}[X]$ and $\mathbb{E}[\phi(X)]$ both exist and are finite, then*

$$\phi(\mathbb{E}[X]) \;\le\; \mathbb{E}[\phi(X)].$$

If ϕ is strictly convex, equality holds if and only if $X = \mathbb{E}[X]$ almost surely (i.e., X is degenerate at its mean).

Proof. Since ϕ is convex, at every point $x_0 \in \mathbb{R}$ there exists a *supporting hyperplane*: a linear function $\ell(x) = \alpha + \beta(x - x_0)$ such that

$$\phi(x) \;\ge\; \ell(x) = \alpha + \beta(x - x_0) \quad \text{for all } x \in \mathbb{R},$$

with $\phi(x_0) = \ell(x_0) = \alpha$. (When ϕ is differentiable, $\beta = \phi'(x_0)$ is the derivative and ℓ is the tangent line; the subdifferential argument handles the non-differentiable case.)

Set $x_0 = \mathbb{E}[X]$. Then for all x,

$$\phi(x) \;\ge\; \phi(\mathbb{E}[X]) + \beta\,(x - \mathbb{E}[X]).$$

Substituting $x = X$ (a random variable) and taking expectations of both sides,

$$\mathbb{E}[\phi(X)] \;\ge\; \mathbb{E}[\phi(\mathbb{E}[X]) + \beta\,(X - \mathbb{E}[X])] = \phi(\mathbb{E}[X]) + \beta\,\mathbb{E}[X - \mathbb{E}[X]] = \phi(\mathbb{E}[X]) + \beta \cdot 0 = \phi(\mathbb{E}[X]),$$

which is the desired inequality. ∎

Logic and Intuition. The key insight is geometric: a convex function curves upward, so the chord between any two points on its graph lies above the graph itself. Jensen's inequality is simply this chord-above-curve property, applied to a probability-weighted average (the expectation) rather than a simple two-point average. Figure 4 illustrates this geometry.

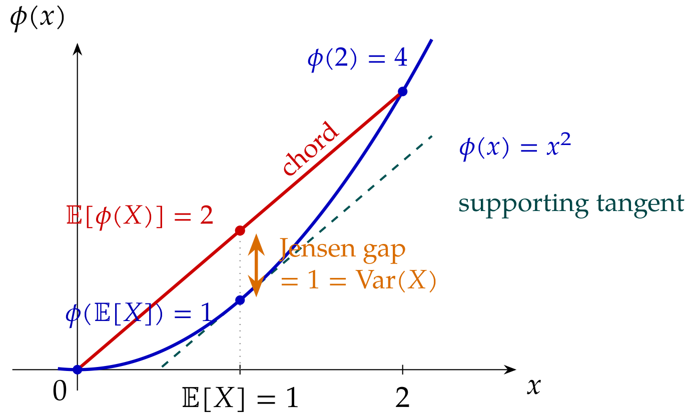

Figure 4: **The geometry of Jensen's inequality.** The convex function $\phi(x) = x^2$ (blue) is drawn together with the two-point random variable that takes the values 0 and 2 equally often. The chord (red) joining $(0, \phi(0))$ and $(2, \phi(2))$ lies above the graph; its midpoint is $(\mathbb{E}[X], \mathbb{E}[\phi(X)]) = (1, 2)$. The point on the curve at the mean is $(\mathbb{E}[X], \phi(\mathbb{E}[X])) = (1, 1)$. The orange vertical gap between these two heights is the Jensen gap $\mathbb{E}[\phi(X)] - \phi(\mathbb{E}[X]) = \mathrm{Var}(X) = 1$. The dashed supporting tangent at $x = \mathbb{E}[X]$ lies everywhere below ϕ and meets the curve at $\phi(\mathbb{E}[X])$, which is the algebraic mechanism of the proof.

The supporting tangent line (or subgradient) at $\mathbb{E}[X]$ provides the algebraic mechanism: it lies *everywhere below* ϕ, so averaging $\phi(X)$ over all realisations of X yields something at least as large as averaging the linear lower bound, which by linearity of expectation collapses to $\phi(\mathbb{E}[X])$. The non-linearity of ϕ is what prevents equality: if X is not degenerate, its realisations spread around the mean, and the curvature of ϕ ensures that the high values of ϕ outweigh the low values enough to push $\mathbb{E}[\phi(X)]$ above $\phi(\mathbb{E}[X])$.

A concrete example fixes the idea. Take $\phi(x) = x^2$ (strictly convex) and suppose X takes values 0 and 2 with equal probability. Then $\mathbb{E}[X] = 1$ and $\phi(\mathbb{E}[X]) = 1$, but $\mathbb{E}[\phi(X)] = \mathbb{E}[X^2] = (0 + 4)/2 = 2 > 1$. The spreading of X around its mean inflates the squared expectation relative to the square of the expectation, and this inflation is exactly the variance: $\mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \mathrm{Var}(X) \geq 0$.

Example 6.6 (Jensen's Inequality and Risk Aversion)**.** A smallholder farmer in Tanzania receives a random harvest income Y with $\mathbb{E}[Y] = \mu$. Under a concave utility function u (e.g., $u(y) = \log y$, which exhibits diminishing marginal utility), Jensen's inequality applied to $-u$ (which is convex) gives

$$u(\mathbb{E}[Y]) = u(\mu) \;\geq\; \mathbb{E}[u(Y)].$$

The farmer strictly prefers a certain income of μ to the risky income Y with the same mean: the expected utility of the risky income is *lower* than the utility of the certain equivalent. This is the mathematical foundation of risk aversion, and it emerges directly from Jensen's inequality applied to a concave utility function.

Remark (Jensen's Inequality in Econometrics). Jensen's inequality appears throughout econometric theory. Three applications are especially important.

Bias of non-linear transformations. If $\hat{\theta}$ is an unbiased estimator of θ and g is a non-linear function, then $g(\hat{\theta})$ is generally a biased estimator of $g(\theta)$. If g is convex, Jensen's inequality gives $\mathbb{E}[g(\hat{\theta})] \geq g(\mathbb{E}[\hat{\theta}]) = g(\theta)$, so the bias is upward. This matters, for example, when estimating variance components or nonlinear functions of regression coefficients.

Log-linearisation. In models involving logarithms — such as log-linear demand or production functions, or models for trade flows (gravity equations) — Jensen's inequality implies $\mathbb{E}[\log Y] \leq \log \mathbb{E}[Y]$, since log is concave. OLS estimation on the log-transformed model estimates $\mathbb{E}[\log Y|X]$, not $\log \mathbb{E}[Y|X]$; the gap between these two objects is a function of the conditional variance, and ignoring it leads to biased predictions of the level Y.

Information inequalities. Many information-theoretic quantities used in econometrics — Kullback–Leibler divergence, likelihood-ratio statistics, entropy — are derived from Jensen's inequality applied to convex functions of likelihood ratios or log-likelihood functions.

6.4 Lyapunov's Inequality

Theorem 6.7 (Lyapunov's Inequality)**.** *Let X be a random variable and let $0 < r < s < \infty$. If $\mathbb{E}[|X|^s] < \infty$, then*

$$(\mathbb{E}[|X|^r])^{1/r} \;\leq\; (\mathbb{E}[|X|^s])^{1/s}.$$

Equivalently, defining the L^p norm $\|X\|_p = (\mathbb{E}[|X|^p])^{1/p}$, we have $\|X\|_r \leq \|X\|_s$ whenever $r \leq s$.

Proof. Apply Jensen's inequality with the convex function $\phi(t) = t^{s/r}$ (convex because $s/r > 1$) to the non-negative random variable $|X|^r$. Jensen's inequality gives

$$\phi(\mathbb{E}[|X|^r]) \;\leq\; \mathbb{E}[\phi(|X|^r)],$$

i.e.,

$$(\mathbb{E}[|X|^r])^{s/r} \;\leq\; \mathbb{E}[|X|^s].$$

Taking both sides to the power $1/s > 0$ (a monotone operation) yields

$$(\mathbb{E}[|X|^r])^{1/r} \;\leq\; (\mathbb{E}[|X|^s])^{1/s}.$$

■

Logic and Intuition. Lyapunov's inequality says that higher-order moments, once properly normalised by their order, are larger than lower-order moments. The reason is that raising $|X|$ to a higher power amplifies large values of $|X|$ disproportionately: if $|X|$ occasionally takes large values, those values contribute much more to $\mathbb{E}[|X|^s]$ (relative to its normalisation) than to $\mathbb{E}[|X|^r]$. The inequality formalises the fact that the L^p "size" of a random variable is a non-decreasing function of the exponent p.

A concrete illustration: take X equal to 0 or 10 with equal probability. Then $\mathbb{E}[X] = 5$, $\mathbb{E}[X^2] = 50$, so $(\mathbb{E}[X^2])^{1/2} = \sqrt{50} \approx 7.07 > 5 = \mathbb{E}[X]^1$. The large value 10 contributes a squared term of 100, which when averaged and square-rooted dominates the simple average. This is the spirit of Lyapunov's inequality: the amplification of extreme values by higher powers prevents the normalised higher moment from falling below the normalised lower moment.

The proof relies on Jensen's inequality in a particularly elegant way: the map $t \mapsto t^{s/r}$ is convex for $s > r$, so Jensen applied to the random variable $|X|^r$ immediately gives the comparison of moments. Thus Lyapunov's inequality is, at its core, a consequence of convexity — the same geometric property that underlies Jensen's inequality itself.

Remark (Lyapunov's Inequality in Econometrics). Lyapunov's inequality has two principal roles in econometric asymptotic theory.

Checking moment conditions. Many regularity conditions for laws of large numbers and central limit theorems require that certain moments are finite. Lyapunov's inequality provides a simple way to establish finiteness of lower moments from finiteness of higher ones: if $\mathbb{E}[|X|^s] < \infty$ for some $s > r$, then $\mathbb{E}[|X|^r] < \infty$ automatically. This is used routinely when verifying that estimators satisfy the moment conditions of limit theorems.

Lyapunov's CLT condition. The most direct application is Lyapunov's Central Limit Theorem. For a sequence of independent (not necessarily identically distributed) random variables (X_i) with $\mathbb{E}[X_i] = \mu_i$ and $\mathrm{Var}(X_i) = \sigma_i^2$, let $s_n^2 = \sum_{i=1}^n \sigma_i^2$. Lyapunov's CLT states that if there exists $\delta > 0$ such that

$$\frac{1}{s_n^{2+\delta}} \sum_{i=1}^{n} \mathbb{E}[|X_i - \mu_i|^{2+\delta}] \longrightarrow 0,$$

then $(S_n - \sum \mu_i)/s_n \xrightarrow{d} \mathcal{N}(0,1)$. This is a standard CLT for independent heterogeneous observations. The Lyapunov condition ensures that no single observation dominates the sum. It requires direct control of the $(2+\delta)$-th absolute central moments; Lyapunov's inequality cannot bound these higher moments from the variance. Rather, it shows that an assumed finite $(2+\delta)$-th moment also guarantees the finiteness of all lower moments, including the variance.

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7 Convergence Concepts in Econometrics

The study of convergence is central to both probability theory and econometrics. Estimators are defined as functions of samples of increasing size, and the validity of asymptotic inference rests on precise statements about how sequences of random quantities behave as the sample grows without bound. Before analysing the convergence of stochastic sequences — sequences of random variables — it is essential to understand the convergence of deterministic sequences of real numbers. The deterministic case provides the precise ε-N language that is then imported, with appropriate modification, into all the stochastic convergence concepts that underpin econometric theory.

7.1 Sequences of Real Numbers

7.1.1 Formal Definition

Definition 7.1 (Sequence of Real Numbers)**.** A **sequence of real numbers** is a function

$$a_n : \mathbb{N} \longrightarrow \mathbb{R}, \qquad n \longmapsto a(n) =: a_n.$$

That is, the function assigns to each positive integer n a unique real number, its value $a(n)$, which we write with a subscript as a_n and call the **n-th term** of the sequence. The sequence is denoted $(a_n)_{n=1}^{\infty}$, or simply (a_n) when the index set is clear from context.

Remark (Sequences as Ordered Lists). A sequence (a_n) is an *ordered* collection: the index n carries information about position. Two sequences that assign the same values to every n are identical, but the ordering of terms matters. This distinguishes a sequence from a set, in which elements carry no natural order and repetition is ignored. In econometrics, sequences arise naturally when one indexes observations by time ($t = 1, 2, \dots, T$) or by sample size ($n = 1, 2, \dots$), and the ordering is economically meaningful.

7.1.2 Examples of Sequences

Example 7.2 (A Convergent Sequence: $a_n = 1/n$). Define the sequence by $a_n = 1/n$ for $n \geq 1$. Its first several terms are

$$1, \quad \frac{1}{2}, \quad \frac{1}{3}, \quad \frac{1}{4}, \quad \frac{1}{5}, \quad \dots$$

The terms are strictly positive and strictly decreasing, and they approach zero as n increases. This sequence is a canonical example of convergence: the terms become arbitrarily small for large n. In econometrics, sequences of the form c/n (for a constant $c > 0$) arise naturally as the rate at which estimation error or variance shrinks with sample size.

Example 7.3 (A Divergent Sequence: $a_n = (-1)^n$). Define the sequence by $a_n = (-1)^n$ for $n \geq 1$. Its terms alternate between -1 and $+1$:

$$-1, \quad 1, \quad -1, \quad 1, \quad -1, \quad \dots$$

The sequence does not settle near any fixed value: for any candidate limit L, infinitely many terms satisfy $|a_n - L| \geq 1$. This sequence therefore has *no limit*; it is said to **diverge**. Oscillating or non-convergent sequences serve as useful counterexamples when verifying that a proposed convergence result does not hold without its stated hypotheses.

Example 7.4 (A Sequence Converging to e: $a_n = (1 + 1/n)^n$). Define the sequence by $a_n = (1 + 1/n)^n$ for $n \geq 1$. Its first several terms are

$$2, \quad 2.25, \quad 2.370\dots, \quad 2.441\dots, \quad 2.488\dots, \quad \dots$$

The terms are strictly increasing and bounded above. It can be shown that

$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e \approx 2.71828\dots,$$

where e is Euler's number. This sequence illustrates that a limit need not be a "simple" number, and that sequences defined by products or powers can converge to transcendental constants. Sequences of this form appear in continuous-time finance (compound interest) and in the derivation of the Poisson distribution as a limit of binomial distributions.

7.1.3 The Limit of a Sequence

The informal idea of a sequence "approaching" a value is made precise through the following classical definition.

Definition 7.5 (Limit of a Sequence)**.** Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. A real number L is the **limit** of (a_n), written

$$\lim_{n\to\infty} a_n = L \qquad \text{or} \qquad a_n \to L \text{ as } n \to \infty,$$

if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$n > N \implies |a_n - L| < \varepsilon.$$

If such an L exists, the sequence is said to **converge** to L; otherwise it **diverges**.

Remark (Reading the ε-N Definition). The definition says the following: no matter how tight a tolerance band $(-\varepsilon, +\varepsilon)$ one draws around L, all terms of the sequence from some point N onward fall inside that band. The quantifier order is critical: ε is chosen *first* (by an adversary, if one likes), and then N must be found in response. If N can always be found, no matter how small ε is, then the sequence genuinely closes in on L and does not merely pass near it finitely many times. This ε-N template is the direct precursor of the ε-δ definition in calculus and the ε-based definitions of convergence in probability and almost sure convergence in probability theory.

Example 7.6 (Limit of $a_n = 1/n$)**. Claim:** $\lim_{n\to\infty} \frac{1}{n} = 0$.

Verification. Let $\varepsilon > 0$ be given. By the Archimedean property of $\mathbb{R}$, there exists $N \in \mathbb{N}$ with $N > 1/\varepsilon$. Then for every $n > N$,

$$\left|\frac{1}{n} - 0\right| = \frac{1}{n} < \frac{1}{N} < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, the definition is satisfied, and the limit is 0. □

Example 7.7 (Non-existence of a limit: $a_n = (-1)^n$)**. Claim:** The sequence $a_n = (-1)^n$ has no limit.

Verification. Suppose for contradiction that $a_n \to L$ for some $L \in \mathbb{R}$. Take $\varepsilon = 1/2$. By the definition of convergence, there exists N such that $|a_n - L| < 1/2$ for all $n > N$. In particular, $|a_n - L| < 1/2$ for two consecutive indices $n = 2k$ and $n = 2k + 1$ (for large k), giving $a_{2k} = 1$ and $a_{2k+1} = -1$. But then

$$2 = |1 - (-1)| = |a_{2k} - a_{2k+1}| \leq |a_{2k} - L| + |L - a_{2k+1}| < \frac{1}{2} + \frac{1}{2} = 1,$$

a contradiction. Hence no limit L exists. □

Example 7.8 (Limit of $a_n = (1 + 1/n)^n$)**.** As stated in Example 7.3, the sequence $a_n = (1 + 1/n)^n$ is increasing and bounded above by 3 (which can be shown via the binomial theorem). By the Monotone Convergence Theorem for real sequences, every bounded monotone sequence converges, so (a_n) has a limit. That limit is defined to be Euler's number:

$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e \approx 2.71828.$$

A rigorous ε-N verification requires bounding the sequence and using monotonicity; the full argument is a standard exercise in real analysis. □

7.1.4 Uniqueness of the Limit

Theorem 7.9 (Uniqueness of the Limit)**.** *Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. If (a_n) converges, then its limit is unique. That is, if $a_n \to L$ and $a_n \to M$ as $n \to \infty$, then $L = M$.*

Proof. Suppose $a_n \to L$ and $a_n \to M$. Let $\varepsilon > 0$ be arbitrary. By the definition of convergence:

- since $a_n \to L$, there exists $N_1 \in \mathbb{N}$ such that $n > N_1 \implies |a_n - L| < \dfrac{\varepsilon}{2}$;
- since $a_n \to M$, there exists $N_2 \in \mathbb{N}$ such that $n > N_2 \implies |a_n - M| < \dfrac{\varepsilon}{2}$.

Set $N = \max(N_1, N_2)$. For any $n > N$, both conditions hold simultaneously, so by the triangle inequality,

$$|L - M| = |L - a_n + a_n - M| \le |L - a_n| + |a_n - M| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since $\varepsilon > 0$ was chosen arbitrarily, we have shown that $|L - M| < \varepsilon$ for *every* $\varepsilon > 0$. Because $|L - M|$ is a fixed non-negative real number and it is strictly less than every positive number, it must equal zero:

$$|L - M| = 0 \implies L = M.$$

■

Logic and Intuition. The proof exploits a single fundamental insight: the definition of convergence places *both* candidate limits L and M in an arbitrarily tight grip around the terms of the sequence simultaneously. Because every tail of the sequence must lie within $\varepsilon/2$ of L and within $\varepsilon/2$ of M, the triangle inequality forces L and M to be within ε of *each other* — and this holds for every $\varepsilon > 0$, no matter how small.

Think of it geometrically. Imagine two points L and M on the real line, separated by some distance $d = |L - M|$. Now draw a band of radius $d/3$ around each. If the sequence eventually stays inside both bands simultaneously, then every term in the tail is simultaneously within $d/3$ of L and within $d/3$ of M. But two balls of radius $d/3$ centred at points separated by d do not overlap: a point cannot be within $d/3$ of both L and M if $d > 0$. This contradiction forces $d = 0$, i.e. $L = M$.

Remark (Why Uniqueness Matters in Econometrics). The uniqueness theorem guarantees that when we say an estimator $\hat{\theta}_n$ is *consistent*, the value it converges to is unambiguously defined. Without uniqueness, consistency could, in principle, point to two different targets simultaneously, making the concept incoherent. The same logic carries over intact to every stochastic convergence concept. If $X_n \xrightarrow{p} L$ and $X_n \xrightarrow{p} M$, then for any $\varepsilon > 0$,

$$\mathbb{P}(|L - M| > \varepsilon) \;\leq\; \mathbb{P}(|X_n - L| > \tfrac{\varepsilon}{2}) + \mathbb{P}(|X_n - M| > \tfrac{\varepsilon}{2}) \to 0,$$

so $|L - M| = 0$ almost surely. The structure of the argument is identical to the deterministic proof: two convergence assumptions, the triangle inequality, and an arbitrarily small tolerance. Uniqueness of probability limits, almost sure limits, and L^p limits all follow from the same template, underscoring once more how the deterministic theory of real sequences is the direct precursor of stochastic convergence theory.

7.1.5 Relevance to Convergence Concepts in Econometrics

Remark (From Deterministic Sequences to Stochastic Convergence). The study of deterministic sequences and their limits is not merely a mathematical preliminary; it is the conceptual template on which every notion of convergence in probability theory and econometrics is built.

In econometrics, one works not with sequences of fixed numbers but with sequences of *random variables* $(X_n)_{n=1}^{\infty}$ — for instance, a sequence of estimators $\hat{\theta}_n$ based on a sample of size n, or a sequence of partial sums $\bar{X}_n = n^{-1} \sum_{i=1}^{n} X_i$. The central question is whether, and in what sense, such a sequence settles near a fixed target as $n \to \infty$. Three distinct answers correspond to three notions of stochastic convergence, each of which is a direct probabilistic analogue of the ε-N definition above.

Convergence in probability ($X_n \xrightarrow{p} X$) asks that, for every $\varepsilon > 0$, the probability of the deviation $|X_n - X|$ exceeding ε tends to zero:

$$\mathbb{P}(|X_n - X| > \varepsilon) \to 0 \quad \text{as } n \to \infty.$$

Comparing this with the deterministic definition: the role of the *condition* $|a_n - L| < \varepsilon$ being true is now played by the *probability* of this condition being true approaching one. Consistency of an estimator is precisely convergence in probability of the estimator to the true parameter.

Almost sure convergence ($X_n \xrightarrow{\text{a.s.}} X$) is the pathwise version: it requires that the sequence of realisations $X_n(\omega)$ converges to $X(\omega)$ in the deterministic sense for almost every $\omega \in \Omega$, i.e.

$$\mathbb{P}(\{\omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\}) = 1.$$

For each fixed ω in a set of probability one, the deterministic sequence $(X_n(\omega))$ converges to $X(\omega)$ in the ε-N sense. The Strong Law of Large Numbers asserts almost sure convergence of the sample mean to the population mean.

Convergence in distribution ($X_n \xrightarrow{d} X$) weakens the requirement further: it asks only that the sequence of cumulative distribution functions F_{X_n} converges pointwise to F_X at continuity points of F_X. Here the underlying deterministic sequence being studied is the sequence of real numbers $(F_{X_n}(x))_{n=1}^{\infty}$ for each fixed x. The Central Limit Theorem, the cornerstone result justifying normal approximations in large samples, is a theorem about convergence in distribution.

In each case, understanding what it means for a *deterministic* sequence of real numbers to converge, and being able to verify or refute convergence using the ε-N definition, is the prerequisite skill. The stochastic convergence concepts that drive econometric asymptotic theory are obtained by inserting a probability measure into the same logical structure that governs sequences of real numbers.

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7.2 Almost Sure Convergence

Definition 7.10 (Almost Sure Convergence)**.** A sequence of random variables $(X_n)_{n=1}^{\infty}$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ **converges almost surely** (a.s.) to a random variable X, written $X_n \xrightarrow{\text{a.s.}} X$, if

$$\mathbb{P}\Big(\Big\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\Big\}\Big) = 1.$$

Equivalently, there exists a set $N \in \mathcal{F}$ with $\mathbb{P}(N) = 0$ such that for every $\omega \notin N$, $X_n(\omega) \to X(\omega)$ in the deterministic ε-N sense.

Logic and Intuition. Almost sure convergence is the most demanding of the stochastic convergence concepts, because it is a *pathwise* statement. The sample space Ω can be thought of as a collection of possible histories of the world. For each history ω, the sequence $X_1(\omega), X_2(\omega), X_3(\omega), \ldots$ is an ordinary sequence of real numbers. Almost sure convergence requires that this deterministic sequence converges to $X(\omega)$ in the classical sense for *almost every* history — that is, for all histories except possibly a set that the probability measure assigns total mass zero. It is the most direct probabilistic analogue of pointwise convergence of functions.

The contrast with weaker convergence concepts is instructive. Convergence in probability only requires that, at each fixed n, the random variable X_n is close to X with high probability — but it permits deviations to recur infinitely often across different values of n, provided they become individually rare. Almost sure convergence rules this out: it requires that the entire trajectory $\{X_n(\omega)\}_{n=1}^{\infty}$ ultimately stays close to $X(\omega)$ and never strays again.

Example 7.11 (Sample mean of i.i.d. variables)**.** Let $X_1, X_2, \ldots$ be i.i.d. random variables with $\mathbb{E}[X_1] = \mu < \infty$. The Strong Law of Large Numbers (see §7.7) states that

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{\text{a.s.}} \mu.$$

For almost every realised sequence of observations $x_1(\omega), x_2(\omega), \ldots$, the running average $n^{-1}\sum_{i=1}^{n} x_i(\omega)$ converges to μ in the classical deterministic sense. This is a path-by-path statement: almost every possible history of the random experiment produces a sequence of sample means that converges.

Remark (Almost Sure Convergence in Econometrics). Almost sure convergence is the strongest consistency property one can prove for an estimator, and it underwrites the strongest form of the Law of Large Numbers. In econometrics, the key results that rely on almost sure convergence include the following.

The **Strong Law of Large Numbers** (SLLN) asserts $\bar{X}_n \xrightarrow{\text{a.s.}} \mu$ whenever $\mathbb{E}[|X_1|] < \infty$ for an i.i.d. sequence, a result that requires no finite-variance condition. This is important when working with heavy-tailed variables — incomes, wealth, or trade flows that may have infinite variance.

The **Ergodic Theorem** is the time-series counterpart of the SLLN: for a strictly stationary and ergodic process $\{X_t\}$, the time average $T^{-1}\sum_{t=1}^{T} X_t$ converges almost surely to $\mathbb{E}[X_t]$. Ergodicity is the standard condition invoked in time-series econometrics to justify treating a single long time series as providing consistent estimates of population moments.

Almost sure convergence also enters the proofs of pointwise consistency of extremum estimators (M-estimators, maximum likelihood) via the Uniform Law of Large Numbers, which asserts that the sample objective function converges almost surely to its population counterpart uniformly in the parameter.

7.3 Convergence in Probability

Definition 7.12 (Convergence in Probability)**.** A sequence of random variables $(X_n)_{n=1}^{\infty}$ **converges in probability** to a random variable X, written $X_n \xrightarrow{p} X$, if for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0.$$

Logic and Intuition. Convergence in probability is a statement about the *distribution* of the deviation $|X_n - X|$ at each n, not about the path $\omega \mapsto X_n(\omega)$. It says that for any fixed tolerance $\varepsilon > 0$, the probability that X_n and X differ by more than ε shrinks to zero as n grows. The sequence need not converge path by path; it is permitted that individual paths wander, provided that the probability mass attached to large deviations evaporates in the limit.

A useful way to build intuition is to picture the probability distribution of X_n as a cloud of mass on the real line. Convergence in probability says that, as $n \to \infty$, the cloud concentrates around X: for any ball of radius ε around X, the fraction of probability mass lying outside that ball tends to zero. The cloud need not collapse to a point along any particular trajectory — it only needs to concentrate in the distributional sense.

Example 7.13 (Sample mean: WLLN)**.** Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. The sample mean $\bar{X}_n = n^{-1}\sum_{i=1}^{n} X_i$ satisfies $\mathbb{E}[\bar{X}_n] = \mu$ and $\text{Var}(\bar{X}_n) = \sigma^2/n$. By Chebyshev's inequality, for any $\varepsilon > 0$,

$$\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \;\leq\; \frac{\sigma^2}{n\varepsilon^2} \;\longrightarrow\; 0.$$

Hence $\bar{X}_n \xrightarrow{p} \mu$. The variance of the sample mean shrinks at rate $1/n$, pulling ever more probability mass into any fixed neighbourhood of μ.

Remark (Convergence in Probability in Econometrics). Convergence in probability is the standard notion of **consistency** in econometrics. An estimator $\hat{\theta}_n$ is consistent for the true parameter θ_0 if $\hat{\theta}_n \xrightarrow{p} \theta_0$. This is the minimal requirement for a useful large-sample estimator: it ensures that, with sufficiently many observations, the estimator is arbitrarily close to the truth with arbitrarily high probability.

Convergence in probability is also the form of convergence asserted by the Weak Law of Large Numbers (§7.7), which underpins the consistency of method-of-moments estimators, sample analogue estimators, and the first step of many two-step estimators (such as feasible GLS and control function approaches). In addition, convergence in probability is what the Continuous Mapping Theorem (§7.6) and Slutsky's Theorem (§7.6) operate on when deriving the large-sample distributions of functions of consistent estimators.

7.4 Convergence in Mean Square

Definition 7.14 (Convergence in Mean Square)**.** A sequence of random variables $(X_n)_{n=1}^{\infty}$ with $\mathbb{E}[X_n^2] < \infty$ for all n **converges in mean square** (or in L^2) to a random variable X with $\mathbb{E}[X^2] < \infty$, written $X_n \xrightarrow{L^2} X$ or $X_n \xrightarrow{\text{m.s.}} X$, if

$$\lim_{n\to\infty} \mathbb{E}\big[(X_n - X)^2\big] = 0.$$

Logic and Intuition. Mean square convergence requires that the *expected squared deviation* between X_n and X vanishes. Because $\mathbb{E}[(X_n - X)^2] = \text{Var}(X_n - X) + [\mathbb{E}(X_n - X)]^2$, mean square convergence jointly demands that both the bias $\mathbb{E}[X_n - X]$ and the variance $\text{Var}(X_n - X)$ of the deviation tend to zero. It is therefore a statement about the second moment of the error, not merely about its probability.

The relationship to convergence in probability is immediate from Markov's inequality applied to $(X_n - X)^2$: for any $\varepsilon > 0$,

$$\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}\big((X_n - X)^2 > \varepsilon^2\big) \leq \frac{\mathbb{E}[(X_n - X)^2]}{\varepsilon^2} \longrightarrow 0.$$

Hence L^2 **convergence implies convergence in probability**. The converse is false: convergence in probability does not require the second moment to exist, let alone to vanish.

Example 7.15 (Sample mean)**.** Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. Then $\mathbb{E}[(\bar{X}_n - \mu)^2] = \text{Var}(\bar{X}_n) = \sigma^2/n \to 0$, so $\bar{X}_n \xrightarrow{L^2} \mu$. The mean square error of the sample mean as an estimator of μ decays at the rate $1/n$.

Example 7.16 (A sequence converging in probability but not in L^2)**.** Let X_n take the value n with probability $1/n$ and the value 0 with probability $1 - 1/n$. For any $\varepsilon > 0$, $\mathbb{P}(|X_n| > \varepsilon) = 1/n \to 0$, so $X_n \xrightarrow{p} 0$. However, $\mathbb{E}[X_n^2] = n^2 \cdot (1/n) = n \to \infty$, so X_n does not converge in L^2 to 0. The occasional large value n inflates the second moment even though it becomes increasingly unlikely.

Remark (Mean Square Convergence in Econometrics). Mean square convergence is central to linear estimation theory. The Gauss–Markov theorem characterises the best linear unbiased estimator (BLUE) as the estimator minimising the mean square error in the class of linear unbiased estimators. In time-series econometrics, least-squares projections and prediction theory are formulated in terms of L^2 distance: the conditional expectation $\mathbb{E}[Y|X]$ is the L^2 projection of Y onto the space of measurable functions of X, meaning it minimises $\mathbb{E}[(Y-g(X))^2]$ over all functions g.

Mean square convergence also appears in the proof of consistency for time-series estimators under summable autocovariances (§7.7), where the variance of the sample mean is shown to tend to zero, and in the theory of L^2-stationary (covariance-stationary) processes, where spectral representations and Hilbert space methods rely on the L^2 structure.

7.5 Convergence in Distribution

Definition 7.17 (Convergence in Distribution)**.** A sequence of random variables $(X_n)_{n=1}^{\infty}$ with cumulative distribution functions F_{X_n} **converges in distribution** (or **weakly**) to a random variable X with CDF F_X, written $X_n \xrightarrow{d} X$, if

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$$

at every continuity point x of F_X.

Logic and Intuition. Convergence in distribution is the weakest of the four convergence concepts. It does not require X_n and X to be defined on the same probability space, and it makes no claim about how close the *realisations* $X_n(\omega)$ are to $X(\omega)$ for any individual ω. Instead, it asks only that the *probability distributions* of X_n converge, in the sense that probabilities assigned to half-lines $(-\infty, x]$ converge to those assigned by the limiting distribution. The CDFs, viewed as sequences of real-valued functions of x, converge pointwise at every continuity point.

The phrase "at every continuity point" is not a technicality to be overlooked. If F_X has a jump at some x^* (as discrete distributions do), the CDF values $F_{X_n}(x^*)$ are not required to converge to $F_X(x^*)$. This is because the limiting mass at the point x^* can be approached from either side, and requiring convergence at a jump point would impose a convention about how the limiting distribution assigns mass to that point, which the definition wisely avoids.

A concrete example makes the point vivid. Let X_n be degenerate at the value $a_n = 1/n$, so that all of its probability mass sits at a single point that slides toward the origin as n grows. Intuitively X_n converges to the constant 0, and indeed $X_n \xrightarrow{d} X$ where X is degenerate at 0. Each F_{X_n} is a step function that jumps from 0 to 1 at $a_n = 1/n$, while the limiting CDF F_X jumps at 0. At every point $x \neq 0$ the approximating CDFs eventually agree with the limit: for $x < 0$ all values are 0, and for $x > 0$ they are eventually 1. But exactly at the jump point $x^* = 0$ we have $F_{X_n}(0) = 0$ for *every* n (the mass sits just to the right, at $1/n > 0$), whereas $F_X(0) = 1$. The gap between $F_{X_n}(0)$ and $F_X(0)$ never closes — it equals 1 for all n — even though $X_n \xrightarrow{d} 0$ in every meaningful sense. Figure~5 illustrates this situation. Requiring convergence at x^* would perversely declare this natural sequence non-convergent, which is precisely why the definition restricts attention to continuity points.

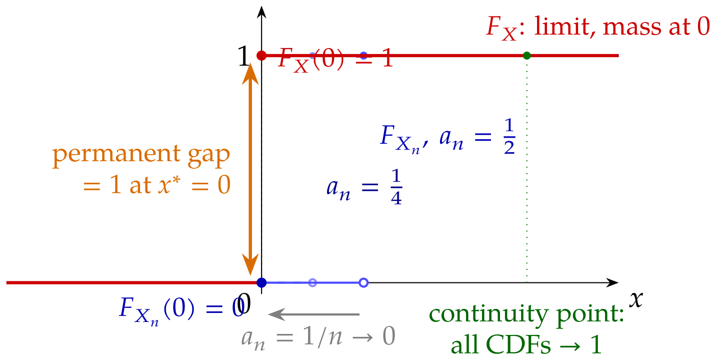

Figure 5: **Why convergence in distribution is required only at continuity points.** The random variable X_n places all its mass at $a_n = 1/n$, so its CDF F_{X_n} (blue) is a step that jumps from 0 to 1 at $1/n$; as $n \to \infty$ the jump slides left toward the origin. The limit X is degenerate at 0, with CDF F_X (red) jumping at $x^* = 0$. At every $x \neq 0$ the blue curves eventually coincide with the red one — for instance at the continuity point marked in green, all CDFs equal 1. But exactly at the jump point $x^* = 0$ the approximating values stay pinned at $F_{X_n}(0) = 0$ while $F_X(0) = 1$, leaving a gap of 1 that never closes. Insisting on convergence there would wrongly brand this perfectly sensible sequence non-convergent; restricting the requirement to continuity points resolves the paradox.

Example 7.18 (Central Limit Theorem)**.** Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{E}[X_i] = \mu$ and $0 < \mathrm{Var}(X_i) = \sigma^2 < \infty$. The standardised sample mean

$$Z_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}$$

has mean 0 and variance 1 for every n. The Central Limit Theorem (§7.8) states that $Z_n \xrightarrow{d} \mathcal{N}(0, 1)$: the CDF of Z_n converges to the standard normal CDF Φ at every point. The individual realisations of Z_n need not be close to any fixed normal random variable; only the distributional shape converges.

Remark (Convergence in Distribution in Econometrics)*.* Convergence in distribution is the foundation of asymptotic inference in econometrics. Since estimators are typically not exactly normally distributed in finite samples, large-sample theory establishes that a suitably normalised estimator converges in distribution to a known limit (usually normal or chi-squared), and this limiting distribution is then used to approximate finite-sample distributions and construct tests and confidence intervals.

Three results that rely on convergence in distribution are especially prominent. The **Central Limit Theorem** (§7.8) establishes the asymptotic normality of sample means and, by extension, of OLS, IV, and GMM estimators under standard regularity conditions. The **delta method** extends asymptotic normality to smooth nonlinear functions of estimators: if $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, V)$ and g is differentiable at θ_0, then $\sqrt{n}(g(\hat{\theta}) - g(\theta_0)) \xrightarrow{d} \mathcal{N}(0, [g'(\theta_0)]^2 V)$. **Asymptotic pivot statistics** — t-statistics, F-statistics, Wald, LR, and LM test statistics — are functions of estimators whose limiting distributions are known (standard normal, chi-squared), and their validity as test statistics rests on convergence in distribution results.

7.6 Relationships Among Convergence Concepts

The four convergence concepts introduced above are not independent: they form a strict hierarchy, with almost sure convergence being the strongest and convergence in distribution the weakest. Understanding these relationships is essential for knowing which convergence results carry over to others, and which do not.

Theorem 7.19 (Hierarchy of Convergence)**.** *Let (X_n) be a sequence of random variables and X a random variable, all defined on the same probability space. Then:*

1. $X_n \xrightarrow{\text{a.s.}} X \implies X_n \xrightarrow{p} X$.
2. $X_n \xrightarrow{L^2} X \implies X_n \xrightarrow{p} X$.
3. $X_n \xrightarrow{p} X \implies X_n \xrightarrow{d} X$.

None of these implications reverses in general.

Proof. **(1) Almost sure implies in probability.** Fix $\varepsilon > 0$. Define the event $A_n = \{|X_n - X| > \varepsilon\}$ and let $B = \{\omega : X_n(\omega) \not\to X(\omega)\}$. Since $X_n \xrightarrow{\text{a.s.}} X$, we have $\mathbb{P}(B) = 0$. Now $\limsup_n A_n \subseteq B$ (if $|X_n(\omega) - X(\omega)| > \varepsilon$ infinitely often, then $X_n(\omega)$ does not converge to $X(\omega)$), so $\mathbb{P}(\limsup_n A_n) \leq \mathbb{P}(B) = 0$. Since $\mathbb{P}(\limsup_n A_n) \geq \limsup_n \mathbb{P}(A_n) \geq 0$ by Fatou's lemma for probabilities, we conclude $\limsup_n \mathbb{P}(A_n) = 0$, hence $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$. □

(2) L^2 implies in probability. This follows directly from Markov's inequality: for any $\varepsilon > 0$,

$$\mathbb{P}(|X_n - X| > \varepsilon) \leq \frac{\mathbb{E}[(X_n - X)^2]}{\varepsilon^2} \to 0. \quad \square$$

(3) In probability implies in distribution. Fix a continuity point x of F_X and let $\varepsilon > 0$. For any $\delta > 0$,

$$F_{X_n}(x) = \mathbb{P}(X_n \leq x) \leq \mathbb{P}(X \leq x + \delta) + \mathbb{P}(|X_n - X| > \delta) = F_X(x + \delta) + \mathbb{P}(|X_n - X| > \delta).$$

Similarly, $F_X(x - \delta) \leq F_{X_n}(x) + \mathbb{P}(|X_n - X| > \delta)$. Since $X_n \xrightarrow{p} X$, the probability term $\to 0$. Since x is a continuity point, $F_X(x \pm \delta) \to F_X(x)$ as $\delta \to 0$. Letting $n \to \infty$ and then $\delta \to 0$ gives $F_{X_n}(x) \to F_X(x)$. □ ■

Remark (Counterexamples for the Reversed Implications). Each implication in Theorem 7.1 is strict; the converses fail.

In probability does not imply a.s. The classic counterexample is the "moving bump": for $n = 2^k + j$ with $0 \leq j < 2^k$, let $X_n = \mathbf{1}_{[j/2^k, (j+1)/2^k]}(U)$ where $U \sim \text{Uniform}[0, 1]$. Then $\mathbb{P}(|X_n| > \varepsilon) = 2^{-k} \to 0$, but for every ω, $X_n(\omega) = 1$ infinitely often and $X_n(\omega) = 0$ infinitely often, so the sequence does not converge for any ω.

In distribution does not imply in probability. Let $X \sim \mathcal{N}(0, 1)$ and $X_n = -X$ for all n. Then $X_n \xrightarrow{d} X$ (same distribution), but X_n does not converge in probability to X because $\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}(2|X| > \varepsilon) \not\to 0$.

Remark (Convergence in Distribution to a Constant). An important special case bridges convergence in distribution and convergence in probability: if the limiting distribution is degenerate at a constant c (i.e. $X = c$ almost surely), then $X_n \xrightarrow{d} c$ if and only if $X_n \xrightarrow{p} c$. This equivalence is used repeatedly in econometrics when showing that a consistent estimator, which converges in probability to the true parameter, has a degenerate limiting distribution at that parameter.

7.6.1 The Continuous Mapping Theorem

A key tool for deriving the large-sample distributions of functions of estimators is the Continuous Mapping Theorem (CMT), which states that continuous transformations preserve convergence.

Theorem 7.20 (Continuous Mapping Theorem). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $\{X_n\}$, X be random vectors taking values in $\mathbb{R}^k$. Let $g : \mathbb{R}^k \to \mathbb{R}^m$ be a Borel-measurable function, and let $C \subseteq \mathbb{R}^k$ be the set of points at which g is continuous. Suppose $\mathbb{P}(X \in C) = 1$. Then:*

1. *If $X_n \xrightarrow{\text{a.s.}} X$, then $g(X_n) \xrightarrow{\text{a.s.}} g(X)$.*
2. *If $X_n \xrightarrow{p} X$, then $g(X_n) \xrightarrow{p} g(X)$.*
3. *If $X_n \xrightarrow{d} X$, then $g(X_n) \xrightarrow{d} g(X)$.*

Logic and Intuition. The CMT says that continuous functions cannot create new limiting behaviour: if the inputs converge (in any of the three senses), the outputs converge to the image of the limit. The reason is that continuity at $X(\omega)$ means that for any $\varepsilon > 0$ there exists $\delta > 0$ such that $\|x - X(\omega)\| < \delta \implies \|g(x) - g(X(\omega))\| < \varepsilon$. Whenever $X_n(\omega)$ is within δ of $X(\omega)$, $g(X_n(\omega))$ is automatically within ε of $g(X(\omega))$. The condition $\mathbb{P}(X \in C) = 1$ is the precise sense in which g is "essentially continuous": discontinuities are permitted, but they must form a set that the limiting random variable X visits with probability zero.

Proof. We prove the three parts in order of increasing difficulty. Part (ii) is reduced to Part (i) via the subsequence characterisation of convergence in probability; Part (iii) uses the closed-set form of the Portmanteau theorem.

Part (i): Almost sure convergence.

Let $N = \{\omega : X_n(\omega) \not\to X(\omega)\}$ be the null set on which X_n fails to converge almost surely, so $\mathbb{P}(N) = 0$. Let $D = \{\omega : X(\omega) \notin C\}$ be the set on which X lands at a discontinuity of g; by hypothesis, $\mathbb{P}(D) = 0$.

Fix any $\omega \in (N \cup D)^c$, which has probability $\mathbb{P}((N \cup D)^c) = 1$. On this event:

- $X_n(\omega) \to X(\omega)$ (since $\omega \notin N$), and
- g is continuous at $X(\omega)$ (since $\omega \notin D$, so $X(\omega) \in C$).

By the sequential characterisation of continuity, $x_n \to x$ and continuity of g at x together imply $g(x_n) \to g(x)$. Therefore $g(X_n(\omega)) \to g(X(\omega))$ for every $\omega \in (N \cup D)^c$. Since this set has probability 1, $g(X_n) \xrightarrow{\text{a.s.}} g(X)$. $\square$

Part (ii): Convergence in probability.

We use the following standard lemma, which characterises convergence in probability via subsequences:

Lemma. $Y_n \xrightarrow{p} Y$ if and only if every subsequence $(Y_{n_k})_{k\geq 1}$ contains a further subsequence $(Y_{n_{k_j}})_{j\geq 1}$ with $Y_{n_{k_j}} \xrightarrow{\text{a.s.}} Y$.

Assume $X_n \xrightarrow{p} X$. Let (X_{n_k}) be an arbitrary subsequence. Since the full sequence converges in probability to X, so does (X_{n_k}). By the Lemma (forward direction), there exists a further subsequence $X_{n_{k_j}} \xrightarrow{\text{a.s.}} X$. Applying Part (i) to this almost-surely convergent subsequence:

$$g(X_{n_{k_j}}) \xrightarrow{\text{a.s.}} g(X).$$

Since almost sure convergence implies convergence in probability, $g(X_{n_{k_j}}) \xrightarrow{p} g(X)$.

We have shown: every subsequence of $(g(X_n))$ contains a further subsequence that converges in probability (in fact almost surely) to $g(X)$. By the Lemma (reverse direction), $g(X_n) \xrightarrow{p} g(X)$. $\square$

Part (iii): Convergence in distribution.

We use the Portmanteau theorem, which states that $X_n \xrightarrow{d} X$ if and only if $\limsup_{n\to\infty} \mathbb{P}(X_n \in F) \leq \mathbb{P}(X \in F)$ for every closed set $F \subseteq \mathbb{R}^k$.

Let $G \subseteq \mathbb{R}^m$ be an arbitrary closed set. We must show

$$\limsup_{n\to\infty} \mathbb{P}(g(X_n) \in G) \leq \mathbb{P}(g(X) \in G).$$

Since $\mathbb{P}(g(X_n) \in G) = \mathbb{P}(X_n \in g^{-1}(G))$ and $g^{-1}(G) \subseteq \overline{g^{-1}(G)}$ (the closure in $\mathbb{R}^k$), we have

$$\mathbb{P}(g(X_n) \in G) \leq \mathbb{P}(X_n \in \overline{g^{-1}(G)}).$$

Since $\overline{g^{-1}(G)}$ is closed and $X_n \xrightarrow{d} X$, the Portmanteau theorem gives

$$\limsup_{n\to\infty} \mathbb{P}(g(X_n) \in G) \;\leq\; \limsup_{n\to\infty} \mathbb{P}(X_n \in \overline{g^{-1}(G)}) \;\leq\; \mathbb{P}(X \in \overline{g^{-1}(G)}).$$

It remains to show $\mathbb{P}(X \in \overline{g^{-1}(G)}) = \mathbb{P}(X \in g^{-1}(G))$, i.e. that

$$\mathbb{P}\Big(X \in \overline{g^{-1}(G)} \setminus g^{-1}(G)\Big) = 0.$$

We claim $\overline{g^{-1}(G)} \setminus g^{-1}(G) \subseteq C^c$ (the discontinuity set of g). To see this, take any $x \in \overline{g^{-1}(G)} \setminus g^{-1}(G)$. Since x belongs to the closure of $g^{-1}(G)$, there is a sequence $x_n \in g^{-1}(G)$ with $x_n \to x$, so $g(x_n) \in G$ for all n. Yet $x \notin g^{-1}(G)$, meaning $g(x) \notin G$. If g were continuous at x, then

$g(x_n) \to g(x)$, and since G is closed this would force $g(x) \in G$—a contradiction. Hence g must be discontinuous at x, i.e. $x \in C^c$.

Therefore

$$\mathbb{P}\Big(X \in \overline{g^{-1}(G)} \setminus g^{-1}(G)\Big) \;\le\; \mathbb{P}(X \in C^c) = 0,$$

by the hypothesis $\mathbb{P}(X \in C) = 1$. Combining:

$$\limsup_{n\to\infty} \mathbb{P}(g(X_n) \in G) \;\le\; \mathbb{P}(X \in g^{-1}(G)) = \mathbb{P}(g(X) \in G).$$

Since G was an arbitrary closed set, the Portmanteau theorem gives $g(X_n) \xrightarrow{d} g(X)$. ∎

Remark (CMT in Econometrics). The CMT is used constantly in asymptotic theory, but it normally operates together with an LLN. For example, if $\hat\beta \xrightarrow{p} \beta_0$ and $\hat u_i = y_i - x_i'\hat\beta$, consistency of $\hat\sigma^2 = n^{-1}\sum_i \hat u_i^2$ requires an LLN for the relevant sample moments, appropriate finite-moment and model assumptions, and then the CMT to replace $\hat\beta$ by its probability limit inside the limiting expression. The CMT alone does not supply the probability limit of a growing sample average. By contrast, if $Z_n \xrightarrow{d} \mathcal{N}(0,1)$, then $Z_n^2 \xrightarrow{d} \chi_1^2$ directly because squaring is continuous. Test statistics defined as continuous functions of sample moments inherit their asymptotic distributions only after the required convergence of those moments has been established.

7.6.2 Slutsky's Theorem

Theorem 7.21 (Slutsky's Theorem). *Let $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$ for some constant $c \in \mathbb{R}$. Then:*

1. $X_n + Y_n \xrightarrow{d} X + c$.
2. $X_n Y_n \xrightarrow{d} cX$.
3. *If $c \neq 0$, then $X_n / Y_n \xrightarrow{d} X/c$.*

Logic and Intuition. Slutsky's Theorem handles mixed situations where one sequence converges in distribution (to a non-degenerate limit) and another converges in probability (to a constant). The key insight is that the second sequence, because it converges to a constant, becomes *asymptotically non-random*: it behaves like a fixed number in the limit. The joint pair (X_n, Y_n) therefore converges in distribution to (X, c), and continuous functions of the pair converge accordingly.

Remark (Slutsky's Theorem in Econometrics). Slutsky's Theorem is the workhorse for handling estimated normalising constants in t-statistics. Suppose, under the homoskedastic assumptions stated below,

$$\sqrt{n}(\hat\beta - \beta_0) \xrightarrow{d} \mathcal{N}(0, \sigma^2 Q^{-1}), \qquad \hat\sigma^2 \xrightarrow{p} \sigma^2, \qquad \hat Q \xrightarrow{p} Q.$$

For coefficient j, define the estimated asymptotic standard error

$$\widehat{\text{ase}}_j = \hat\sigma\sqrt{(\hat Q^{-1})_{jj}}.$$

Then the fully studentised statistic

$$t_{n,j} = \frac{\sqrt{n}(\hat{\beta}_j - \beta_{0j})}{\hat{\sigma}\sqrt{(\hat{Q}^{-1})_{jj}}} \xrightarrow{d} \mathcal{N}(0,1).$$

Dividing only by $\hat{\sigma}$ would yield asymptotic covariance Q^{-1}, not Q^{-1}/σ^2, and would not generally produce a standard normal coefficient statistic. Slutsky's Theorem justifies replacing the unknown population normalising quantities by consistent estimators without changing the limiting distribution.

7.7 Laws of Large Numbers

Laws of Large Numbers (LLNs) are the results that formalise the intuition that sample averages converge to population means. They are the primary tools for establishing the consistency of estimators and sample analogue statistics. This section presents four versions: the Weak Law for i.i.d. sequences, the Weak Law for independent non-identically distributed sequences, the Strong Law (Kolmogorov), and a Law of Large Numbers for stationary time series.

7.7.1 Weak Law of Large Numbers (i.i.d.)

Theorem 7.22 (WLLN — i.i.d. case)**.** *Let $X_1, X_2, \ldots$ be i.i.d. random variables with $\mathbb{E}[X_1] = \mu$ and* $\text{Var}(X_1) = \sigma^2 < \infty$*. Define the sample mean $\bar{X}_n = n^{-1}\sum_{i=1}^n X_i$. Then*

$$\bar{X}_n \xrightarrow{p} \mu.$$

Logic and Intuition. The theorem says the sample mean collapses onto the population mean as the sample grows. The engine behind this is a single mechanism: the sample mean is an unbiased estimator whose *variance shrinks to zero* at rate $1/n$. Once the spread of a random variable around a fixed centre vanishes, essentially all of its probability mass must pile up arbitrarily close to that centre — and Chebyshev's inequality is exactly the tool that converts "small variance" into "small probability of being far from the mean." The proof therefore has three moves: (i) locate the centre of $\bar{X}_n$ (its mean), (ii) measure its spread (its variance), and (iii) feed both into Chebyshev and let $n \to \infty$.

Proof. Fix an arbitrary tolerance $\varepsilon > 0$; it will remain fixed throughout. Our goal, by the definition of convergence in probability, is to show that $\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \to 0$ as $n \to \infty$.

Step 1: The sample mean is centred at μ (no bias). Using linearity of expectation (Theorem 5.5) and the fact that each X_i has mean μ,

$$\mathbb{E}[\bar{X}_n] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n X_i\right] = \frac{1}{n}\sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n}\sum_{i=1}^n \mu = \frac{1}{n}\cdot n\mu = \mu.$$

Thus $\bar{X}_n$ is an *unbiased* estimator of μ: whatever the sample size, its distribution is centred exactly on the target. Note that linearity alone suffices here — independence is not yet needed.

Step 2: The variance of the sample mean is σ^2/n. Pulling the constant $1/n$ out of the variance (which contributes its square, $1/n^2$) and using the additivity of variance for uncorrelated variables,

$$\operatorname{Var}(\bar{X}_n) = \operatorname{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right).$$

Now expand the variance of the sum. In general,

$$\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n}\operatorname{Var}(X_i) + 2\sum_{1\le i<j\le n}\operatorname{Cov}(X_i, X_j).$$

Because the X_i are independent, every cross-covariance vanishes: $\operatorname{Cov}(X_i, X_j) = 0$ for $i \neq j$. (Independence is precisely the assumption doing the work in this step.) Only the diagonal terms survive, and each equals σ^2, so

$$\operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n}\sigma^2 = n\sigma^2, \qquad \text{hence} \qquad \operatorname{Var}(\bar{X}_n) = \frac{1}{n^2}\cdot n\sigma^2 = \frac{\sigma^2}{n}.$$

This is the crux of the whole argument: the spread of $\bar{X}_n$ around μ decreases like $1/n$, and the finiteness of σ^2 is exactly what guarantees this quantity is well defined and tends to 0.

Step 3: Convert small variance into small probability via Chebyshev. Chebyshev's inequality (established earlier among the moment inequalities) states that for any random variable Y with finite variance and any $\varepsilon > 0$, $\mathbb{P}(|Y - \mathbb{E}[Y]| \ge \varepsilon) \le \operatorname{Var}(Y)/\varepsilon^2$. Apply it to $Y = \bar{X}_n$, whose mean is μ (Step 1) and whose variance is σ^2/n (Step 2):

$$\mathbb{P}(|\bar{X}_n - \mu| \ge \varepsilon) \;\le\; \frac{\operatorname{Var}(\bar{X}_n)}{\varepsilon^2} \;=\; \frac{\sigma^2}{n\,\varepsilon^2}.$$

Step 4: Take the limit. With ε and σ^2 fixed, the right-hand side is a constant divided by n, so it vanishes as the sample grows:

$$\lim_{n\to\infty}\frac{\sigma^2}{n\,\varepsilon^2} = 0.$$

Since a probability is never negative, we have the sandwich

$$0 \;\le\; \mathbb{P}(|\bar{X}_n - \mu| \ge \varepsilon) \;\le\; \frac{\sigma^2}{n\,\varepsilon^2},$$

and both outer bounds tend to 0. By the squeeze principle, the middle term tends to 0 as well:

$$\lim_{n\to\infty}\mathbb{P}(|\bar{X}_n - \mu| \ge \varepsilon) = 0.$$

Because $\varepsilon > 0$ was arbitrary, this holds for every tolerance, which is exactly the definition of convergence in probability. Therefore $\bar{X}_n \xrightarrow{p} \mu$. ■

Remark (On the assumptions and their econometric meaning). Two features of this proof deserve emphasis. First, *finite variance is a convenience, not a necessity*: the WLLN in fact holds under the weaker assumption that only the mean $\mathbb{E}|X_1| < \infty$ exists (Khinchin's theorem), though that proof requires characteristic functions or a truncation argument rather than the one-line Chebyshev bound used here. The variance-based proof is preferred in a first course precisely because it is transparent. Second, the mechanism "unbiased + variance → 0" is the template for nearly every consistency proof in econometrics. To show an estimator $\hat{\theta}_n$ is consistent for θ_0, one typically shows $\mathbb{E}[\hat{\theta}_n] \to \theta_0$ (asymptotic unbiasedness) and $\text{Var}(\hat{\theta}_n) \to 0$, and then invokes exactly the Chebyshev argument above. The WLLN is thus not merely a result about sample means but the prototype for the large-sample justification of estimators throughout the discipline.

7.7.1.1 Application: Proving the Consistency of Estimators

The practical payoff of the Weak Law of Large Numbers is enormous, and its use in econometrics is remarkably simple and mechanical. Almost every estimator we work with can be written as a smooth function of *sample averages*. The WLLN tells us that each such sample average converges in probability to the corresponding *population expectation*. Establishing consistency then reduces to a three-step recipe:

1. **Write the estimator through sample averages.** Express $\hat{\theta}_n$ as a continuous function of one or more quantities of the form $n^{-1}\sum_{i=1}^{n} h(W_i)$.
2. **Invoke the WLLN.** Replace each sample average by its probability limit: $n^{-1}\sum_{i=1}^{n} h(W_i) \xrightarrow{p} \mathbb{E}[h(W_i)]$.
3. **Apply the Continuous Mapping Theorem and Slutsky's Theorem.** Since the estimator is a continuous function of these averages (at the relevant limiting values), it converges in probability to the same function of the population expectations, which equals the true parameter.

The whole difficulty of a consistency proof is therefore concentrated in a single, familiar tool. Once the estimator is written in terms of averages, one simply "reads off" the limits. The two examples below — the workhorse OLS estimator and the general GMM estimator — show the recipe in action.

Example 7.23 (Consistency of the OLS estimator)**.** Consider the linear model $y_i = x_i'\beta_0 + u_i$, where x_i is a $k \times 1$ vector of regressors and $\{(x_i, u_i)\}_{i=1}^{n}$ are i.i.d. Assume (i) $\mathbb{E}[x_i x_i'] = Q$ is finite and nonsingular, and (ii) $\mathbb{E}[x_i u_i] = 0$ (the regressors are exogenous), with finite second moments so that the WLLN applies to the averages below. The OLS estimator is

$$\hat{\beta}_n = \left(\sum_{i=1}^{n} x_i x_i'\right)^{-1} \sum_{i=1}^{n} x_i y_i = \left(\frac{1}{n}\sum_{i=1}^{n} x_i x_i'\right)^{-1} \left(\frac{1}{n}\sum_{i=1}^{n} x_i y_i\right),$$

where dividing numerator and denominator by n turns both pieces into sample averages. Substituting $y_i = x_i'\beta_0 + u_i$ into the second factor,

$$\frac{1}{n}\sum_{i=1}^{n} x_i y_i = \left(\frac{1}{n}\sum_{i=1}^{n} x_i x_i'\right)\beta_0 + \frac{1}{n}\sum_{i=1}^{n} x_i u_i,$$

so that

$$\hat{\beta}_n = \beta_0 + \left(\frac{1}{n}\sum_{i=1}^{n} x_i x_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n} x_i u_i\right).$$

Now apply the WLLN to each average (element by element, each being an average of i.i.d. terms):

$$\frac{1}{n}\sum_{i=1}^{n} x_i x_i' \xrightarrow{p} \mathbb{E}[x_i x_i'] = Q, \qquad \frac{1}{n}\sum_{i=1}^{n} x_i u_i \xrightarrow{p} \mathbb{E}[x_i u_i] = 0.$$

Because matrix inversion is continuous at the nonsingular matrix Q, the Continuous Mapping Theorem (Theorem 7.20) gives $(n^{-1}\sum x_i x_i')^{-1} \xrightarrow{p} Q^{-1}$, and Slutsky's Theorem lets us multiply the convergent factors:

$$\hat{\beta}_n \xrightarrow{p} \beta_0 + Q^{-1}\cdot 0 = \beta_0.$$

The OLS estimator is therefore consistent. The entire argument rests on two applications of the WLLN; exogeneity ($\mathbb{E}[x_i u_i] = 0$) is exactly what forces the error term to average out.

Example 7.24 (Consistency of the GMM estimator)**.** Generalised Method of Moments estimation starts from a population moment condition: there is a $q \times 1$ vector of moment functions $g(w_i, \theta)$ and a true parameter $\theta_0 \in \Theta \subseteq \mathbb{R}^p$ (with $q \geq p$) such that

$$\mathbb{E}[g(w_i, \theta_0)] = 0,$$

and θ_0 is the *only* value in Θ satisfying this (the identification condition). Given i.i.d. data $\{w_i\}$ and a weight matrix $W_n \xrightarrow{p} W$ (positive definite), the GMM estimator minimises the quadratic form in the sample moments,

$$\hat{\theta}_n = \arg\min_{\theta\in\Theta} \bar{g}_n(\theta)' W_n \bar{g}_n(\theta), \qquad \bar{g}_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} g(w_i, \theta).$$

Here the WLLN again does the essential work. For each fixed θ, $\bar{g}_n(\theta)$ is a sample average of the i.i.d. vectors $g(w_i, \theta)$, so

$$\bar{g}_n(\theta) \xrightarrow{p} \mathbb{E}[g(w_i, \theta)] \equiv m(\theta).$$

Consequently the sample objective converges to its population counterpart, $\bar{g}_n(\theta)' W_n \bar{g}_n(\theta) \xrightarrow{p} m(\theta)' W m(\theta) \equiv Q_0(\theta)$. Under identification, $m(\theta) = 0$ if and only if $\theta = \theta_0$, so $Q_0(\theta)$ is uniquely minimised at θ_0 (where it equals 0; elsewhere it is strictly positive since W is positive definite). Provided the convergence of the sample objective to Q_0 is *uniform* in θ (a uniform WLLN, guaranteed under standard regularity conditions such as a compact Θ and continuous, dominated g), the minimiser of the sample objective converges to the minimiser of the population objective:

$$\hat{\theta}_n \xrightarrow{p} \theta_0.$$

Once more the substantive engine is the WLLN, which delivers $\bar{g}_n(\theta) \xrightarrow{p} \mathbb{E}[g(w_i, \theta)]$; identification and uniform convergence merely upgrade this pointwise statement to convergence of the argmin.

In the *just-identified* case $q = p$, the estimator solves $\bar{g}_n(\hat{\theta}_n) = 0$ directly and the argument becomes as transparent as the OLS case above. For instance, the instrumental-variables estimator arises from the moment condition $\mathbb{E}[z_i(y_i - x_i'\beta_0)] = 0$, giving $\hat{\beta}_n^{\mathrm{IV}} = (n^{-1}\sum z_i x_i')^{-1}(n^{-1}\sum z_i y_i)$, which is consistent by exactly the two-WLLN argument used for OLS, with z_i replacing x_i in the averages.

Example 7.25 (Consistency of the linear GMM estimator)**.** The general GMM argument becomes fully explicit — with a closed-form estimator, exactly like OLS — when the moment functions are *linear* in the parameter. This is the case that covers instrumental-variables estimation with possibly more instruments than regressors, and it is the workhorse of applied econometrics.

Consider again the linear model $y_i = x_i'\beta_0 + u_i$, where x_i is $p \times 1$, but now suppose we have a $q \times 1$ vector of instruments z_i with $q \geq p$, satisfying the moment condition

$$\mathbb{E}[z_i\, u_i] = \mathbb{E}\big[z_i\,(y_i - x_i'\beta_0)\big] = 0.$$

The corresponding sample moment function is linear in β,

$$\bar{g}_n(\beta) = \frac{1}{n}\sum_{i=1}^{n} z_i(y_i - x_i'\beta) = S_{zy} - S_{zx}\,\beta,$$

where we have introduced the sample averages $S_{zx} = n^{-1}\sum_i z_i x_i'$ (of dimension $q \times p$) and $S_{zy} = n^{-1}\sum_i z_i y_i$ (of dimension $q \times 1$). Given a weight matrix $W_n \xrightarrow{p} W$ (symmetric positive definite), the GMM estimator minimises the quadratic form $\bar{g}_n(\beta)' W_n\, \bar{g}_n(\beta)$. Because $\bar{g}_n(\beta)$ is linear in β, this is a quadratic minimisation with the explicit solution

$$\hat{\beta}_n^{\text{GMM}} = (S_{zx}' W_n\, S_{zx})^{-1} S_{zx}' W_n\, S_{zy}.$$

To establish consistency, substitute $y_i = x_i'\beta_0 + u_i$ into S_{zy}, which gives $S_{zy} = S_{zx}\beta_0 + n^{-1}\sum_i z_i u_i$. Writing $S_{zu} = n^{-1}\sum_i z_i u_i$ and inserting this into the formula above, the β_0 term passes through the estimator unchanged, leaving

$$\hat{\beta}_n^{\text{GMM}} = \beta_0 + (S_{zx}' W_n\, S_{zx})^{-1} S_{zx}' W_n\, S_{zu}.$$

Now every ingredient is a sample average to which the WLLN applies:

$$S_{zx} \xrightarrow{p} \mathbb{E}[z_i x_i'] = Q_{zx}, \qquad S_{zu} \xrightarrow{p} \mathbb{E}[z_i u_i] = 0, \qquad W_n \xrightarrow{p} W.$$

Assume the *rank condition* that Q_{zx} has full column rank p (instrument relevance); together with W positive definite this makes the $p \times p$ matrix $Q_{zx}' W Q_{zx}$ nonsingular, so matrix inversion is continuous at the limit. Applying the Continuous Mapping Theorem (Theorem 7.20) and Slutsky's Theorem term by term,

$$\hat{\beta}_n^{\text{GMM}} \xrightarrow{p} \beta_0 + (Q_{zx}' W Q_{zx})^{-1} Q_{zx}' W \cdot 0 = \beta_0.$$

The linear GMM estimator is therefore consistent. The proof is once more just the two-WLLN recipe — one law for S_{zx} (which must be well behaved and full rank) and one for S_{zu} (which must average to zero by instrument exogeneity). This single estimator nests the earlier cases: taking $z_i = x_i$ recovers OLS (for any W), and taking $q = p$ recovers the just-identified IV estimator, in which the weight matrix drops out entirely because S_{zx} is square and invertible.

7.7.2 Weak Law of Large Numbers (Non-i.i.d.)

Theorem 7.26 (WLLN — Independent, Non-Identically Distributed)**.** *Let $X_1, X_2, \ldots$ be* independent (*not necessarily identically distributed*) *random variables with $\mu_i = \mathbb{E}[X_i]$ and $\sigma_i^2 = \mathrm{Var}(X_i) < \infty$. If*

$$\frac{1}{n^2}\sum_{i=1}^{n} \sigma_i^2 \longrightarrow 0,$$

then

$$\frac{1}{n}\sum_{i=1}^{n} X_i - \frac{1}{n}\sum_{i=1}^{n} \mu_i \xrightarrow{p} 0.$$

In particular, if the variances are uniformly bounded ($\sigma_i^2 \le C < \infty$ for all i), the condition holds since $n^{-2}\sum \sigma_i^2 \le C/n \to 0$.

Proof. Let $Z_n = n^{-1}\sum_{i=1}^{n}(X_i - \mu_i)$. Then $\mathbb{E}[Z_n] = 0$ and, by independence, $\mathrm{Var}(Z_n) = n^{-2}\sum_{i=1}^{n} \sigma_i^2 \to 0$. Chebyshev's inequality gives $\mathbb{P}(|Z_n| > \varepsilon) \le \mathrm{Var}(Z_n)/\varepsilon^2 \to 0$. ∎

Example 7.27 (Stratified survey: urban and rural households in Tanzania)**.** Consider a stratified sample of n households from Tanzania, with $n_U = \lfloor \pi n \rfloor$ urban draws and $n_R = n - n_U$ rural draws, all independent. Urban incomes have mean μ_U and variance σ_U^2; rural incomes have mean μ_R and variance σ_R^2. The combined sample mean

$$\bar{X}_n = \frac{1}{n}\left(\sum_{i=1}^{n_U} X_i^{(U)} + \sum_{j=1}^{n_R} X_j^{(R)}\right)$$

has variance

$$\mathrm{Var}(\bar{X}_n) = \frac{n_U\sigma_U^2 + n_R\sigma_R^2}{n^2} \sim \frac{\pi\sigma_U^2 + (1-\pi)\sigma_R^2}{n} \to 0,$$

so by the WLLN for non-i.i.d. variables, $\bar{X}_n \xrightarrow{p} \pi\mu_U + (1-\pi)\mu_R$. Despite heterogeneity across strata, the sample mean consistently estimates the population mixture mean.

7.7.3 Strong Law of Large Numbers

Theorem 7.28 (Strong Law of Large Numbers — Kolmogorov)**.** *Let $X_1, X_2, \ldots$ be i.i.d. random variables. Then*

$$\bar{X}_n \xrightarrow{\text{a.s.}} \mu$$

if and only if $\mathbb{E}[|X_1|] < \infty$, in which case $\mu = \mathbb{E}[X_1]$.

Remark (On the Proof and Significance of the SLLN)*.* The proof of Kolmogorov's SLLN is substantially more involved than the WLLN, requiring the Borel–Cantelli lemma and truncation arguments (or, alternatively, martingale methods). Its significance relative to the WLLN lies in two respects. First, it requires only $\mathbb{E}[|X_1|] < \infty$, with no finite variance assumption — a meaningful relaxation when modelling heavy-tailed economic variables such as wealth or firm size. Second, it yields almost sure convergence rather than convergence in probability, which is the stronger pathwise statement needed in proofs of consistency for extremum estimators via the Uniform Law of Large Numbers.

7.7.4 Law of Large Numbers for Stationary Time Series

In time-series settings, observations are dependent, so the i.i.d. LLNs do not apply directly. Consistency of time-series estimators requires a version of the LLN that accommodates dependence.

Theorem 7.29 (LLN under Summable Autocovariances)**.** *Let* $\{X_t\}_{t\in\mathbb{Z}}$ *be a covariance-stationary time series with* $\mathbb{E}[X_t] = \mu$*,* $\mathrm{Var}(X_t) = \gamma(0) < \infty$*, and autocovariance function* $\gamma(h) = \mathrm{Cov}(X_t, X_{t-h})$*. If*

$$\sum_{h=-\infty}^{\infty} |\gamma(h)| < \infty,$$

then

$$\bar{X}_T = \frac{1}{T}\sum_{t=1}^{T} X_t \xrightarrow{L^2} \mu \quad \textit{and hence} \quad \bar{X}_T \xrightarrow{p} \mu.$$

Proof. Since $\mathbb{E}[\bar{X}_T] = \mu$, we have $\mathbb{E}[(\bar{X}_T - \mu)^2] = \mathrm{Var}(\bar{X}_T)$. By covariance stationarity,

$$\mathrm{Var}(\bar{X}_T) = \frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\gamma(t-s) = \frac{1}{T}\sum_{h=-(T-1)}^{T-1}\left(1 - \frac{|h|}{T}\right)\gamma(h).$$

Taking absolute values and using the summability condition,

$$\mathrm{Var}(\bar{X}_T) \le \frac{1}{T}\sum_{h=-\infty}^{\infty}|\gamma(h)| = \frac{C}{T} \to 0,$$

where $C = \sum_h |\gamma(h)| < \infty$. Hence $\mathbb{E}[(\bar{X}_T - \mu)^2] \to 0$, giving L^2 convergence, which implies convergence in probability. ■

Remark (Summable Autocovariances and Weak Dependence). The condition $\sum_h |\gamma(h)| < \infty$ is a *weak dependence* condition: it requires that the serial correlation of the process decays fast enough that the sum of all autocovariances is finite. This holds, for instance, when autocovariances decay at a geometric rate (as in stationary ARMA processes) or at a polynomial rate faster than $|h|^{-1}$. Processes with very long memory (such as ARFIMA models with $d \ge 1/2$) violate this condition, and the sample mean may converge more slowly than $T^{-1/2}$, requiring different normalisation in inference.

A stronger result uses the Ergodic Theorem: for a strictly stationary and *ergodic* process with $\mathbb{E}[|X_t|] < \infty$, $\bar{X}_T \xrightarrow{\text{a.s.}} \mu$. Ergodicity means that every event invariant under the time-shift operator has probability zero or one. It is this absence of non-trivial invariant components that makes time averages equal ensemble expectations almost surely. Tail triviality—the triviality of events determined by the arbitrarily distant future—is a distinct and generally stronger property and is not the definition of ergodicity [@davidson2021stochastic; @white2000asymptotic].

7.8 Central Limit Theorems

While Laws of Large Numbers establish that sample averages converge to their population means, they say nothing about the *rate* of convergence or the *shape* of the sampling distribution for large n. Central Limit Theorems (CLTs) supply both: they show that, after scaling by $\sqrt{n}$, the deviation $\bar{X}_n - \mu$ converges in distribution to a normal random variable, regardless of the distribution of the individual observations. This normal approximation is the foundation of virtually all asymptotic inference in econometrics.

7.8.1 Classical CLT (Lindeberg–Lévy)

Theorem 7.30 (Lindeberg–Lévy Central Limit Theorem)**.** *Let $X_1, X_2, \ldots$ be i.i.d. random variables with $\mathbb{E}[X_i] = \mu$ and $0 < \operatorname{Var}(X_i) = \sigma^2 < \infty$. Then*

$$\sqrt{n}\,\frac{\bar{X}_n - \mu}{\sigma} \xrightarrow{d} \mathcal{N}(0,1),$$

or equivalently, $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0,\sigma^2)$.

Logic and Intuition. The CLT is remarkable because the normal limit emerges regardless of the shape of the distribution of the individual X_i: symmetric or skewed, bounded or unbounded, discrete or continuous. The explanation lies in the convolution structure of the sum $S_n = \sum X_i$. Each term independently adds a "contribution" to the sum, and as n grows, the combined distribution of these contributions smooths out, with no single term dominating. The characteristic function proof makes this precise: the characteristic function of the standardised sum converges to $e^{-t^2/2}$, the characteristic function of $\mathcal{N}(0,1)$, because the second-order Taylor expansion of $\log \phi_X(t/\sqrt{n})$ captures the variance term and higher-order contributions vanish.

Remark (CLT in Econometrics). The Lindeberg–Lévy CLT helps underpin the asymptotic normality of the OLS estimator. For i.i.d. observations, assume $\mathbb{E}[u_i \mid x_i] = 0$, $\mathbb{E}[\|x_i\|^2] < \infty$, $Q = \mathbb{E}[x_i x_i']$ is nonsingular, and conditional homoskedasticity $\mathbb{E}[u_i^2 \mid x_i] = \sigma^2$, together with sufficient higher-moment conditions for the CLT. Then

$$\sqrt{n}(\hat{\beta}_{\text{OLS}} - \beta_0) \xrightarrow{d} \mathcal{N}(0,\, \sigma^2 Q^{-1}),$$

where $Q = \operatorname{plim} n^{-1} X'X$ is the population moment matrix of the regressors. Under conditional heteroskedasticity, the covariance becomes the sandwich matrix

$$Q^{-1}\Omega Q^{-1}, \qquad \Omega = \mathbb{E}[u_i^2 x_i x_i'],$$

and heteroskedasticity-robust standard errors are required. With serial dependence, the middle matrix is a long-run covariance and HAC methods are appropriate. These results justify large-sample tests and confidence intervals only under their corresponding assumptions [@hansen2022probability; @white2000asymptotic].

7.8.2 Lyapunov CLT (Non-Identically Distributed)

Theorem 7.31 (Lyapunov Central Limit Theorem)**.** *Let $X_1, X_2, \ldots$ be independent random variables with $\mathbb{E}[X_i] = \mu_i$, $\mathrm{Var}(X_i) = \sigma_i^2 > 0$, and let $s_n^2 = \sum_{i=1}^n \sigma_i^2$. Suppose there exists $\delta > 0$ such that the **Lyapunov condition** holds:*

$$\frac{1}{s_n^{2+\delta}} \sum_{i=1}^n \mathbb{E}\big[|X_i - \mu_i|^{2+\delta}\big] \to 0.$$

Then

$$\frac{1}{s_n} \sum_{i=1}^n (X_i - \mu_i) \xrightarrow{d} \mathcal{N}(0, 1).$$

Logic and Intuition. The Lyapunov condition is a *negligibility* condition: it requires the aggregate $(2+\delta)$-th absolute central moment, scaled by the total spread $s_n^{2+\delta}$, to vanish. This ensures that no individual observation dominates the sum. The condition must be checked using higher-moment bounds; variance information alone is insufficient. One convenient sufficient condition is

$$\sup_i \mathbb{E}[|X_i - \mu_i|^{2+\delta}] < \infty \quad \text{and} \quad s_n^2 \geq cn$$

for some $c > 0$ and all sufficiently large n. The numerator is then $O(n)$ while the denominator is at least of order $n^{1+\delta/2}$, so their ratio tends to zero. Lyapunov's inequality (§6.4) works in the other direction: a finite higher moment guarantees finite lower moments.

Remark (Lyapunov CLT in Econometrics)*.* The Lyapunov CLT is the version of the CLT most commonly invoked in heterogeneous panel data and cross-sectional econometrics, where observations are drawn from different distributions. In stratified surveys, treatment effect estimations with heterogeneous units, and panel regressions with varying regressors, the i.i.d. assumption is implausible, and the Lyapunov CLT provides the theoretical basis for asymptotic normality. It is also used in the asymptotic theory of GMM estimators when the moment conditions involve heterogeneous observations.

7.8.3 CLT for Dependent Sequences

The i.i.d. and Lyapunov CLTs require independence. In time-series settings, where observations are serially correlated, a different set of CLTs applies. The key condition is that the dependence is "weak" enough that the central limit phenomenon persists.

Theorem 7.32 (CLT for Stationary Mixing Sequences — informal statement)**.** *Let $\{X_t\}$ be a strictly stationary and strong-mixing sequence with $\mathbb{E}[X_t] = \mu$, $\mathbb{E}[X_t^2] < \infty$, and mixing coefficients $\alpha(k) \to 0$ sufficiently fast. Define the long-run variance*

$$\Sigma^2 = \sum_{h=-\infty}^{\infty} \mathrm{Cov}(X_0, X_h) = \gamma(0) + 2\sum_{h=1}^{\infty} \gamma(h).$$

If $\Sigma^2 > 0$, then

$$\frac{\sqrt{T}(\bar{X}_T - \mu)}{\Sigma} \xrightarrow{d} \mathcal{N}(0, 1).$$

Remark (Long-Run Variance and HAC Estimation). The critical difference from the i.i.d. case is that the asymptotic variance is the *long-run variance* $\Sigma^2 = \sum_h \gamma(h)$, not merely $\gamma(0) = \mathrm{Var}(X_t)$. Positive serial correlation ($\gamma(h) > 0$ for $h > 0$) inflates Σ^2 above $\gamma(0)$: the sample mean is more variable than it would be under independence, because observations carry redundant information. Negative serial correlation deflates it. In practice, Σ^2 must be estimated from the data; this is the role of **heteroskedasticity and autocorrelation consistent (HAC)** standard errors (e.g. Newey–West), which consistently estimate Σ^2 under general forms of serial correlation and thereby make valid inference possible in time-series regressions.

7.9 Review Exercises

1. Let $\Omega = \{\text{drought}, \text{normal}, \text{flood}\}$. Construct a σ-algebra that distinguishes drought from non-drought conditions but does not distinguish normal rainfall from flooding. Define a probability measure on it.
2. Give an example of two random variables that are uncorrelated but not independent, and verify both claims.
3. State the integrability condition under which $\mathbb{E}[X]$ is finite. Explain why the expectation of a standard Cauchy random variable is not defined.
4. Prove Chebyshev's inequality from Markov's inequality and identify the step that requires a finite variance.
5. Construct a sequence that converges in probability but not almost surely. Explain which part of the definition fails.
6. In the linear model $y_i = x_i'\beta_0 + u_i$, list separately the assumptions needed for consistency and those added for asymptotic normality.
7. Derive the homoskedastic OLS asymptotic covariance and then show where the sandwich middle matrix $\Omega = \mathbb{E}[u_i^2 x_i x_i']$ enters under heteroskedasticity.
8. For a heterogeneous independent triangular array, propose conditions that make the Lyapunov ratio vanish and verify their rate implications.

8 Monte Carlo Illustrations

Mathematical theorems tell us what must happen in the limit; Monte Carlo simulation lets us watch it happen. In this section we use the computer to draw random samples, compute sample statistics, and visualise how those statistics behave as sample size grows. The two results we simulate — the Law of Large Numbers and the Central Limit Theorem — are the cornerstones of asymptotic econometrics established in Section 7. Seeing them emerge from raw simulation, across distributions with very different shapes, builds an intuition that no formal proof alone can provide.

The R code that produces every figure in this section is collected in Appendix B, where it is presented with full annotations. The figures appear here; the mechanics are there.

8.1 The Logic of Monte Carlo Simulation

A Monte Carlo simulation replaces a difficult analytical calculation with a large number of random draws. The idea rests directly on the Law of Large Numbers: if we repeat an experiment a very large number of times B and average the results, the average converges to the true expected value. We therefore use the computer as a probability engine — instructing it to draw pseudo-random numbers from a specified distribution, compute whatever statistic interests us, and repeat the process thousands of times to trace out the statistic's sampling distribution.

For our purposes, the workflow is:

1. **Specify a distribution** with known mean μ and variance σ^2 (e.g. Exponential, Uniform, Bernoulli).
2. **Draw a sample** of size n and compute the sample mean $\bar{X}_n$.
3. **Repeat** step 2 independently $B = 5{,}000$ times, collecting B values of $\bar{X}_n$.
4. **Examine** how the collection of $\bar{X}_n$ values behaves as n changes.

This procedure lets us verify the LLN (the values of $\bar{X}_n$ converge toward μ) and the CLT (the standardised values $\sqrt{n}(\bar{X}_n - \mu)/\sigma$ converge in distribution to $\mathcal{N}(0,1)$) empirically, across distributions of very different shapes.

8.2 Simulating the Law of Large Numbers

8.2.1 Logic and Intuition

The Weak Law of Large Numbers states that for any $\varepsilon > 0$, $\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \to 0$ as $n \to \infty$. The Strong Law strengthens this to almost sure convergence: for almost every realised sequence of draws, the running average $n^{-1}\sum_{i=1}^{n} X_i$ converges to μ as a deterministic sequence of real numbers.

To see this visually, we simulate multiple independent *paths* — each path is one realised sequence of draws — and plot the running mean after each observation. If the LLN holds, every path should eventually settle near μ, regardless of how erratically it begins.

The key insight the simulation reveals is that convergence is a property of the long run, not the short run. In early observations the running mean can wander substantially from μ, especially for distributions with large variance or heavy tails. But as n grows, the probability that the mean deviates from μ by more than any fixed tolerance shrinks to zero. The variance of the sample mean is σ^2/n, which falls at rate $1/n$; the simulation makes this shrinkage visible.

We simulate four distributions with different shapes and moments to show that the LLN is a universal result, not an artefact of normality:

- $\mathcal{N}(0,1)$: symmetric, thin-tailed, $\mu = 0$.
- Exponential(1): right-skewed, $\mu = 1$.
- Uniform$(0,1)$: bounded and flat, $\mu = 0.5$.
- Bernoulli(0.3): discrete, $\mu = 0.3$.

For each distribution, five independent paths are shown.

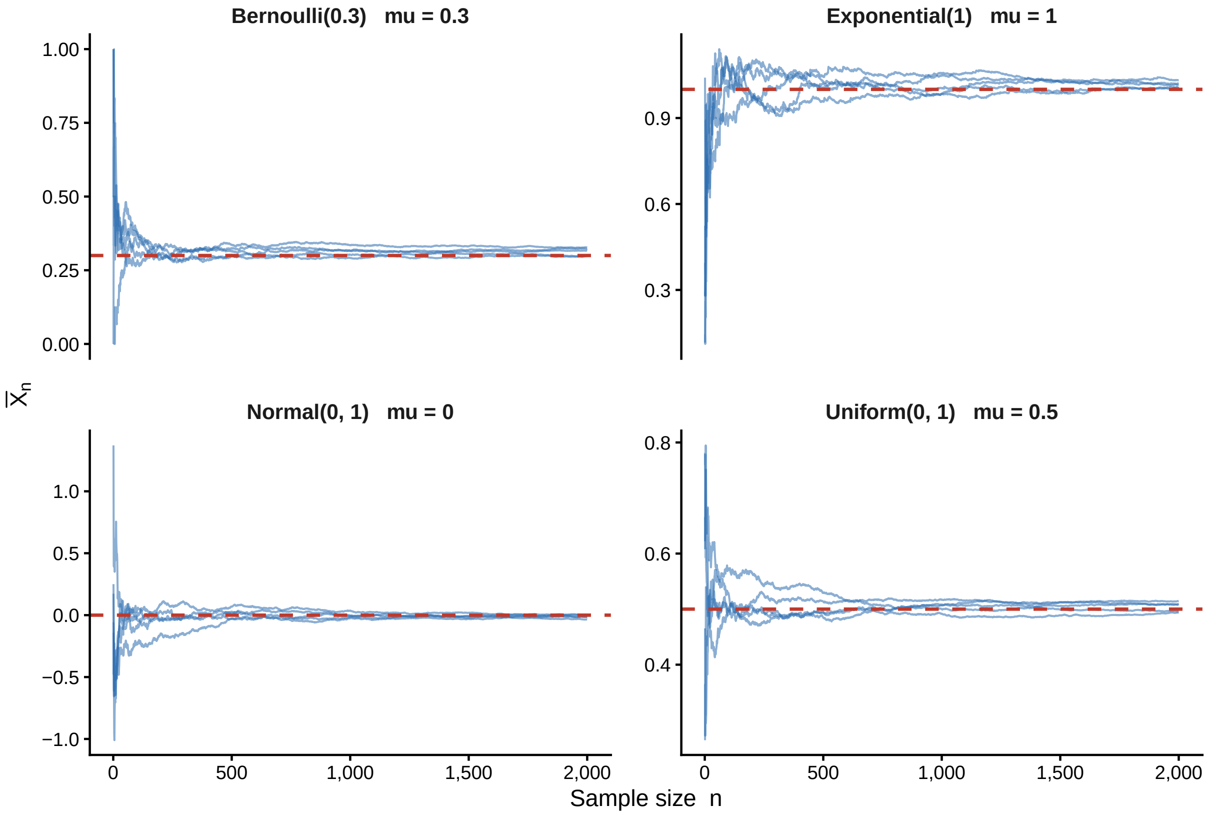

Figure 6: **Law of Large Numbers simulation.** Each panel shows five independent realisations of the running sample mean $\bar{X}_n = n^{-1} \sum_{i=1}^{n} X_i$ (blue lines) plotted against sample size n, for four distributions with markedly different shapes. The red dashed line marks the true population mean μ in each case. Regardless of the distribution, every path converges toward μ as n grows, illustrating the universality of the LLN.

8.2.2 Reading the Figure

Several features of Figure 6 are worth noting.

Early-sample volatility. All four panels show substantial path-to-path variability when n is small. This is not a failure of the LLN; it is exactly what probability theory predicts. $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$ is large when n is small, so the sample mean is genuinely noisy. The LLN is an asymptotic result — it describes the eventual behaviour of the average, not its behaviour at any finite n.

Universality across distributions. The convergence pattern is qualitatively identical across all four distributions, despite their very different shapes. A binary Bernoulli(0.3) variable that only ever takes the values 0 and 1 produces a running mean that converges to 0.3 just as surely as a continuous Normal$(0, 1)$ variable converges to 0. This universality is the content of the LLN.

Rate of convergence. The paths settle toward μ at roughly the same rate across distributions for similar variance. The Exponential(1), which has $\sigma^2 = 1$, converges at about the same rate as the Normal(0,1), also with $\sigma^2 = 1$. This confirms the theoretical result $\text{Var}(\bar{X}_n) = \sigma^2/n$: it is the variance σ^2, not the shape of the distribution, that governs how quickly the mean concentrates.

8.3 Simulating the Central Limit Theorem

8.3.1 Logic and Intuition

The Central Limit Theorem asserts that the standardised sample mean

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}$$

converges in distribution to $\mathcal{N}(0,1)$ as $n \to \infty$, regardless of the distribution of the X_i (provided $\sigma^2 < \infty$). This is a statement about the *shape* of the sampling distribution of $\bar{X}_n$: it becomes increasingly bell-shaped as n grows, no matter how non-normal the original data.

The simulation makes this concrete by repeating the following experiment $B = 5{,}000$ times for each combination of distribution and sample size: draw n observations, compute Z_n, and record its value. After B repetitions, the histogram of the recorded Z_n values approximates the true sampling distribution. Overlaying the $\mathcal{N}(0,1)$ density curve reveals how quickly the approximation becomes accurate.

We use three deliberately non-normal distributions to stress-test the CLT:

- Exponential(1): strongly right-skewed, $\mu = 1$, $\sigma = 1$.
- Uniform$(0,1)$: bounded and perfectly flat, $\mu = 0.5$, $\sigma = 1/\sqrt{12}$.
- Bernoulli(0.3): discrete, heavily asymmetric, $\mu = 0.3$, $\sigma = \sqrt{0.21}$.

For each distribution we examine four sample sizes: $n \in \{1, 5, 30, 100\}$, arranged as columns in the figure.

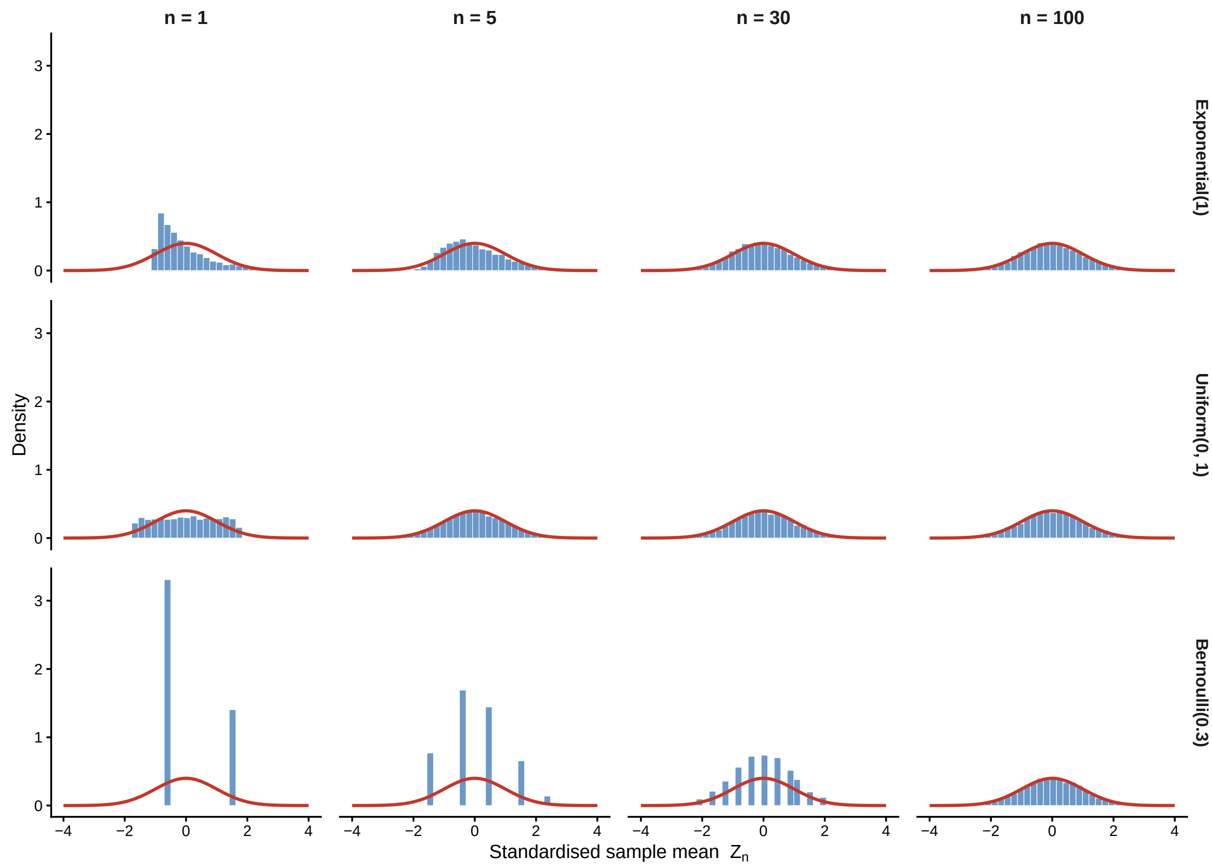

Figure 7: **Central Limit Theorem simulation.** Each panel shows the simulated sampling distribution of the standardised sample mean $Z_n = \sqrt{n}(\bar{X}_n - \mu)/\sigma$ (blue histogram, $B = 5{,}000$ replications) overlaid with the standard normal density (red curve). Rows correspond to three non-normal parent distributions; columns to four sample sizes. The progressive convergence of each histogram toward the bell curve illustrates the CLT.

8.3.2 Reading the Figure

Figure 7 rewards careful examination row by row and column by column.

Along the columns (increasing n). At $n = 1$, the histogram simply reflects the shape of the parent distribution: right-skewed for the Exponential, flat for the Uniform, two-spike for the Bernoulli. By $n = 5$ some bell-shaping is already visible for the Uniform but the Exponential and Bernoulli remain noticeably asymmetric. By $n = 30$ all three distributions are well-approximated by the normal curve, and at $n = 100$ the match is nearly exact in every case. This progression is the CLT in action.

Across the rows (different distributions). The speed of convergence varies with how far the parent distribution is from normality. The Uniform, being symmetric and bounded, approximates normality very quickly — even at $n = 5$ the fit is good. The Exponential, being right-skewed, takes longer; at $n = 30$ there is still a faint rightward tilt. The Bernoulli, being discrete and asymmetric, is the slowest but still well-approximated by $n = 30$. This confirms the theoretical result: the CLT holds for all three, but the rate of convergence depends on the distribution.

The role of standardisation. Notice that the histograms are plotted for the standardised statistic $Z_n = \sqrt{n}(\bar{X}_n - \mu)/\sigma$, not the raw sample mean $\bar{X}_n$. Without standardisation, the mean's distribution would shrink toward a point (by the LLN), giving a trivial degenerate limit. Standardising by $\sqrt{n}$ inflates the scale just enough to keep the distribution non-degenerate, revealing its shape — and it is that shape which converges to $\mathcal{N}(0, 1)$.

Remark (From Simulation to Asymptotic Inference). These simulations reveal why the CLT is the engine of asymptotic inference in econometrics. A fully studentised statistic for a regression coefficient is asymptotically analogous to Z_n. Under the relevant sampling, moment, identification, and dependence conditions, asymptotic normality can justify t-tests and confidence intervals without assuming normally distributed data. The figures also show that convergence can be much slower for skewed or discrete populations. They should not be read as establishing universal sample-size rules: whether $n = 30, 100$, or more is adequate depends on tail behaviour, heterogeneity, dependence, leverage, and the statistic being studied. Simulation calibrated to the empirical application is therefore preferable to a fixed rule of thumb.

9 Appendix

9.1 Lebesgue Measure and Lebesgue Integration

This appendix introduces two foundational ideas of modern mathematics — **Lebesgue measure** and the **Lebesgue integral** — from the ground up, with intuition and pictures rather than formal proofs. No prior exposure to real analysis is assumed. By the end, you should understand why economists and probability theorists use these tools, what they actually do, and how they connect to the expectation operator that appears throughout this text.

Think of this appendix as a guided tour, not a textbook. We will stop at the key landmarks, look at them from multiple angles, and move on without inspecting every brick.

9.1.1 A.1 — The Problem We Are Trying to Solve

Start with something familiar. You know how to compute the average income in a population: add up everyone's income and divide by the number of people. In continuous models — where income Y is drawn from a probability distribution with density f — the average is the integral $\int y f(y)\, dy$. This is just an infinite version of the weighted average, with weights given by the density.

So far, so good. The integral you learned in calculus (the Riemann integral) handles this perfectly well as long as the density f is reasonably smooth. But economics and probability routinely produce objects that are not smooth — distributions with atoms (discrete mass at a point), distributions that are partly discrete and partly continuous, and limiting objects in asymptotic theory that arise as limits of sequences of functions. The Riemann integral struggles or fails with all of these.

The specific failure. The Riemann integral is built by slicing the *horizontal* axis into thin strips, approximating the function over each strip by a constant, and summing up the resulting rectangles. This works beautifully for smooth functions. But consider the function $1_{\mathbb{Q}}$ that equals 1 at every rational number and 0 at every irrational number. Over the interval $[0, 1]$, every horizontal strip contains both rationals and irrationals, so the function oscillates wildly between 0 and 1 inside every strip, no matter how thin you make it. The Riemann integral cannot be defined.

Yet intuitively, the rational numbers in $[0, 1]$ are "vanishingly few" compared to the irrationals — there are only countably many of them, while the irrationals are uncountable. The integral should be 0. The Lebesgue integral agrees: it gives $\int 1_{\mathbb{Q}}\, d\lambda = 0$. The Riemann integral cannot even get started.

Why does this matter for economics? In econometrics, we regularly take limits of sequences of random variables and need to interchange the limit and the integral (i.e., swap lim and $\mathbb{E}[\cdot]$). The theorems that justify this swap — the Monotone Convergence Theorem, Dominated Convergence Theorem, and Fatou's Lemma — are proved using the Lebesgue integral. Without them, asymptotic theory would have no rigorous foundation.

9.1.2 A.2 — Measuring the "Size" of Sets

Before we can integrate, we need a reliable way to measure how large a set is. This is the job of Lebesgue measure.

What does "size" mean? For an interval $[a, b]$, size clearly means length $b - a$. For a rectangle in the plane, it means area. For a solid object in space, it means volume. Lebesgue measure generalises all of these to arbitrary subsets of $\mathbb{R}$ (or $\mathbb{R}^n$), providing a consistent notion of "how much space a set occupies."

The key properties we want any reasonable measure of size to satisfy are:

- **Non-negativity:** every set has a size of at least zero.
- **Normalisation:** a single point has size zero.
- **Additivity:** if two sets do not overlap, their combined size equals the sum of their individual sizes.
- **Consistency with length:** for intervals, size should equal length.

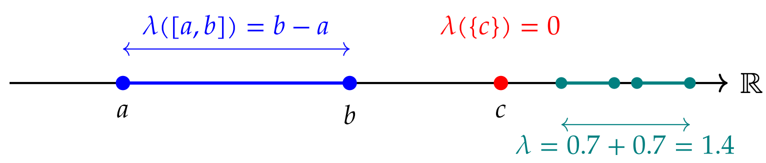

Figure 8: Lebesgue measure assigns length to intervals, zero to single points, and adds lengths of disjoint pieces. An interval $[a, b]$ has measure $b - a$; a single point $\{c\}$ has measure zero; two disjoint intervals have measures that add.

Points have zero size. A single point on the real line occupies no space: $\lambda(\{c\}) = 0$. This might seem obvious, but it has a powerful consequence: countably many points, taken together, also have zero size. The rational numbers $\mathbb{Q}$ are countable (you can list them: $0, 1, -1, \frac{1}{2}, -\frac{1}{2}, 2, \ldots$), so $\lambda(\mathbb{Q}) = 0$. This is true even though the rationals are *dense* on the real line — between any two real numbers there is always a rational. "Dense" does not mean "large in measure." The reals are so much more numerous than the rationals that the rationals take up no measurable space at all.

9.1.3 A.3 — Building Lebesgue Measure: The Covering Idea

Now we need to extend "size" from simple intervals to arbitrary complicated sets. The central technique is **covering**: we approximate the size of a set by asking how efficiently we can cover it with intervals.

Outer measure: wrapping a set in intervals. Given any set E, imagine throwing a collection of open intervals over it like a net, making sure every point of E is inside at least one interval. The total length of the net is an upper bound on the size of E. The **outer measure** $\lambda^*(E)$ is the smallest total length achievable over all possible coverings:

$$\lambda^*(E) = \inf\left\{ \sum_{n=1}^{\infty} |I_n| \;\middle|\; E \subseteq \bigcup_{n=1}^{\infty} I_n,\ I_n \text{ open intervals} \right\}.$$

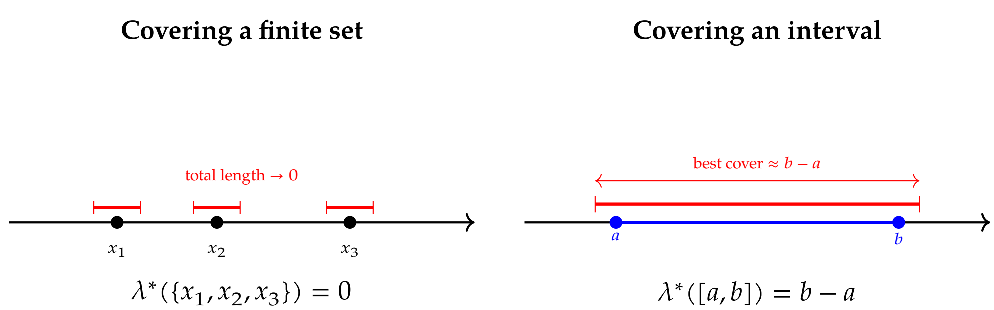

Figure 9: The outer measure $\lambda^*(E)$ is found by covering E with open intervals (shown in red) and minimising the total length of the cover. Isolated points can be covered by arbitrarily thin intervals, so their outer measure is zero. An interval $[a, b]$ can only be covered by something at least as long as $b - a$.

Measurable sets. Outer measure works for any set, but some pathological sets behave inconsistently. A set E is called **Lebesgue measurable** if it cleanly splits every other set A into two non-overlapping pieces — those inside E and those outside — with sizes that add up correctly:

$$\lambda^*(A) = \lambda^*(A \cap E) + \lambda^*(A \cap E^c) \quad \text{for every set } A.$$

This is Carathéodory's criterion. Think of it as a consistency check: a measurable set is one that "plays nicely" with all other sets. For such sets, we write $\lambda(E)$ for the outer measure and call it the Lebesgue measure.

In practice, every set you will ever encounter in economics and probability — open sets, closed sets, intervals, their countable unions and intersections, the support of any distribution — is Lebesgue measurable. The sets that fail the criterion (called non-measurable sets) are mathematical curiosities that require the Axiom of Choice to construct and have no economic interpretation.

Summary: what Lebesgue measure gives us. The Lebesgue measure λ is a function that assigns a non-negative real number (or $+\infty$) to every measurable set, satisfying:

- $\lambda([a, b]) = b - a$ (consistent with ordinary length),
- $\lambda(\bigcup_n E_n) = \sum_n \lambda(E_n)$ for pairwise disjoint measurable sets E_n (countable additivity),
- $\lambda(E + t) = \lambda(E)$ for every real number t (translation invariance: shifting a set doesn't change its size).

9.1.4 A.4 — The Riemann Integral and Its Limits

Before introducing the Lebesgue integral, it helps to understand exactly what the Riemann integral does and why it is not enough for probability theory.

How the Riemann integral works. To compute $\int_a^b f(x)\,dx$ in the Riemann sense, you partition the horizontal axis $[a, b]$ into n thin sub-intervals $[x_{k-1}, x_k]$, approximate f over each sub-interval by a constant (say, the value at the left endpoint), and sum the resulting rectangles. As the sub-intervals get thinner ($n \to \infty$), the sum converges to the integral — provided the function is well-behaved enough.

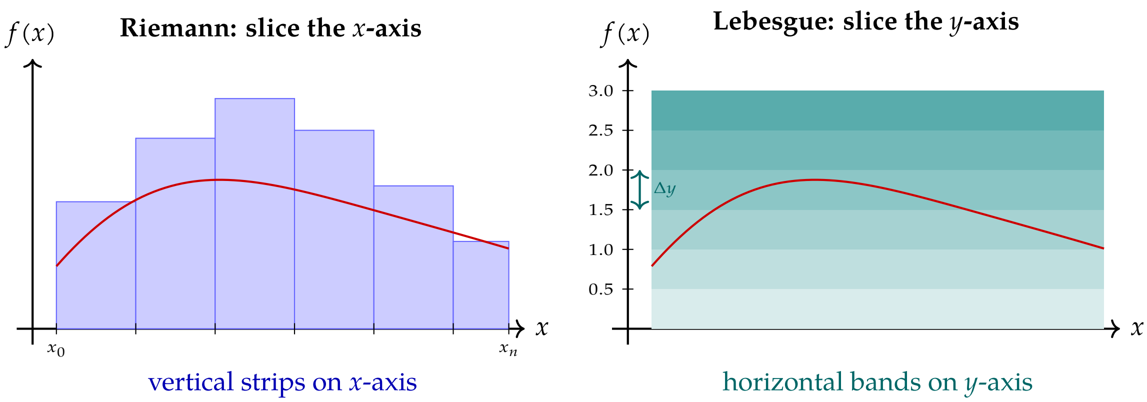

Figure 10: **Riemann vs. Lebesgue integration.** Riemann slices the domain (the x-axis) into thin vertical strips and sums rectangles of width Δx and height $f(x)$ (left). Lebesgue slices the range (the y-axis) into thin horizontal bands and asks: *how much of the x-axis maps into this band?* (right). Both approaches give the same answer for smooth functions, but the Lebesgue approach works for a far larger class of functions.

Where Riemann fails. The Riemann integral requires that, as sub-intervals shrink, the upper and lower approximations to the function converge to the same limit. For the function $1_{\mathbb{Q}}$ (equal to 1 on rationals, 0 on irrationals), every sub-interval contains both rationals and irrationals, so the upper approximation is always 1 and the lower approximation is always 0. They never meet, so the Riemann integral does not exist.

More generally, the Riemann integral has difficulty with functions that have too many discontinuities, with limits of sequences of integrable functions, and with integrals over abstract spaces (such as the probability space Ω) that have no natural notion of "interval." The Lebesgue integral overcomes all of these.

9.1.5 A.5 — The Lebesgue Integral: A Better Way to Average

The key idea: slice the range, not the domain. Instead of partitioning the x-axis into strips and approximating the function from above and below, the Lebesgue integral partitions the y-axis (the set of output values) and asks: *for each output level, how much of the input produces that output?* That "how much" is answered by Lebesgue measure.

Think of it this way. You want to compute the total wage bill for a group of workers. The Riemann approach says: go through each worker one by one (x-axis = workers), multiply their wage by their hours, and add up. The Lebesgue approach says: group workers by pay grade (y-axis = wages), compute how many workers are in each pay grade (measure of the pre-image), multiply by the wage, and add up. Both approaches give the same total, but the second is more powerful because it can handle groups of any structure, not just ordered lists.

Step 1: Simple functions. The simplest non-constant functions are **simple functions** — functions that take only finitely many distinct values. A simple function looks like a staircase: it equals a constant a_i on each of finitely many measurable sets E_i.

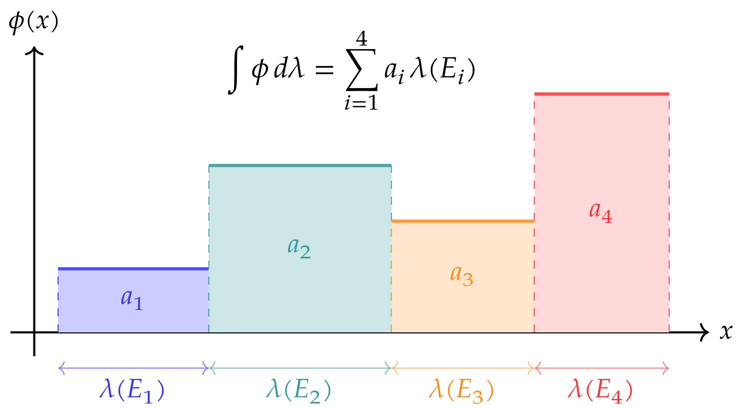

Figure 11: A **simple function** takes finitely many values a_1, a_2, a_3, a_4 on disjoint measurable sets E_1, E_2, E_3, E_4. Its Lebesgue integral is the weighted sum $\sum_i a_i \lambda(E_i)$: value times measure-of-the-set-where-that-value-is-taken. Think of a_i as a wage rate and $\lambda(E_i)$ as the number of workers at that rate; the integral is the total wage bill.

For a simple function $\phi = \sum_{i=1}^{n} a_i 1_{E_i}$, the Lebesgue integral is defined as:

$$\int \phi\, d\lambda = \sum_{i=1}^{n} a_i \lambda(E_i).$$

This is completely natural: multiply each value by the measure (size) of the set where the function takes that value, and sum up. It is a weighted average where the weights are the "sizes" of the regions, not just their count.

Step 2: General non-negative functions. Any non-negative measurable function f can be approximated by an increasing sequence of simple functions $\phi_1 \le \phi_2 \le \cdots \le f$ that converge upward to f at every point. The Lebesgue integral of f is defined as the limit of the integrals of the approximations:

$$\int f\, d\lambda = \lim_{n\to\infty} \int \phi_n\, d\lambda.$$

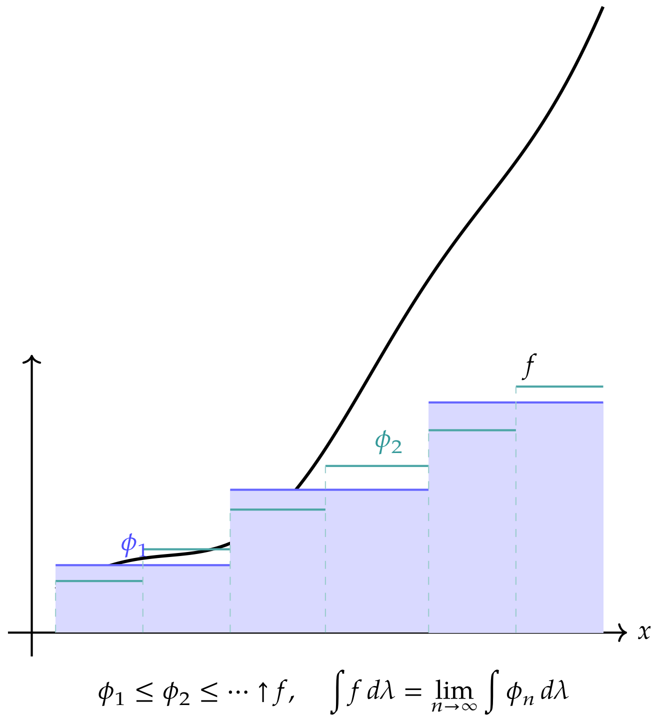

Figure 12: A general non-negative function f (black curve) is approximated from below by an increasing sequence of simple functions: a coarse staircase ϕ_1 (blue steps) and a finer one ϕ_2 (teal steps). As the steps get thinner and more numerous, the staircase rises toward f, and the integrals converge to the Lebesgue integral of f.

Step 3: Functions with positive and negative parts. For a general function f that can be negative, we split it into its positive part $f^+ = \max(f, 0)$ and negative part $f^- = \max(-f, 0)$, integrate each separately using Step 2, and subtract:

$$\int f\, d\lambda = \int f^+\, d\lambda - \int f^-\, d\lambda.$$

If both integrals are finite, f is called **Lebesgue integrable** and we write $f \in L^1(\lambda)$.

9.1.6 A.6 — Three Advantages That Matter for Probability

The Lebesgue integral agrees with the Riemann integral whenever both are defined — nothing is lost from what you already know. But the Lebesgue integral does three things that the Riemann integral cannot.

Advantage 1: Handles discontinuous functions. The Lebesgue integral can handle functions with far more discontinuities than the Riemann integral. In particular, it handles the mixed discrete-continuous distributions common in economics. A distribution with a mass point at c (e.g., a corner solution where a household has exactly zero savings) is neither purely discrete nor purely continuous. The Lebesgue integral can integrate over it seamlessly, because the "weight" of the mass point at c is simply $\mathbb{P}(\{c\}) > 0$, which Lebesgue measure assigns correctly. The Riemann integral cannot handle this without special cases and workarounds.

Advantage 2: "Almost everywhere" is the right standard. Under Lebesgue integration, two functions that agree except on a set of measure zero are treated as identical. Two functions f and g are said to be equal **almost everywhere** (written $f = g$ a.e.) if the set where they differ has measure zero:

$$\lambda(\{x : f(x) \neq g(x)\}) = 0.$$

This is important in probability because we typically do not care about events that have probability zero. Modifying a density function at finitely many isolated points changes nothing — those points form a set of measure zero and contribute zero to any integral.

Advantage 3: Limits and integrals can be swapped. This is the most important advantage for econometric theory. The three key results are:

- **Monotone Convergence Theorem (MCT):** If $0 \leq f_1 \leq f_2 \leq \cdots$ and $f_n \to f$ pointwise, then $\int f_n \, d\lambda \to \int f \, d\lambda$.
- **Dominated Convergence Theorem (DCT):** If $f_n \to f$ pointwise and $|f_n| \leq g$ for an integrable function g, then $\int f_n \, d\lambda \to \int f \, d\lambda$.
- **Fatou's Lemma:** $\int \liminf_n f_n \, d\lambda \leq \liminf_n \int f_n \, d\lambda$ for non-negative f_n.

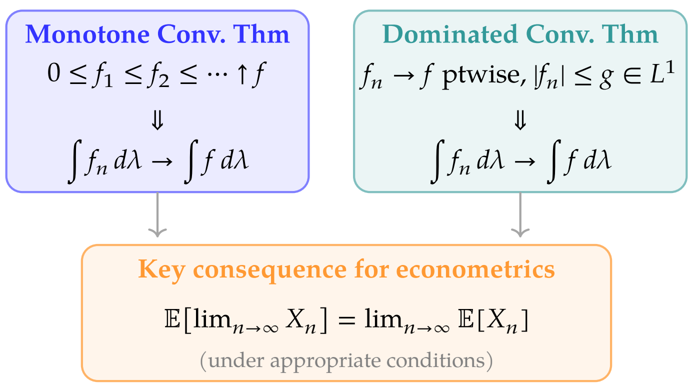

Figure 13: The Monotone and Dominated Convergence Theorems both justify swapping a limit and a Lebesgue integral — equivalently, swapping a limit and an expectation. These interchanges are used constantly in econometric proofs of consistency, asymptotic normality, and the validity of moment conditions.

These theorems are used throughout econometric theory. Consistency proofs often need to show that $\mathbb{E}[g(X_n)] \to \mathbb{E}[g(X)]$ for some function g; the Dominated Convergence Theorem justifies moving the limit inside the expectation. Derivations of moment conditions for GMM and proofs of the existence of moments for asymptotic distributions also invoke these results repeatedly.

9.1.7 A.7 — From Lebesgue to Probability: Expectation as an Integral

Everything above extends naturally from the real line to an abstract probability space $(\Omega, \mathcal{F}, \mathbb{P})$. A probability measure $\mathbb{P}$ is simply a Lebesgue-type measure on the abstract space Ω, with the single additional requirement that the total mass equals one: $\mathbb{P}(\Omega) = 1$.

A random variable $X : \Omega \to \mathbb{R}$ is a measurable function from the abstract outcome space Ω to the real line. Its **expected value** is the Lebesgue integral of X with respect to $\mathbb{P}$:

$$\mathbb{E}[X] = \int_{\Omega} X(\omega)\, d\mathbb{P}(\omega).$$

This single expression unifies three cases that are usually treated separately in elementary statistics:

Type of X	Formula for $\mathbb{E}[X]$
Discrete (X takes values x_i)	$\sum_i x_i\, \mathbb{P}(X = x_i)$
Continuous (density f)	$\int_{-\infty}^{\infty} x f(x)\, dx$
General (any distribution)	$\int_{\Omega} X(\omega)\, d\mathbb{P}(\omega)$

Lebesgue integral

All three are special cases of the same Lebesgue integral $\int X\, d\mathbb{P}$.

Figure 14: The Lebesgue integral $\int X\, d\mathbb{P}$ unifies the three ways of computing expectations. The discrete and continuous formulae you learned in introductory probability are both special cases. The Lebesgue formulation works for any random variable on any probability space, including mixed discrete-continuous distributions and limits of sequences of random variables.

This unification is not merely aesthetic. It means that every theorem proved using the Lebesgue integral — including the convergence theorems of Section A.6 — applies simultaneously to discrete, continuous, and mixed random variables, without separate case-by-case arguments. In econometrics, where sample spaces and distributions can be exotic (panel data with fixed effects, treatment variables with mass points, generated regressors), this generality is indispensable.

The connection to the rest of this text. Every time you see $\mathbb{E}[X]$ in the main text, you are seeing the Lebesgue integral $\int X\, d\mathbb{P}$ in shorthand. Every convergence theorem in Section 7 — the Laws of Large Numbers, the Central Limit Theorems, the limit-swap results used in their proofs — rests on the measure-theoretic framework built here. Lebesgue measure and Lebesgue integration are not abstract generalisations for their own sake; they are the precise, reliable machinery that makes modern probability theory — and therefore modern econometrics — work.

For a thorough graduate-level treatment, see Billingsley, Probability and Measure (3rd ed.); Royden and Fitzpatrick, Real Analysis (4th ed.); and Folland, Real Analysis: Modern Techniques and Their Applications (2nd ed.).

9.2 R Code: Monte Carlo Simulations

This section contains the fully annotated R code that produces the Law of Large Numbers figure (Figure 6) and the Central Limit Theorem figure (Figure 7) in Section 8. Both simulations use base R for random number generation and `ggplot2` with `theme_classic()` for all graphics. No additional packages beyond `ggplot2` and `scales` are required.

9.2.1 Simulating the Law of Large Numbers

This program plots the running sample mean $\bar{X}_k = k^{-1} \sum_{i=1}^{k} X_i$ against the sample size k, for five independent samples drawn from each of four distributions, and overlays the true population mean as a reference line. If the LLN holds, every path should settle onto that line as k grows. The code is organised into four stages, presented and explained one at a time below.

Stage 1 — Load the library, fix the seed, set the parameters. Loading `ggplot2` makes the plotting functions available. The call to `set.seed(42)` fixes the state of R's random-number generator so that re-running the script reproduces exactly the same figure — essential for a result that others must be able to replicate. The two parameters control how much data we generate: `n_max` is the largest sample size shown on the horizontal axis, and `n_paths` is how many independent repetitions we draw per distribution (five, so the reader can see that *all* paths converge, not just a lucky one).

```
library(ggplot2)    # plotting functions
set.seed(42)        # reproducible random numbers

n_max   <- 2000     # longest sample path (x-axis: 1 .. n_max)
n_paths <- 5        # independent repetitions per distribution
```

Stage 2 — A helper that builds the running-mean paths for one distribution. The function `make_paths()` takes three arguments: `rfun`, a function that draws `n` observations from the target distribution; `mu_true`, that distribution's known population mean (used later for the reference line); and `dist_label`, the text shown above each panel. For each of the `n_paths` repetitions it draws one full sample `x` of length `n_max`, then computes the running mean at every step. The key expression is `cumsum(x) / seq_len(n_max)`: `cumsum(x)` is the vector of partial sums (x_1, $x_1 + x_2$, ...), and dividing element-wise by $(1, 2, 3, \ldots)$ yields the sequence of sample means $\bar{X}_1, \bar{X}_2, \ldots$ The results for the five repetitions are stacked into one data frame by `do.call(rbind, ...)`.

```r
# Build the running means of n_paths independent samples
# from one distribution and stack them in a data frame.
#   rfun       : function(n) drawing n observations
#   mu_true    : population mean (for the reference line)
#   dist_label : label shown on the panel strip
make_paths <- function(rfun, mu_true, dist_label) {
  do.call(rbind, lapply(seq_len(n_paths), function(p) {
    x <- rfun(n_max)                # one sample of size n_max
    data.frame(
      n            = seq_len(n_max),
      # running mean X-bar_k for every k = 1 .. n_max
      running_mean = cumsum(x) / seq_len(n_max),
      path         = factor(p),     # which repetition
      distribution = dist_label,    # panel label
      mu_true      = mu_true        # true mean
    )
  }))
}
```

Stage 3 — Run the helper for four distributions. We call `make_paths()` four times, once per distribution, and `rbind` the results into a single long data frame `lln_data`. The four distributions are deliberately varied — a symmetric continuous law, a skewed one, a bounded one, and a discrete one — to show that the LLN does not depend on the shape of the underlying distribution. Each is paired with its known mean. The last line extracts the distinct (`distribution`, `mu_true`) pairs into `mu_ref`, which supplies one dashed reference line per panel.

```r
lln_data <- rbind(
  make_paths(function(n) rnorm(n, 0, 1),
             mu_true = 0,   dist_label = "Normal(0,1), mu=0"),
  make_paths(function(n) rexp(n, rate = 1),
             mu_true = 1,   dist_label = "Exponential(1), mu=1"),
  make_paths(function(n) runif(n, 0, 1),
             mu_true = 0.5, dist_label = "Uniform(0,1), mu=0.5"),
  make_paths(function(n) rbinom(n, 1, 0.3),
             mu_true = 0.3, dist_label = "Bernoulli(0.3), mu=0.3")
)

# one reference row (true mean) per distribution
mu_ref <- unique(lln_data[, c("distribution", "mu_true")])
```

Stage 4 — Draw the plot. The `ggplot()` call maps sample size `n` to the horizontal axis and `running_mean` to the vertical axis, with `group = path` so that each repetition is drawn as its own line. `geom_line()` renders the five paths as translucent blue curves; `geom_hline()` adds the dashed red line at the true mean, read from `mu_ref`; and `facet_wrap()` splits the plot into one panel per distribution, each with its own vertical scale (`scales = "free_y"`). The remaining lines label the axes and apply the classic (unshaded) theme.

```r
ggplot(lln_data, aes(x = n, y = running_mean, group = path)) +
  # the five sample-mean paths (translucent blue)
  geom_line(colour = "#2b6cb0", alpha = 0.55,
            linewidth = 0.45) +
  # the true population mean (dashed red)
  geom_hline(data = mu_ref, aes(yintercept = mu_true),
             colour = "#c0392b", linetype = "dashed",
             linewidth = 0.75) +
  # one panel per distribution, independent y-scales
  facet_wrap(~ distribution, scales = "free_y", nrow = 2) +
  scale_x_continuous(labels = scales::comma) +
  labs(x = "Sample size  n", y = expression(bar(X)[n])) +
  theme_classic(base_size = 11) +
  theme(
    strip.background = element_blank(),
    strip.text       = element_text(face = "bold", size = 10),
    axis.title       = element_text(size = 11),
    panel.spacing    = unit(1.1, "lines")
  )
```

9.2.2 Simulating the Central Limit Theorem

This program approximates the *sampling distribution* of the standardised sample mean by brute force: for each distribution and each sample size n, it draws $B = 5{,}000$ independent samples, computes and standardises the mean of each, and plots a histogram of the 5,000 standardised values against the $\mathcal{N}(0, 1)$ density. As n grows, the histograms should approach the bell curve regardless of the parent distribution. The code again proceeds in stages.

Stage 1 — Parameters. `B` is the number of Monte Carlo replications: the more samples we draw, the smoother and more accurate the estimated sampling distribution. `sample_sizes` lists the four values of n we examine; these become the columns of the final figure and let us watch the approximation improve as n increases.

```r
library(ggplot2)
set.seed(42)                          # reproducibility

B            <- 5000                  # Monte Carlo replications
sample_sizes <- c(1, 5, 30, 100)      # sample sizes examined
```

Stage 2 — Describe the three distributions. Each distribution is stored as a small list holding four items: a display `name`, a draw function `rfun`, the true mean `mu`, and the true standard deviation `sigma`. The mean and standard deviation are needed because standardising a sample mean requires the *exact* population values, not estimates. The three closed-form standard deviations used are $\sigma = 1$ for Exp(1), $\sigma = \sqrt{1/12}$ for Uniform$(0, 1)$, and $\sigma = \sqrt{p(1-p)} = \sqrt{0.3 \cdot 0.7}$ for Bernoulli(0.3).

```r
# Each entry: name, draw function, true mean, true sd.
dist_specs <- list(
  list(name  = "Exponential(1)",
       rfun  = function(n) rexp(n, rate = 1),
       mu    = 1,
       sigma = 1),                    # sd of Exp(1)
  list(name  = "Uniform(0, 1)",
       rfun  = function(n) runif(n, 0, 1),
       mu    = 0.5,
       sigma = sqrt(1 / 12)),         # sd of Unif(0,1)
  list(name  = "Bernoulli(0.3)",
       rfun  = function(n) rbinom(n, 1, 0.3),
       mu    = 0.3,
       sigma = sqrt(0.3 * 0.7))       # sd of Bern(0.3)
)
```

Stage 3 — The simulation loop (the heart of the program). Two nested loops (written with `lapply`) run over every distribution `d` and every sample size `n`. For a given pair, the line

```r
means <- replicate(B, mean(d$rfun(n)))
```

does the core work: `d$rfun(n)` draws one sample of size `n`, `mean(...)` collapses it to a single sample mean, and `replicate(B, ...)` repeats this `B` times, yielding a vector of 5,000 sample means. The next line standardises each mean,

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma / \sqrt{n}},$$

which is exactly the transformation appearing in the CLT. The standardised values, together with labels identifying the sample size and distribution, are stored in a data frame; `do.call(rbind, ...)` stacks all the pieces into one long table `clt_data`.

```r
clt_data <- do.call(rbind, lapply(dist_specs, function(d) {
  do.call(rbind, lapply(sample_sizes, function(n) {
    # B independent sample means of size n
    means <- replicate(B, mean(d$rfun(n)))
    # standardise each mean to Z_n
    z <- (means - d$mu) / (d$sigma / sqrt(n))
    data.frame(
      z            = z,
      n_label      = paste0("n = ", n),
      distribution = d$name
    )
  }))
}))
```

Stage 4 — Fix the panel order and build the reference curve. By default R orders factor levels alphabetically, which would scramble the panels. Converting `n_label` and `distribution` to factors with explicit `levels` forces the columns to appear in increasing n and the rows in the intended order. The last two lines tabulate the standard normal density on a fine grid from -4 to 4; this is the red curve overlaid on every panel.

```
# put panels in the intended order (not alphabetical)
clt_data$n_label <- factor(
  clt_data$n_label,
  levels = paste0("n = ", sample_sizes))
clt_data$distribution <- factor(
  clt_data$distribution,
  levels = sapply(dist_specs, `[[`, "name"))

# N(0,1) density for the reference curve
norm_df <- data.frame(z = seq(-4, 4, length.out = 300))
norm_df$density <- dnorm(norm_df$z)
```

Stage 5 — Draw the plot. `geom_histogram()` draws the empirical sampling distribution, with `aes(y = after_stat(density))` rescaling the bars to integrate to one so they are directly comparable to a density. `geom_line()` overlays the $\mathcal{N}(0,1)$ curve from `norm_df`. `facet_grid(distribution ~ n_label)` arranges the panels in a grid — one row per distribution, one column per sample size — so the eye can track convergence across each row. `coord_cartesian(xlim = c(-4, 4))` trims the view to the region of interest.

```
ggplot(clt_data, aes(x = z)) +
  # empirical sampling distribution (density-scaled)
  geom_histogram(aes(y = after_stat(density)),
                 bins = 55, fill = "#2b6cb0",
                 colour = "white", alpha = 0.70,
                 linewidth = 0.15) +
  # N(0,1) reference density (red)
  geom_line(data = norm_df, aes(x = z, y = density),
            colour = "#c0392b", linewidth = 0.80,
            inherit.aes = FALSE) +
  # rows = distributions, columns = sample sizes
  facet_grid(distribution ~ n_label) +
  coord_cartesian(xlim = c(-4, 4)) +
  labs(x = expression("Standardised mean  " * Z[n]),
       y = "Density") +
  theme_classic(base_size = 10) +
  theme(
    strip.background = element_blank(),
    strip.text.x     = element_text(face = "bold", size = 10),
    strip.text.y     = element_text(face = "bold", size = 9),
    panel.spacing    = unit(0.7, "lines"),
    axis.title       = element_text(size = 10)
  )
```

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